Sampling Design and Survey Practice

Lecture Notes (Ch. 1–11)

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Chapter 1. Introduction and Elementary Concepts

1 Collection of Survey Data

1.1 Primary and secondary data

Definition 1.1

- **Primary data**: data collected by the investigator directly from the source.
- **Secondary data**: data that has already been collected by some agency or individual and is available in published or unpublished records.

Primary data are collected for the purpose at hand, so they fit the target of the study; secondary data are cheaper but were collected for some other purpose and may not match the target exactly. Primary data can be collected by

- direct personal interviews, or interviews by enumerators,
- questionnaires sent by mail,
- telephone interviews.

1.2 Framing a questionnaire

The person framing the questionnaire should know the field of inquiry in detail. Points to keep in mind:

- The surveyor introduces themselves and states the objective of the survey.
- Questions are clear, unambiguous and to the point; leading questions are avoided.
- Questions affecting the prestige or sentiments of respondents, and questions requiring calculations, are avoided.
- Precise instructions for filling in the questionnaire, including units of measurement, are given.
- The questionnaire is concise. Every question costs response quality, so the number of questions should be kept close to the number of quantities (parameters) one actually wants to estimate.
- The layout is visually appealing.

1.3 Technical terms

Definition 1.2

- An **element** is a unit for which information is sought.
- The **population** (universe) is the aggregate of elements about which inference is to be made.
- **Sampling units** are non-overlapping collections of population elements.
- The **sampling frame** is a list of all sampling units in the population.
- A **sample** is a subset of the population selected from the frame to draw inferences about

population characteristics.

Collecting information on every unit of the population is a **complete enumeration** or **census**. Often a census is impossible (the target population is not fully accessible in practice), and a sample survey has advantages over it: greater speed, more detailed information, reduced cost and, perhaps surprisingly, greater accuracy, since a small survey can be run with trained staff and careful follow-up, so non-sampling errors are smaller.

1.4 Probability and non-probability sampling

The method used to select the sample is the **sampling procedure**.

- **Probability sampling**: units are selected by a probability mechanism. Only these procedures are considered in this course.
- **Non-probability sampling**: units are selected without a probability mechanism (convenience, quota, judgement); such procedures are subject to human bias.

1.5 With and without replacement

Definition 1.3

- **With replacement (WR)**: units are drawn one by one, and the unit selected at each draw is returned to the population before the next draw. If order matters, there are N^n possible samples.
- **Without replacement (WOR)**: the selected unit is not returned before the next draw. There are $\binom{N}{n}$ possible (unordered) samples.

Example 1.4: Birth weights

Four children A, B, C, D have birth weights $5.5, 8.0, 6.5, 7.0$ pounds. The WR samples of size 2 (ordered) are the $4^2 = 16$ pairs

$$(A,A),(A,B),(A,C),(A,D),(B,A),\ldots,(D,D),$$

whereas WOR gives the $\binom{4}{2} = 6$ samples $\{A,B\},\{A,C\},\{A,D\},\{B,C\},\{B,D\},\{C,D\}$.

In R: `sample(n)` is a random permutation of $1,\ldots,n$; `sample(x)` permutes x; `sample(x, n)` draws n items WOR; `sample(x, n, replace=T)` draws WR; `sample(x, replace=T)` is a bootstrap sample.

1.6 Planning and execution of a sample survey

Objectives → population to be studied → sampling unit → sampling frame → sample selection (sample size and procedure) → method of collecting information → pilot survey → handling of nonresponse → organization of fieldwork → analysis of data and report.

2 Elementary Concepts

2.1 Expectation, variance, covariance

If x takes values $x_1, \dots, x_k$ with probabilities $p_1, \dots, p_k$,

$$\mathrm{E}(x) = \sum_{i=1}^{k} p_i x_i, \qquad \mathrm{Var}(x) = \mathrm{E}[x - \mathrm{E}(x)]^2 = \mathrm{E}(x^2) - [\mathrm{E}(x)]^2,$$

$$\mathrm{Cov}(x, y) = \mathrm{E}[(x - \mathrm{E}x)(y - \mathrm{E}y)] = \mathrm{E}(xy) - \mathrm{E}(x)\mathrm{E}(y),$$

$$\mathrm{Var}\Big(\sum_{i=1}^{k} a_i x_i\Big) = \sum_{i=1}^{k} a_i^2 \mathrm{Var}(x_i) + 2\sum_{i<j} a_i a_j \mathrm{Cov}(x_i, x_j).$$

2.2 Parameters and statistics

Definition 1.5

A **parameter** is a real-valued function of the variable values of all population units. A **statistic** is a real-valued function of the sample values; a statistic used to estimate a parameter is an **estimator**.

For population values $Y_1, \dots, Y_N$ and sample values $y_1, \dots, y_n$:

$$\mu = \frac{1}{N}\sum_{i=1}^{N} Y_i, \quad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(Y_i - \mu)^2, \quad S^2 = \frac{1}{N-1}\sum_{i=1}^{N}(Y_i - \mu)^2 = \frac{N}{N-1}\sigma^2,$$

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2.$$

The finite-population variance S^2 (divisor $N-1$) is the natural target of s^2 under SRS without replacement (Ch. 2), and $(N-1)S^2 = N\sigma^2$.

2.3 Bias, variance and MSE

Definition 1.6

- $\hat{\theta}$ is **unbiased** for θ if $\mathrm{E}(\hat{\theta}) = \theta$. Otherwise its **bias** is $B(\hat{\theta}) = \mathrm{E}(\hat{\theta}) - \theta$ and its **relative bias** is $RB(\hat{\theta}) = B(\hat{\theta})/\theta$.
- Sampling variance $\mathrm{Var}(\hat{\theta}) = \mathrm{E}[\hat{\theta} - \mathrm{E}\hat{\theta}]^2$; standard error $SE(\hat{\theta}) = \sqrt{\mathrm{Var}(\hat{\theta})}$.
- Mean squared error $\mathrm{MSE}(\hat{\theta}) = \mathrm{E}(\hat{\theta} - \theta)^2$.
- The **relative efficiency** of $\hat{\theta}_2$ with respect to $\hat{\theta}_1$ is $\mathrm{RE}(\hat{\theta}_1, \hat{\theta}_2) = \mathrm{MSE}(\hat{\theta}_1)/\mathrm{MSE}(\hat{\theta}_2)$; if it exceeds 1, $\hat{\theta}_2$ is more efficient.

Theorem 1.7: Bias–variance decomposition

$\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + [B(\hat{\theta})]^2$.

Proof.

$$\mathrm{E}(\hat{\theta} - \theta)^2 = \mathrm{E}\big[(\hat{\theta} - \mathrm{E}\hat{\theta}) + (\mathrm{E}\hat{\theta} - \theta)\big]^2$$

$$= \mathrm{E}(\hat\theta - \mathrm{E}\hat\theta)^2 + (\mathrm{E}\hat\theta - \theta)^2 + 2(\mathrm{E}\hat\theta - \theta)\,\mathrm{E}(\hat\theta - \mathrm{E}\hat\theta),$$

and the last expectation is zero. ■

2.4 Confidence intervals and the bound on the error

For a desired probability $1-\alpha$ we look for a bound B with

$$P(|\hat\theta - \theta| \le B) = 1-\alpha \iff P(\hat\theta - B \le \theta \le \hat\theta + B) = 1-\alpha,$$

so $(\hat\theta - B, \hat\theta + B)$ is a confidence interval with confidence coefficient $1-\alpha$.

- **Normal case**: if $\hat\theta$ is (approximately) normal and unbiased, $B = z_{\alpha/2}\sigma_{\hat\theta}$. For $1-\alpha = 0.95$, $z_{0.025} = 1.96 \approx 2$, so $B = 2\sigma_{\hat\theta}$, the "2-SD bound on the error of estimation" used throughout the course.
- **Any distribution**: by Chebyshev's inequality $P(|\hat\theta - \theta| \le 2\sigma_{\hat\theta}) \ge 1 - \frac{1}{4} = 0.75$.

2.5 Sample size determination

Only the investigator can say how much precision is needed. It is expressed as $P(|\hat\theta - \theta| \le B) = 1-\alpha$, where B is the **tolerable error** (margin of error). For surveys of people measuring a proportion, $B = 0.03$ and $\alpha = 0.05$ are typical.

Example 1.8: Opinion polls

With the normal approximation and $p = 1/2$ (the worst case),

$$B = 1.96\sqrt{\frac{p(1-p)}{n}} \le 1.96\sqrt{\frac{1}{4n}}.$$

Setting $B = 0.03$ gives $n = (1.96)^2/(4\cdot 0.03^2) = 1067.1$, so about 1,068 respondents; this is why national polls typically report a margin of error of $\pm 3\%$ with about a thousand respondents.

3 Selected exercises

Motivation. These items practise the basic tools of the chapter: the probability guarantee behind the 2-SD bound, an exact sampling distribution computed by enumeration, and the fact that a biased estimator can beat an unbiased one in MSE.

Exercise 1.1 (Chebyshev guarantee for the bound on the error)

Let $\hat\theta$ be an estimator of θ with finite variance, and let $k > 0$.
(a) If $\hat\theta$ is unbiased with standard error $\sigma_{\hat\theta} > 0$, show that $P(|\hat\theta - \theta| \le k\sigma_{\hat\theta}) \ge 1 - 1/k^2$.
(b) Without assuming unbiasedness, show that $P(|\hat\theta - \theta| \le k\sqrt{\mathrm{MSE}(\hat\theta)}) \ge 1 - 1/k^2$ whenever $\mathrm{MSE}(\hat\theta) > 0$, and explain why the 2-SD bound of a biased estimator should be built from $\sqrt{\mathrm{MSE}}$ rather than from the standard error.

Solution. [Supplement] Both parts follow from one inequality.

Step 1. (Markov's inequality.) For a random variable $U \ge 0$ and $a > 0$, $a\,\mathbf{1}\{U \ge a\} \le U$ pointwise; taking expectations gives $P(U \ge a) \le \mathrm{E}(U)/a$.

Step 2. Apply Step 1 to $U = (\hat\theta - \theta)^2$ and $a = k^2\mathrm{E}(\hat\theta-\theta)^2 = k^2\mathrm{MSE}(\hat\theta) > 0$:

$$P(|\hat\theta - \theta| > k\sqrt{\mathrm{MSE}(\hat\theta)}) \le P((\hat\theta-\theta)^2 \ge k^2\mathrm{MSE}(\hat\theta)) \le \frac{\mathrm{MSE}(\hat\theta)}{k^2\mathrm{MSE}(\hat\theta)} = \frac{1}{k^2}.$$

Taking complements proves (b).

(a) If $\mathrm{E}\hat\theta = \theta$, the bias–variance decomposition gives $\mathrm{MSE}(\hat\theta) = \mathrm{Var}(\hat\theta) = \sigma^2_{\hat\theta}$, so (a) is the special case of (b). With $k = 2$ the probability is at least $3/4$, the distribution-free statement made for confidence intervals above.

For a biased estimator, $\mathrm{MSE} = \sigma^2_{\hat\theta} + B^2 > \sigma^2_{\hat\theta}$, and the event $|\hat\theta - \theta| \le 2\sigma_{\hat\theta}$ need not have probability $3/4$: if $|B| > 2\sigma_{\hat\theta}$, then, e.g. for a normal $\hat\theta$, it has probability below $1/2$. The guarantee is restored only by measuring the error with $\sqrt{\mathrm{MSE}}$. ■

Exercise 1.2 (Birth weights: exact sampling distribution of $\bar y$)

For the four birth weights $5.5, 8.0, 6.5, 7.0$ of the example on sampling with and without replacement, let $\bar y$ be the mean of a sample of size $n = 2$.
(a) Under WOR, list the six sample means and compute $\mathrm{E}(\bar y)$ and $\mathrm{Var}(\bar y)$ by enumeration.
(b) Under WR (ordered pairs, each with probability $1/16$), compute $\mathrm{Var}(\bar y)$ and the ratio of the two variances. Compare with σ^2/n and with $\frac{N-n}{N-1}\frac{\sigma^2}{n}$.

Solution. The population mean and variance are $\mu = 27/4 = 6.75$ and

$$\sigma^2 = \tfrac{1}{4}[(-1.25)^2 + 1.25^2 + (-0.25)^2 + 0.25^2] = \tfrac{3.25}{4} = 0.8125.$$

(a) The six samples, each with probability $1/6$, have means

$\{A,B\}$	$\{A,C\}$	$\{A,D\}$	$\{B,C\}$	$\{B,D\}$	$\{C,D\}$
6.75	6.00	6.25	7.25	7.50	6.75

Their sum is 40.5, so $\mathrm{E}(\bar y) = 40.5/6 = 6.75 = \mu$. The squared deviations from 6.75 are 0, 0.5625, 0.25, 0.25, 0.5625, 0, so

$$\mathrm{Var}(\bar y) = \frac{1.625}{6} = 0.2708.$$

(b) Under WR the two draws are independent, each uniform on the four values, so $\mathrm{Var}(\bar y) = \frac{1}{4}[\mathrm{Var}(y_1) + \mathrm{Var}(y_2)] = \sigma^2/2 = 0.40625$. (Enumerating the 16 ordered pairs gives the same number.) The ratio is

$$\frac{0.40625}{0.2708} = 1.5 = \frac{N-1}{N-n} = \frac{3}{2}.$$

Thus the WR variance equals σ^2/n, and the WOR variance equals $\frac{N-n}{N-1}\frac{\sigma^2}{n} = \frac{2}{3}\cdot 0.40625$, in agreement with the general formulas of Ch. 2. Sampling without replacement is 50% more efficient here because half of the population is observed. ■

Exercise 1.3 (Shrinking the sample mean)

Let $y_1, \dots, y_n$ be iid with mean $\mu \neq 0$ and variance $\sigma^2 > 0$, and consider $\hat\theta_c = c\,\bar y$ for a constant c.
(a) Find $\mathrm{MSE}(\hat\theta_c)$ as a function of c.
(b) Find the minimizing c^* and show that $\mathrm{MSE}(\hat\theta_{c^*}) < \mathrm{MSE}(\bar y)$.
(c) Compute $\mathrm{RE}(\bar y, \hat\theta_{c^*})$ and discuss why $\hat\theta_{c^*}$ is not used in practice.

Solution. [Supplement]

(a) $\mathrm{E}(\hat{\theta}_c) = c\mu$ and $\mathrm{Var}(\hat{\theta}_c) = c^2\sigma^2/n$, so by the bias–variance decomposition

$$\mathrm{MSE}(c) = c^2\frac{\sigma^2}{n} + (c-1)^2\mu^2.$$

(b) $\mathrm{MSE}(c)$ is a quadratic in c with leading coefficient $\sigma^2/n + \mu^2 > 0$, and $\mathrm{MSE}'(c) = 2c\sigma^2/n + 2(c-1)\mu^2 = 0$ gives

$$c^* = \frac{\mu^2}{\mu^2 + \sigma^2/n} \in (0,1).$$

Substituting, with $v = \sigma^2/n$,

$$\mathrm{MSE}(c^*) = \frac{\mu^4 v + v^2\mu^2}{(\mu^2+v)^2} = \frac{\mu^2 v}{\mu^2 + v} < v = \mathrm{MSE}(\bar{y}),$$

the inequality being strict because $v > 0$.

(c) $\mathrm{RE}(\bar{y}, \hat{\theta}_{c^*}) = \mathrm{MSE}(\bar{y})/\mathrm{MSE}(c^*) = (\mu^2+v)/\mu^2 = 1 + \sigma^2/(n\mu^2) > 1$, so the shrunken estimator is more efficient, and the gain is large when the coefficient of variation σ/μ is large relative to $\sqrt{n}$. However, c^* depends on the unknown ratio σ^2/μ^2; plugging in estimates gives a random c for which the bound of (b) is no longer guaranteed. The example shows that unbiasedness is a convention, not an optimality property: the MSE is the criterion that matters. ■

Chapter 2. Simple Random Sampling

1 Introduction

Definition 2.1

Simple random sampling (SRS) is the procedure in which every possible sample of size n has the same chance of being selected; in particular every population unit has the same chance of being in the sample. Without replacement each of the $\binom{N}{n}$ samples has probability $1/\binom{N}{n}$.

A simple random sample can be drawn by the lottery method or with random number tables / computer programs (`sample(N, n)` in R).

Setting. The population values are $u_1, \dots, u_N$ with

$$\mu = \frac{1}{N}\sum_{i=1}^{N} u_i, \qquad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(u_i-\mu)^2, \qquad S^2 = \frac{1}{N-1}\sum_{i=1}^{N}(u_i-\mu)^2,$$

and $y_1, \dots, y_n$ is an SRS drawn without replacement (WOR). The **sampling fraction** is $f = n/N$ and $1-f$ is the **finite population correction** (fpc).

2 Estimation of the Population Mean

Motivation. In survey sampling the population is a fixed finite list and the only randomness comes from which units we select. Under simple random sampling the sample mean is unbiased, and its variance carries the finite population correction $(N-n)/N$, which vanishes when we sample everyone.

Theorem 2.2: Sample mean under SRS (WOR)

Let $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ and $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i-\bar{y})^2$. Then

(i) $\mathrm{E}(\bar{y}) = \mu$;

(ii) $\mathrm{Var}(\bar{y}) = \dfrac{N-n}{N-1}\dfrac{\sigma^2}{n} = \dfrac{N-n}{N}\dfrac{S^2}{n} = \dfrac{S^2}{n}(1-f)$;

(iii) $\mathrm{E}(s^2) = S^2$, hence $\widehat{\mathrm{Var}}(\bar{y}) = \dfrac{s^2}{n}(1-f)$ is unbiased for $\mathrm{Var}(\bar{y})$.

Derivation: proof of the theorem

Marginal distribution of one draw. By symmetry each y_i equals u_k with probability $1/N$, so

$$\mathrm{E}(y_i) = \sum_{k=1}^{N} u_k \frac{1}{N} = \mu, \qquad \mathrm{Var}(y_i) = \sum_{k=1}^{N}(u_k-\mu)^2\frac{1}{N} = \sigma^2 \quad \text{for all } i.$$

(i) follows: $\mathrm{E}(\bar{y}) = \frac{1}{n}\sum \mathrm{E}(y_i) = \mu$.

Covariance of two draws. For $i \neq j$ the pair (y_i, y_j) is (u_k, u_l), $k \neq l$, with probability $\frac{1}{N(N-1)}$. Hence

$$\begin{aligned}\operatorname{Cov}(y_i, y_j) &= \mathrm{E}(y_i y_j) - \mu^2 = \frac{1}{N(N-1)} \sum_{k \neq l} u_k u_l - \frac{1}{N^2}\Big(\sum_k u_k\Big)^2 \\ &= \frac{1}{N(N-1)}\Big[\Big(\sum_k u_k\Big)^2 - \sum_k u_k^2\Big] - \frac{1}{N^2}\Big(\sum_k u_k\Big)^2 \\ &= \frac{1}{N(N-1)}\Big[\frac{1}{N}\Big(\sum_k u_k\Big)^2 - \sum_k u_k^2\Big] = -\frac{1}{N(N-1)}\Big(\sum_k u_k^2 - N\mu^2\Big) \\ &= -\frac{1}{N(N-1)} \sum_k (u_k - \mu)^2 = -\frac{\sigma^2}{N-1}.\end{aligned}$$

Variance of the mean.

$$\begin{aligned}\operatorname{Var}(\bar{y}) &= \frac{1}{n^2}\Big[\sum_i \operatorname{Var}(y_i) + 2\sum_{i<j} \operatorname{Cov}(y_i, y_j)\Big] = \frac{1}{n^2}\Big[n\sigma^2 - n(n-1)\frac{\sigma^2}{N-1}\Big] \\ &= \frac{\sigma^2}{n}\Big(1 - \frac{n-1}{N-1}\Big) = \frac{\sigma^2}{n}\frac{N-n}{N-1} = \frac{S^2}{n}\frac{N-n}{N},\end{aligned}$$

using $N\sigma^2 = (N-1)S^2$.

Unbiasedness of s^2. Using $\sum (y_i - \bar{y})^2 = \sum (y_i - \mu)^2 - n(\bar{y} - \mu)^2$,

$$\mathrm{E}(s^2) = \frac{1}{n-1}[n\sigma^2 - n\operatorname{Var}(\bar{y})] = \frac{\sigma^2}{n-1}\Big(n - \frac{N-n}{N-1}\Big) = \frac{\sigma^2}{n-1} \cdot \frac{N(n-1)}{N-1} = \frac{N}{N-1}\sigma^2 = S^2.$$

Therefore $\mathrm{E}[\widehat{\operatorname{Var}}(\bar{y})] = \frac{N-n}{Nn}\mathrm{E}(s^2) = \operatorname{Var}(\bar{y})$.

Remark 2.3: SRS with replacement

If the draws are made with replacement, the y_i are iid with mean μ and variance σ^2. Then $\operatorname{Cov}(y_i, y_j) = 0$ (e.g. for 0/1 values $\mathrm{E}(y_i y_j) = P(y_i = 1)P(y_j = 1) = \frac{N_1}{N}\frac{N_1}{N}$), so

$$\operatorname{Var}(\bar{y}) = \frac{\sigma^2}{n}, \qquad \mathrm{E}(s^2) = \frac{1}{n-1}\Big(n\sigma^2 - n\frac{\sigma^2}{n}\Big) = \sigma^2, \qquad \widehat{\operatorname{Var}}(\bar{y}) = \frac{s^2}{n}.$$

Without replacement the variance is smaller by the factor $\frac{N-n}{N-1}$; the fpc matters only when f is not small.

Confidence interval and bound. A $100(1-\alpha)\%$ confidence interval for μ is

$$\bar{y} \pm z_{\alpha/2}\sqrt{1-f}\,\frac{s}{\sqrt{n}},$$

and with $\alpha = 0.05$ we use the 2-SD bound $B = 2\sqrt{\widehat{\operatorname{Var}}(\bar{y})} = 2\sqrt{\frac{s^2}{n}(1-f)}$.

Example 2.4: Tractors per village

Estimate the average number of tractors per village; $N = 69$, $n = 10$, $\bar{y} = 16.7$, $s^2 = 81.344$.

$$\widehat{\operatorname{Var}}(\bar{y}) = \frac{s^2}{n} \cdot \frac{N-n}{N} = \frac{81.344}{10} \cdot \frac{59}{69} = 6.956, \qquad 2\sqrt{6.956} = 5.27,$$

so the interval is $16.7 \pm 5.27 = [11.43,\ 21.97]$.

3 Estimation of the Population Total

Motivation. Often the quantity of interest is a total (total income, total harvest). Scaling the mean by N gives an unbiased estimator whose variance is N^2 times that of the mean.

The population total is $\tau = N\mu$, and with N known the natural estimator is $\hat{\tau} = N\bar{y}$.

Theorem 2.5

$\mathrm{E}(\hat{\tau}) = \tau$ and

$$\mathrm{Var}(\hat{\tau}) = N^2 \frac{N-n}{N-1}\frac{\sigma^2}{n} = N^2 \frac{N-n}{N}\frac{S^2}{n}, \qquad \widehat{\mathrm{Var}}(\hat{\tau}) = N^2 \frac{N-n}{N}\frac{s^2}{n} = N^2 \frac{s^2}{n}(1-f).$$

A $100(1-\alpha)\%$ interval is $\hat{\tau} \pm z_{\alpha/2}\sqrt{\widehat{\mathrm{Var}}(\hat{\tau})}$, typically $\hat{\tau} \pm 2\sqrt{\widehat{\mathrm{Var}}(\hat{\tau})}$.

Example 2.6: Total number of tractors

$\hat{\tau} = 69 \times 16.7 = 1152.3$, $\widehat{\mathrm{Var}}(\hat{\tau}) = 69^2 \times 6.956 = 33{,}117$, and $2\sqrt{33117} = 364.0$. The interval is $1152.3 \pm 364.0 = [788, 1516]$.

4 Sample Size for Estimating μ and τ

To estimate μ with a bound B, set $2\sqrt{\mathrm{Var}(\bar{y})} = B$:

$$B = 2\sqrt{\frac{N-n}{N-1}\frac{\sigma^2}{n}} \;\Longrightarrow\; n = \frac{N\sigma^2}{(N-1)D + \sigma^2} = \frac{NS^2}{ND + S^2}, \qquad D = \frac{B^2}{4}.$$

Derivation

Write $D = B^2/4$. Then $D\,n(N-1) = (N-n)\sigma^2$, i.e. $n[(N-1)D + \sigma^2] = N\sigma^2$. The second form follows from $\sigma^2 = \frac{N-1}{N}S^2$.

For the total, $B = 2\sqrt{N^2\frac{N-n}{N}\frac{S^2}{n}}$ gives

$$n = \frac{NS^2}{ND + S^2}, \qquad D = \frac{B^2}{4N^2}.$$

In practice S^2 (or σ^2) is unknown and replaced by a prior estimate or a pilot-sample s^2.

Example 2.7: Tractors, continued

With $s^2 = 81.344$ and $B = 3$ ($D = 2.25$),

$$n = \frac{69 \times 81.344}{69 \times 2.25 + 81.344} = \frac{5612.7}{236.6} = 23.72 \;\Rightarrow\; n = 24,$$

so 14 more villages are needed beyond the 10 already sampled.

5 Estimation of a Population Proportion

Let $y_i = 1$ if the ith unit has the characteristic and 0 otherwise. Then the population proportion is the population mean of the 0/1 values, $p = \mu$, and

$$\hat{p} = \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad \hat{q} = 1 - \hat{p}.$$

For 0/1 values $\sigma^2 = pq$ and $S^2 = \frac{N}{N-1}pq$.

Theorem 2.8: Proportion under SRS

$$\mathrm{E}(\hat{p}) = p, \qquad \mathrm{Var}(\hat{p}) = \frac{pq}{n}\frac{N-n}{N-1}, \qquad \widehat{\mathrm{Var}}(\hat{p}) = \frac{\hat{p}\hat{q}}{n-1}\frac{N-n}{N} = \frac{\hat{p}\hat{q}}{n-1}(1-f),$$

and $\mathrm{E}[\widehat{\mathrm{Var}}(\hat{p})] = \mathrm{Var}(\hat{p})$.

Derivation: why the divisor is $n-1$

$$\mathrm{E}(\hat{p}\hat{q}) = \mathrm{E}(\hat{p}) - \mathrm{E}(\hat{p}^2) = p - \left[\mathrm{Var}(\hat{p}) + p^2\right] = pq - \frac{pq}{n}\frac{N-n}{N-1}$$
$$= pq\,\frac{n(N-1) - (N-n)}{n(N-1)} = \frac{N(n-1)}{(N-1)n}\,pq.$$

Hence

$$\mathrm{E}\Big[\frac{\hat{p}\hat{q}}{n-1}\frac{N-n}{N}\Big] = \frac{N-n}{N(n-1)}\cdot\frac{N(n-1)}{(N-1)n}pq = \frac{pq}{n}\frac{N-n}{N-1} = \mathrm{Var}(\hat{p}).$$

By contrast the plug-in $\frac{\hat{p}\hat{q}}{n}(1-f)$ is biased downward: $\mathrm{E}\big[\frac{\hat{p}\hat{q}}{n}(1-f)\big] \neq \mathrm{Var}(\hat{p})$. This is the same phenomenon as s^2 versus the divisor-n variance, since $s^2 = \frac{n}{n-1}\hat{p}\hat{q}$ for 0/1 data.

Example 2.9: Student vote

An SRS of $n = 120$ from $N = 2400$ students; 72 support candidate A. Then $\hat{p} = 0.6$ and

$$\widehat{\mathrm{Var}}(\hat{p}) = \Big(1 - \frac{120}{2400}\Big)\frac{0.6\times 0.4}{119} = 0.001916, \qquad 2\sqrt{0.001916} = 0.0875,$$

so the interval is $0.6 \pm 0.0875 = (0.512,\ 0.688)$.

6 Sample Size for Estimating p

Replace σ^2 by pq in the formula for μ:

$$n = \frac{Npq}{(N-1)D + pq}, \qquad D = \frac{B^2}{4} \quad (\text{or } D = B^2/1.96^2 \text{ with the exact normal quantile}).$$

With no prior information take $p = 1/2$, which maximizes pq and gives a conservative n.

Example 2.10

For $N = 2400$ and $B = 0.05$: $D = 0.05^2/4 = 0.000625$ and

$$n = \frac{2400 \times 0.25}{2399 \times 0.000625 + 0.25} = \frac{600}{1.7494} = 342.98 \;\Rightarrow\; n = 343.$$

7 Towards Stratified Sampling

Consider estimating the total first-hospitalization cost of kidney-stone patients. Two regions with very different incidence rates are surveyed separately:

	Carolinas	Rockies
n	363	258
$\bar{y}$	1350	1150

What is the total cost? Pooling all 621 observations into one mean ignores the fact that the regions were sampled separately and have different population sizes. The answer is to weight each regional mean by its population size, which is the stratified estimator of Chapter 3.

8 Selected exercises

Motivation. The first item rederives the SRS variance from inclusion indicators, the device behind all unequal-probability designs; the second gives the covariance of two sample means, which Ch. 5 uses for ratio estimators; the third is a complete computation for a proportion.

Exercise 2.1 (Inclusion indicators)

Draw an SRS of size n without replacement from units $1, \dots, N$ with values $u_1, \dots, u_N$, and let $Z_k = 1$ if unit k is in the sample and $Z_k = 0$ otherwise.
(a) Show that $P(Z_k = 1) = n/N$ and, for $k \neq l$, $P(Z_k = Z_l = 1) = \frac{n(n-1)}{N(N-1)}$.
(b) Writing $\bar{y} = \frac{1}{n}\sum_{k=1}^{N} Z_k u_k$, rederive $\mathrm{E}(\bar{y}) = \mu$ and $\mathrm{Var}(\bar{y}) = \frac{S^2}{n}(1-f)$.

Solution. [Supplement]

(a) All $\binom{N}{n}$ samples are equally likely. The samples containing unit k are obtained by choosing the other $n-1$ units from the remaining $N-1$, so $P(Z_k = 1) = \binom{N-1}{n-1}/\binom{N}{n} = n/N$. Similarly the samples containing both k and l number $\binom{N-2}{n-2}$, and $\binom{N-2}{n-2}/\binom{N}{n} = \frac{n(n-1)}{N(N-1)}$.

(b)

Step 1. (Moments of the indicators.) By (a), $\mathrm{E}Z_k = f$, $\mathrm{Var}Z_k = f(1-f)$ (as $Z_k^2 = Z_k$), and for $k \neq l$

$$\mathrm{Cov}(Z_k, Z_l) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = \frac{n}{N}\Big(\frac{n-1}{N-1} - \frac{n}{N}\Big) = -\frac{f(1-f)}{N-1}.$$

Step 2. (Mean.) Only the Z_k are random, so $\mathrm{E}\bar{y} = \frac{1}{n}\sum_k f\,u_k = \frac{1}{N}\sum_k u_k = \mu$.

Step 3. (Variance.) Since $\sum_k Z_k = n$ is constant, $\bar{y} - \mu = \frac{1}{n}\sum_k Z_k(u_k - \mu)$; write $d_k = u_k - \mu$, so $\sum_k d_k = 0$ and $\sum_k d_k^2 = (N-1)S^2$. Then

$$n^2\mathrm{Var}(\bar{y}) = \sum_k d_k^2 \mathrm{Var}Z_k + \sum_{k\neq l} d_k d_l \mathrm{Cov}(Z_k, Z_l) = f(1-f)\Big[\sum_k d_k^2 - \frac{1}{N-1}\sum_{k\neq l} d_k d_l\Big].$$

Step 4. Because $\sum_{k\neq l} d_k d_l = \left(\sum_k d_k\right)^2 - \sum_k d_k^2 = -\sum_k d_k^2$, the bracket equals $\sum_k d_k^2(1+\frac{1}{N-1}) = \frac{N}{N-1}(N-1)S^2 = NS^2$. Hence

$$\mathrm{Var}(\bar{y}) = \frac{f(1-f)NS^2}{n^2} = \frac{S^2}{n}(1-f).$$

The same computation with arbitrary inclusion probabilities π_k leads to the Horvitz–Thompson estimator $\sum_k Z_k u_k/\pi_k$ of τ. ■

Exercise 2.2 (Covariance of two sample means)

Each unit carries two values (x_k, y_k), $k = 1, \ldots, N$, with population means μ_x, μ_y and

$$S_{xy} = \frac{1}{N-1}\sum_{k=1}^{N}(x_k - \mu_x)(y_k - \mu_y).$$

For an SRS of n pairs without replacement, show that
(a) $\mathrm{Cov}(\bar{x}, \bar{y}) = \frac{1-f}{n} S_{xy}$;
(b) $s_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$ satisfies $\mathrm{E}(s_{xy}) = S_{xy}$.

Solution. [Supplement] Apply the theorem on the sample mean under SRS to the auxiliary variable $z_k = x_k + y_k$, whose sample mean is $\bar{z} = \bar{x} + \bar{y}$.

(a)

Step 1. Expanding the square, $S_z^2 = \frac{1}{N-1}\sum_k[(x_k - \mu_x) + (y_k - \mu_y)]^2 = S_x^2 + S_y^2 + 2S_{xy}$.

Step 2. By the theorem, $\mathrm{Var}(\bar{z}) = \frac{1-f}{n} S_z^2$, $\mathrm{Var}(\bar{x}) = \frac{1-f}{n} S_x^2$ and $\mathrm{Var}(\bar{y}) = \frac{1-f}{n} S_y^2$.

Step 3. Since $\mathrm{Var}(\bar{z}) = \mathrm{Var}(\bar{x}) + \mathrm{Var}(\bar{y}) + 2\mathrm{Cov}(\bar{x}, \bar{y})$,

$$2\mathrm{Cov}(\bar{x}, \bar{y}) = \frac{1-f}{n}(S_z^2 - S_x^2 - S_y^2) = \frac{1-f}{n}\, 2S_{xy}.$$

(b) The sample analogue of Step 1 is $s_z^2 = s_x^2 + s_y^2 + 2s_{xy}$. Taking expectations and using $\mathrm{E}(s_z^2) = S_z^2$, $\mathrm{E}(s_x^2) = S_x^2$, $\mathrm{E}(s_y^2) = S_y^2$ from the theorem,

$$2\mathrm{E}(s_{xy}) = S_z^2 - S_x^2 - S_y^2 = 2S_{xy}.$$

This polarization argument turns every variance statement about SRS means into a covariance statement; it is the step used for the bias of the ratio estimator in Ch. 5. ■

Exercise 2.3 (Hunters of game birds)

An SRS of $n = 1000$ of the $N = 99{,}000$ licensed hunters of a province is taken, and 430 of them hunt game birds.
(a) Estimate the proportion p of hunters seeking game birds and place a bound on the error of estimation.
(b) How large a sample is needed to estimate p with a bound $B = 0.02$, using the estimate of (a) as prior value?

Solution.

(a) $\hat{p} = 430/1000 = 0.43$ and, by the proportion theorem,

$$\widehat{\mathrm{Var}}(\hat{p}) = \frac{\hat{p}\hat{q}}{n-1}\cdot\frac{N-n}{N} = \frac{0.43\times 0.57}{999}\cdot\frac{98{,}000}{99{,}000} = 0.000243, \qquad 2\sqrt{0.000243} = 0.031,$$

so p is estimated by 0.43 ± 0.031.

(b) With $D = B^2/4 = 0.0001$ and $pq = 0.43 \times 0.57 = 0.2451$,

$$n = \frac{Npq}{(N-1)D + pq} = \frac{99{,}000 \times 0.2451}{98{,}999 \times 0.0001 + 0.2451} = \frac{24{,}265}{10.145} = 2391.8,$$

so $n = 2392$ hunters are needed. Since N is large, this is close to the infinite-population value $pq/D = 2451$; the fpc saves about 60 interviews. ■

Chapter 3. Stratified Random Sampling

1 Introduction

Definition 3.1

A **stratified random sample** is obtained by separating the population into non-overlapping groups called **strata** and then selecting a simple random sample from each stratum, independently across strata.

Because the strata are sampled independently, each stratum can be treated as a separate population. Stratification pays off when the variation *between* strata is large (the strata are internally homogeneous). Model-wise, SRS uses $y_i = \mu + \varepsilon_i$, while stratified sampling uses $y_{ij} = \mu_j + \varepsilon_{ij}$: each stratum has its own mean. Once the selection procedure is fixed, the design questions are

- how many strata to construct,
- how to allocate the total sample size to the strata,
- how to choose the strata.

2 Notation

Strata are indexed by $j = 1, \dots, L$ and units within a stratum by i; y_{ij} is the observation on unit i of stratum j. Sampling within each stratum is SRS without replacement.

N_j, n_j	population and sample size of stratum j; $N = \sum N_j$, $n = \sum n_j$
$w_j = N_j/N$	stratum weight
$f_j = n_j/N_j$	sampling fraction in stratum j
$\mu_j = \frac{1}{N_j}\sum_{i=1}^{N_j} Y_{ij}$	true stratum mean
$\bar{y}_j = \frac{1}{n_j}\sum_{i=1}^{n_j} y_{ij}$	sample stratum mean
$\sigma_j^2 = \frac{1}{N_j}\sum_i (Y_{ij} - \mu_j)^2$	true stratum variance
$S_j^2 = \frac{N_j}{N_j - 1}\sigma_j^2$	stratum variance with divisor $N_j - 1$
$s_j^2 = \frac{1}{n_j - 1}\sum_{i=1}^{n_j} (y_{ij} - \bar{y}_j)^2$	sample stratum variance

3 Estimation of the Population Mean

Motivation. If the population splits into groups that differ systematically, sampling each group separately removes the between-group variation from the error. The stratified estimator weights the stratum means by stratum sizes and is never worse than SRS under proportional allocation.

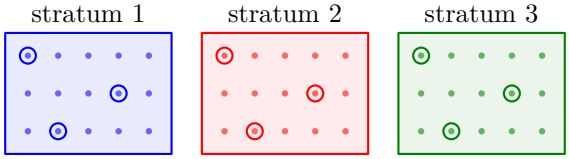


Figure. *Stratified random sampling: the population is split into homogeneous strata and an independent simple random sample (circled units) is drawn within each. Only within-stratum variation enters the variance of* $\bar{y}_{st} = \sum_h W_h \bar{y}_h$.

Theorem 3.2: Stratified estimator of μ

$$\bar{y}_{\text{st}} = \frac{1}{N}\sum_{j=1}^{L} N_j \bar{y}_j = \sum_{j=1}^{L} w_j \bar{y}_j$$

is unbiased for μ, with

$$\text{Var}(\bar{y}_{\text{st}}) = \sum_{j=1}^{L} w_j^2 (1-f_j)\frac{S_j^2}{n_j}, \qquad \widehat{\text{Var}}(\bar{y}_{\text{st}}) = \sum_{j=1}^{L} w_j^2 (1-f_j)\frac{s_j^2}{n_j}, \qquad \text{E}[\widehat{\text{Var}}(\bar{y}_{\text{st}})] = \text{Var}(\bar{y}_{\text{st}}).$$

A $100(1-\alpha)\%$ interval is $\bar{y}_{\text{st}} \pm z_{\alpha/2}\sqrt{\widehat{\text{Var}}(\bar{y}_{\text{st}})}$.

Proof. Each $\bar{y}_j$ is an SRS mean, so by Ch. 2, $\text{E}(\bar{y}_j) = \mu_j$, $\text{Var}(\bar{y}_j) = (1-f_j)S_j^2/n_j$ and $\text{E}(s_j^2) = S_j^2$. Thus

$$\text{E}(\bar{y}_{\text{st}}) = \sum_j w_j \mu_j = \frac{1}{N}(N_1\mu_1 + \cdots + N_L\mu_L) = \frac{1}{N}(\tau_1 + \cdots + \tau_L) = \frac{\tau}{N} = \mu.$$

Since the strata are sampled independently the covariances vanish and $\text{Var}(\bar{y}_{\text{st}}) = \sum_j w_j^2 \text{Var}(\bar{y}_j)$. Unbiasedness of $\widehat{\text{Var}}$ follows from $\text{E}(s_j^2) = S_j^2$. ■

4 Estimation of the Population Total

Motivation. The stratified total is the sum of stratum totals; its variance adds over strata because samples in different strata are independent.

Theorem 3.3

$\hat{\tau}_{\text{st}} = N\bar{y}_{\text{st}} = \sum_{j=1}^{L} N_j \bar{y}_j$ is unbiased for τ, and

$$\text{Var}(\hat{\tau}_{\text{st}}) = N^2 \text{Var}(\bar{y}_{\text{st}}) = \sum_{j=1}^{L} N_j^2 (1-f_j)\frac{S_j^2}{n_j}, \qquad \widehat{\text{Var}}(\hat{\tau}_{\text{st}}) = \sum_{j=1}^{L} N_j^2 (1-f_j)\frac{s_j^2}{n_j}.$$

Example 3.4: TV-viewing time

	N_j	w_j	n_j	$\bar{y}_j$	s_j
Town A	155	0.5	20	33.90	5.95
Town B	62	0.2	8	25.12	15.25
Rural	93	0.3	12	19.00	9.36
Total	310		40		

Mean.

$$\bar{y}_{\text{st}} = \frac{155(33.90) + 62(25.12) + 93(19.00)}{310} = \frac{8578.94}{310} = 27.67.$$

$$\begin{aligned}\widehat{\text{Var}}(\bar{y}_{\text{st}}) &= (0.5)^2\frac{5.95^2}{20}\left(1-\frac{20}{155}\right) + (0.2)^2\frac{15.25^2}{8}\left(1-\frac{8}{62}\right) + (0.3)^2\frac{9.36^2}{12}\left(1-\frac{12}{93}\right) \\ &= 0.385 + 1.013 + 0.572 = 1.97,\end{aligned}$$

so the estimate with a 2-SD bound is $27.67 \pm 2\sqrt{1.97} = 27.67 \pm 2.81$.

Total. $\hat{\tau}_{\text{st}} = 310(27.67) = 8579$, $\widehat{\text{Var}}(\hat{\tau}_{\text{st}}) = 310^2(1.97) \approx 189{,}400$, and the bound is $2\sqrt{189{,}400} \approx 870$.

5 Sample Size for Estimating μ and τ

Setting $2\sqrt{\mathrm{Var}(\bar{y}_{\mathrm{st}})} = B$ alone cannot be solved for n; we must also fix how n is split among the strata. Write the allocation as $n_j = na_j$ with $\sum a_j = 1$.

Theorem 3.5: Sample size for a given allocation

$$n = \frac{\sum_{j=1}^{L} N_j^2 S_j^2 / a_j}{N^2 D + \sum_{j=1}^{L} N_j S_j^2}, \qquad D = \begin{cases} B^2/4 & \text{for } \mu, \\ B^2/(4N^2) & \text{for } \tau, \end{cases}$$

with S_j^2 replaced by prior estimates s_j^2 in practice.

Derivation

For μ let $D = (B/2)^2 = \mathrm{Var}(\bar{y}_{\mathrm{st}}) = \sum_j w_j^2 \frac{S_j^2}{n_j}(1 - \frac{n_j}{N_j})$. Multiply by N^2 and substitute $n_j = na_j$:

$$N^2 D = \sum_j N_j^2 \frac{S_j^2}{n a_j} - \sum_j N_j S_j^2 \implies n\Big(N^2 D + \sum_j N_j S_j^2\Big) = \sum_j \frac{N_j^2 S_j^2}{a_j}.$$

For τ, $\mathrm{Var}(\hat{\tau}_{\mathrm{st}}) = N^2 \mathrm{Var}(\bar{y}_{\mathrm{st}}) = B^2/4$ gives the same formula with $D = B^2/(4N^2)$. (Using 1.96 instead of 2 replaces $D = (B/2)^2$ by $(B/1.96)^2$.)

Example 3.6: Equal allocation

For the TV population ($N_j = 155, 62, 93$) with prior variances $s_1^2 = 25$, $s_2^2 = 225$, $s_3^2 = 100$, equal allocation $a_j = 1/3$ and $B = 2$ for μ ($D = 1$):

$$n = \frac{3(155^2 \cdot 25 + 62^2 \cdot 225 + 93^2 \cdot 100)}{310^2(1) + (155 \cdot 25 + 62 \cdot 225 + 93 \cdot 100)} = \frac{6{,}991{,}275}{123{,}225} = 56.7,$$

so $n = 57$ and $n_1 = n_2 = n_3 = 19$.

6 Allocation of the Sample

Let the survey cost be

$$TC = c_0 + \sum_{j=1}^{L} c_j n_j,$$

where c_0 is the overhead cost and c_j the cost per observation in stratum j.

6.1 Equal allocation

$n_j = n/L$. Then $TC = c_0 + \frac{n}{L}\sum c_j$, so $n = \dfrac{(TC - c_0)L}{\sum_j c_j}$.

6.2 Proportional allocation

$n_j = nw_j = nN_j/N$, i.e. $a_j = w_j$ and all $f_j = f$ are equal. Then $TC = c_0 + n\sum c_j w_j$, so $n = \dfrac{TC - c_0}{\sum_j c_j w_j}$; if $c_j \equiv c$, $n = (TC - c_0)/c$. The estimator becomes self-weighting: $\bar{y}_{\text{st}}$ equals the plain mean of all n observations.

6.3 Optimum allocation

Theorem 3.7: Optimum allocation

For a fixed total cost (equivalently, for a fixed variance) $\text{Var}(\bar{y}_{\text{st}})$ is minimized when

$$n_j \propto \frac{w_j S_j}{\sqrt{c_j}}, \qquad \text{i.e.} \qquad n_j = n\,\frac{w_j S_j/\sqrt{c_j}}{\sum_k w_k S_k/\sqrt{c_k}},$$

and the total sample size under a budget TC is

$$n = \frac{(TC - c_0)\sum_k w_k S_k/\sqrt{c_k}}{\sum_k w_k S_k \sqrt{c_k}}.$$

Derivation: Cauchy–Schwarz argument

Split the variance into the part depending on the allocation and a constant:

$$\text{Var}(\bar{y}_{\text{st}}) = \sum_j w_j^2 \frac{S_j^2}{n_j}\Big(1 - \frac{n_j}{N_j}\Big) = \underbrace{\sum_j \frac{w_j^2 S_j^2}{n_j}}_{V'} - \underbrace{\sum_j \frac{w_j S_j^2}{N}}_{\text{const}}.$$

Let $C' = TC - c_0 = \sum_j c_j n_j$. Minimizing V' for fixed C', or C' for fixed V', is the same as minimizing the product

$$V'C' = \Big(\sum_j \frac{w_j^2 S_j^2}{n_j}\Big)\Big(\sum_j c_j n_j\Big) = \Big(\sum_j a_j^2\Big)\Big(\sum_j b_j^2\Big), \quad a_j = \frac{w_j S_j}{\sqrt{n_j}},\ b_j = \sqrt{c_j n_j}.$$

By the Cauchy–Schwarz inequality

$$V'C' \geq \Big(\sum_j a_j b_j\Big)^2 = \Big(\sum_j w_j S_j \sqrt{c_j}\Big)^2,$$

whose right-hand side does not depend on the n_j. Equality holds iff b_j/a_j is constant:

$$\frac{b_j}{a_j} = \frac{n_j\sqrt{c_j}}{w_j S_j} = \text{const} \iff n_j \propto \frac{w_j S_j}{\sqrt{c_j}}.$$

Substituting $n_j = n\,\frac{w_j S_j/\sqrt{c_j}}{\sum_k w_k S_k/\sqrt{c_k}}$ into $TC = c_0 + \sum c_j n_j$ gives $TC = c_0 + n\alpha$ with $\alpha = \frac{\sum_k w_k S_k \sqrt{c_k}}{\sum_k w_k S_k/\sqrt{c_k}}$, so $n = (TC - c_0)/\alpha$.

We therefore take a larger sample in a stratum that is (1) larger, (2) more variable, or (3) cheaper to sample.

6.4 Neyman allocation

Theorem 3.8: Neyman allocation

If the costs are equal, $c_j \equiv c$, the optimum allocation becomes

$$n_j = n\,\frac{w_j S_j}{\sum_k w_k S_k} = n\,\frac{N_j S_j}{\sum_k N_k S_k}.$$

Example 3.9: Neyman allocation for the TV data

With $n = 40$ and $N_j s_j = 922.3,\ 945.5,\ 870.5$ (sum 2738.2), the Neyman allocation is

$$n_1 = 13.5, \quad n_2 = 13.8, \quad n_3 = 12.7 \ \Rightarrow\ (13, 14, 13),$$

compared with the proportional allocation $(20, 8, 12)$ actually used. The small but very variable Town B gets many more observations. Treating s_j as S_j, the variance of $\bar{y}_{\text{st}}$ drops from 1.97 (proportional) to about 1.66 (Neyman).

7 Estimation of a Population Proportion

Motivation. A proportion is the mean of a 0–1 variable, so the stratified formulas apply directly with $p_h(1-p_h)$ in place of the stratum variances.

With $y_{ij} \in \{0, 1\}$ the stratum mean $\bar{y}_j$ is the stratum proportion $\hat{p}_j$, and

$$\hat{p}_{\text{st}} = \frac{1}{N}(N_1\hat{p}_1 + \cdots + N_L\hat{p}_L) = \sum_j w_j \hat{p}_j, \qquad \mathrm{E}(\hat{p}_{\text{st}}) = \sum_j w_j p_j = p.$$

Theorem 3.10

$$\mathrm{Var}(\hat{p}_{\text{st}}) = \sum_j w_j^2 \frac{p_j q_j}{n_j}\Big(\frac{N_j - n_j}{N_j - 1}\Big), \qquad \widehat{\mathrm{Var}}(\hat{p}_{\text{st}}) = \sum_j w_j^2 \frac{\hat{p}_j \hat{q}_j}{n_j - 1}(1 - f_j),$$

and a $100(1-\alpha)\%$ interval for p is $\hat{p}_{\text{st}} \pm z_{\alpha/2}\sqrt{\widehat{\mathrm{Var}}(\hat{p}_{\text{st}})}$.

These follow from Ch. 2 applied stratum by stratum, the divisor $n_j - 1$ making each term unbiased.

8 Selected exercises

Motivation. The allocation theorems say which allocation is best but not by how much. The first two items give exact expressions for the gains of Neyman over proportional allocation and of proportional allocation over SRS; the third is a full design computation with unequal costs.

Exercise 3.1 (Proportional versus Neyman allocation)

Let the costs be equal and the total sample size n fixed. Denote by $\bar{y}_{\text{prop}}$ and $\bar{y}_{\text{opt}}$ the stratified means under proportional allocation $n_j = nw_j$ and under Neyman allocation $n_j = nw_jS_j/\bar{S}$,

where $\bar{S} = \sum_j w_j S_j$ (assume $n_j \le N_j$). Show that

$$\mathrm{Var}(\bar{y}_{\mathrm{prop}}) - \mathrm{Var}(\bar{y}_{\mathrm{opt}}) = \frac{1}{n}\sum_{j=1}^{L} w_j (S_j - \bar{S})^2,$$

so $\mathrm{Var}(\bar{y}_{\mathrm{prop}}) > \mathrm{Var}(\bar{y}_{\mathrm{opt}})$ unless all S_j are equal.

Solution.

Step 1. (A common constant.) For any allocation,

$$\mathrm{Var}(\bar{y}_{\mathrm{st}}) = \sum_j w_j^2 \frac{S_j^2}{n_j}\Big(1 - \frac{n_j}{N_j}\Big) = \sum_j \frac{w_j^2 S_j^2}{n_j} - \frac{1}{N}\sum_j w_j S_j^2,$$

since $w_j^2/N_j = w_j/N$. The last sum does not depend on the allocation.

Step 2. (Proportional.) With $n_j = n w_j$, $\sum_j w_j^2 S_j^2/n_j = \frac{1}{n}\sum_j w_j S_j^2$.

Step 3. (Neyman.) With $n_j = n w_j S_j/\bar{S}$, $\sum_j w_j^2 S_j^2/n_j = \frac{\bar{S}}{n}\sum_j w_j S_j = \frac{\bar{S}^2}{n}$.

Step 4. (Difference.) Subtracting, the constants of Step 1 cancel and

$$\mathrm{Var}(\bar{y}_{\mathrm{prop}}) - \mathrm{Var}(\bar{y}_{\mathrm{opt}}) = \frac{1}{n}\Big(\sum_j w_j S_j^2 - \bar{S}^2\Big) = \frac{1}{n}\sum_j w_j (S_j - \bar{S})^2,$$

the last step being the identity $\sum_j w_j (S_j - \bar{S})^2 = \sum_j w_j S_j^2 - 2\bar{S}\sum_j w_j S_j + \bar{S}^2 \sum_j w_j = \sum_j w_j S_j^2 - \bar{S}^2$ (recall $\sum_j w_j = 1$).

Step 5. The right side is a weighted variance of the stratum standard deviations, with positive weights; it vanishes iff $S_j = \bar{S}$ for all j, in which case the two allocations coincide. The gain from Neyman allocation is therefore large exactly when the strata differ in variability. ■

Exercise 3.2 (Proportional allocation versus SRS)

Let $\bar{y}_{\mathrm{ran}}$ be the mean of an SRS of size n from the whole population, with variance $\frac{1-f}{n}S^2$. Show that

$$\mathrm{Var}(\bar{y}_{\mathrm{ran}}) - \mathrm{Var}(\bar{y}_{\mathrm{prop}}) = \frac{1-f}{n(N-1)}\Big[\sum_{j=1}^{L} N_j(\mu_j - \mu)^2 - \sum_{j=1}^{L}(1 - w_j)S_j^2\Big],$$

and conclude that proportional allocation is better than SRS unless the stratum means are nearly equal.

Solution. [Supplement]

Step 1. (Analysis of variance of the population.) Adding and subtracting μ_j inside stratum j, the cross term vanishes because $\sum_i (Y_{ij} - \mu_j) = 0$:

$$(N-1)S^2 = \sum_j \sum_i (Y_{ij} - \mu)^2 = \sum_j (N_j - 1)S_j^2 + \sum_j N_j(\mu_j - \mu)^2.$$

Step 2. Under proportional allocation $f_j = f$, so $\mathrm{Var}(\bar{y}_{\mathrm{prop}}) = \sum_j w_j^2 (1-f)\frac{S_j^2}{n w_j} = \frac{1-f}{n}\sum_j w_j S_j^2$. Hence

$$\mathrm{Var}(\bar{y}_{\mathrm{ran}}) - \mathrm{Var}(\bar{y}_{\mathrm{prop}}) = \frac{1-f}{n}\Big(S^2 - \sum_j w_j S_j^2\Big).$$

Step 3. By Step 1,

$$S^2 - \sum_j w_j S_j^2 = \frac{\sum_j N_j(\mu_j - \mu)^2}{N-1} + \sum_j S_j^2\Big(\frac{N_j - 1}{N-1} - \frac{N_j}{N}\Big),$$

and $\frac{N_j-1}{N-1} - \frac{N_j}{N} = \frac{N(N_j-1)-N_j(N-1)}{N(N-1)} = -\frac{N-N_j}{N(N-1)} = -\frac{1-w_j}{N-1}$. Substituting gives the claim.

Step 4. The first sum in the bracket is the between-strata sum of squares, which grows with the stratum sizes N_j; the second is a sum of L within-stratum variances with coefficients below 1. Unless the μ_j are nearly equal, the first dominates and $\mathrm{Var}(\bar{y}_{\mathrm{prop}}) < \mathrm{Var}(\bar{y}_{\mathrm{ran}})$; with large N_j the difference is essentially $\frac{1-f}{n}\sum_j w_j(\mu_j - \mu)^2 \geq 0$. ■

Exercise 3.3 (Allocation with unequal costs)

A county plans to estimate the proportion of families who will use an expanded day-care facility. Families are stratified into users in the city, users in the country, nonusers in the city and nonusers in the country, with $N_j = 97, 43, 145, 68$. About 90% of present users and 50% of present nonusers are expected to use the expanded facility. An observation costs 4 for a user and 8 for a nonuser. Find the approximate sample size and allocation needed to estimate the proportion with a bound $B = 0.05$.

Solution. For 0/1 data $S_j^2 = \frac{N_j}{N_j-1}p_jq_j \approx p_jq_j$, so we use $p_jq_j = 0.09, 0.09, 0.25, 0.25$ in place of S_j^2.

Step 1. (Allocation.) By the optimum-allocation theorem, $n_j \propto w_jS_j/\sqrt{c_j} \propto N_j\sqrt{p_jq_j/c_j}$:

$$N_j\sqrt{p_jq_j/c_j} = 14.55,\ 6.45,\ 25.63,\ 12.02, \qquad \text{sum } 58.65,$$

so $a_j = 0.248,\ 0.110,\ 0.437,\ 0.205$.

Step 2. (Sample size.) With $N = 353$, $D = B^2/4 = 0.000625$ and the sample-size theorem,

$$n = \frac{\sum_j N_j^2 p_jq_j/a_j}{N^2D + \sum_j N_jp_jq_j} = \frac{22{,}595}{77.88 + 65.85} = \frac{22{,}595}{143.73} = 157.2,$$

where $\sum_j N_j^2p_jq_j/a_j = 846.8/0.248 + 166.4/0.110 + 5256.3/0.437 + 1156.0/0.205$ and $\sum_j N_jp_jq_j = 8.73 + 3.87 + 36.25 + 17.00$.

Step 3. Take $n = 158$; then $n_j = na_j = 39.2,\ 17.4,\ 69.0,\ 32.4$, i.e. $(n_1, n_2, n_3, n_4) = (39, 17, 69, 33)$. Nonusers get more observations despite the higher cost because their proportion is more variable ($pq = 0.25$ versus 0.09); the cost factor $1/\sqrt{c_j}$ only partly offsets this. ■

Chapter 4. Systematic Sampling

1 Introduction

To select n names from a long list, a simple way is to pick names at equal intervals, e.g. every tenth name. If the starting point is random, the result is a systematic sample.

Definition 4.1

A **1-in-k systematic sample with a random start** is obtained by randomly selecting one element among the first k elements of the frame and every kth element thereafter. Here n denotes the sample size and k the number of possible systematic samples.

Advantages over SRS:

- the sample is easy to draw and the field work is less prone to mistakes;
- it can provide more information per unit cost, because the sample is spread evenly over the frame.

2 Selection Methods

2.1 Method A: random start among the first k

Example 4.2: $N = 12$, $k = 3$

Pick a random unit among y_1, y_2, y_3 and then every third unit. If y_2 is picked, the sample is y_2, y_5, y_8, y_{11}. The population is cut into $n = N/k = 4$ rows and one unit is taken from each row; the $k = 3$ possible samples are the columns

#1	#2	#3
y_1	y_2	y_3
y_4	y_5	y_6
y_7	y_8	y_9
y_{10}	y_{11}	y_{12}

In general the sample is $j,\ j+k,\ j+2k,\ \ldots,\ j+(n-1)k$ with j uniform on $\{1,\ldots,k\}$, and each systematic sample has probability $1/k$.

If $N \neq nk$, the samples have unequal sizes. For $N = 12$ and $k = 5$ we have $2 < N/k = 2.4 < 3$: the five samples are (y_1, y_6, y_{11}), (y_2, y_7, y_{12}), (y_3, y_8), (y_4, y_9), (y_5, y_{10}), of size 2 or 3, yet each is chosen with probability $1/5$.

2.2 Method B: random start among all N

Select one unit, say the jth, at random from the whole population, and divide j by k: if the remainder is r, the start is y_r (with $r = 0$ meaning y_k). Then take every kth unit.

Example 4.3: $N = 11$, $k = 3$

If $j = 8$, then $8 = 2 \cdot 3 + 2$, so $r = 2$, the start is y_2 and the sample is (y_2, y_5, y_8, y_{11}). The possible samples are

$$\#1 = (y_1, y_4, y_7, y_{10}), \quad \#2 = (y_2, y_5, y_8, y_{11}), \quad \#3 = (y_3, y_6, y_9).$$

Sample #2 is obtained whenever $j \in \{2, 5, 8, 11\}$, so its probability is $4/11$; similarly #1 has probability $4/11$ and #3 has $3/11$.

Remark 4.4: A versus B

Both methods produce the same collection of systematic samples, but the selection probabilities differ: $1/k$ under Method A and n_i/N (proportional to the sample size) under Method B. Method B requires N to be known. When $N = nk$ the two coincide.

3 Estimation of the Population Mean

Let y_{ij} be the jth element of the ith systematic sample ($i = 1, \ldots, k$, $j = 1, \ldots, n_i$), $\bar{y}_i = \frac{1}{n_i}\sum_j y_{ij}$, and $\bar{y}_{\text{sy}}$ the mean of the sample actually selected.

3.1 Unbiasedness

Case 1 (Method A, $N = nk$).

$$\mathrm{E}(\bar{y}_{\text{sy}}) = \frac{1}{k}(\bar{y}_1 + \cdots + \bar{y}_k) = \frac{1}{k}\frac{1}{n}(y_1 + \cdots + y_N) = \mu,$$

so $\bar{y}_{\text{sy}}$ is unbiased.

Example 4.5

Population $6, 3, 4, 9, 2, 5, 1, 7, 8$ with $k = 3$: the samples are $(6, 9, 1)$, $(3, 2, 7)$, $(4, 5, 8)$ with means $16/3$, $12/3$, $17/3$, whose average is $5 = \mu$.

Case 2 (Method A, $N \neq nk$). For $y_1, \ldots, y_7$ and $k = 3$,

$$\mathrm{E}(\bar{y}_{\text{sy}}) = \frac{1}{3}\left[\frac{y_1 + y_4 + y_7}{3} + \frac{y_2 + y_5}{2} + \frac{y_3 + y_6}{2}\right] \neq \frac{y_1 + \cdots + y_7}{7} = \mu$$

in general: $\bar{y}_{\text{sy}}$ is *biased*.

Case 3 (Method B, $N \neq nk$). With probabilities $3/7, 2/7, 2/7$,

$$\mathrm{E}(\bar{y}_{\text{sy}}) = \frac{3}{7}\cdot\frac{y_1 + y_4 + y_7}{3} + \frac{2}{7}\cdot\frac{y_2 + y_5}{2} + \frac{2}{7}\cdot\frac{y_3 + y_6}{2} = \frac{y_1 + \cdots + y_7}{7} = \mu.$$

In general $\mathrm{E}(\bar{y}_{\text{sy}}) = \sum_i \frac{n_i}{N}\bar{y}_i = \frac{1}{N}\sum_i\sum_j y_{ij} = \mu$: Method B is always unbiased.

3.2 Variance of $\bar{y}_{\text{sy}}$ ($N = nk$)

By definition,

$$\mathrm{Var}(\bar{y}_{\text{sy}}) = \frac{1}{k}\sum_{i=1}^{k}(\bar{y}_i - \mu)^2,$$

where $S^2 = \frac{1}{N-1}\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \mu)^2$.

Theorem 4.6: Variance of the systematic mean

$$\operatorname{Var}(\bar{y}_{\text{sy}}) = \frac{N-1}{N} S^2 - \frac{1}{N} \sum_{i=1}^{k} \sum_{j=1}^{n} (y_{ij} - \bar{y}_i)^2.$$

Proof. Decompose the total sum of squares:

$$\sum_i \sum_j (y_{ij} - \mu)^2 = \sum_i \sum_j (y_{ij} - \bar{y}_i)^2 + n \sum_i (\bar{y}_i - \mu)^2 + 2 \sum_i (\bar{y}_i - \mu) \sum_j (y_{ij} - \bar{y}_i),$$

and the cross term vanishes because $\sum_j (y_{ij} - \bar{y}_i) = 0$. The left side is $(N-1)S^2$ and $n \sum_i (\bar{y}_i - \mu)^2 = nk\operatorname{Var}(\bar{y}_{\text{sy}}) = N\operatorname{Var}(\bar{y}_{\text{sy}})$. Divide by N. ■

The first term is the variance of the population as a whole; the second is the pooled variation *within* the k systematic samples. The more heterogeneous the units within a systematic sample, the smaller $\operatorname{Var}(\bar{y}_{\text{sy}})$ becomes.

Remark 4.7: Periodicity

For the population 1 2 3 4 5 | 1 2 3 4 5 | 1 2 3 4 5 with $k = 5$, every systematic sample is constant, e.g. $(2, 2, 2)$. There is no within-sample variation, so $\operatorname{Var}(\bar{y}_{\text{sy}})$ is as large as it can be.

3.3 Intraclass correlation

Homogeneity within a systematic sample is measured by the **intraclass correlation coefficient**

$$\rho = \frac{\mathrm{E}[(y_{ij} - \mu)(y_{ij'} - \mu)]}{\mathrm{E}[(y_{ij} - \mu)^2]}, \qquad j \neq j',$$

where the expectations are averages over the $kn(n-1)/2$ pairs in the same sample and over the N units, respectively.

Theorem 4.8

$$\operatorname{Var}(\bar{y}_{\text{sy}}) = \frac{S^2}{n} \cdot \frac{N-1}{N} [1 + (n-1)\rho], \qquad \rho = \frac{2}{(n-1)(N-1)S^2} \sum_{i=1}^{k} \sum_{j<j'} (y_{ij} - \mu)(y_{ij'} - \mu).$$

Derivation

$$\begin{aligned} \operatorname{Var}(\bar{y}_{\text{sy}}) &= \frac{1}{k} \sum_i \Big(\frac{1}{n} \sum_j y_{ij} - \mu \Big)^2 = \frac{1}{n^2 k} \sum_i \Big(\sum_j (y_{ij} - \mu) \Big)^2 \\ &= \frac{1}{n^2 k} \Big[\sum_i \sum_j (y_{ij} - \mu)^2 + 2 \sum_i \sum_{j<j'} (y_{ij} - \mu)(y_{ij'} - \mu) \Big]. \end{aligned}$$

The denominator of ρ is $\frac{1}{N} \sum_i \sum_j (y_{ij} - \mu)^2 = \frac{N-1}{N} S^2$, and the numerator is $\frac{2}{kn(n-1)} \sum_i \sum_{j<j'} (\cdot)(\cdot)$, so

$$2 \sum_i \sum_{j<j'} (y_{ij} - \mu)(y_{ij'} - \mu) = kn(n-1) \frac{N-1}{N} S^2 \rho = (n-1)(N-1) S^2 \rho,$$

which is the stated formula for ρ. Substituting, with $n^2k = nN$,

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{nN}[(N-1)S^2 + (n-1)(N-1)S^2\rho] = \frac{S^2}{n}\frac{N-1}{N}[1+(n-1)\rho].$$

Compare with SRS, $\mathrm{Var}(\bar{y}_{\mathrm{ran}}) = \frac{S^2}{n}(1-\frac{n}{N})$:

- ρ large and positive (homogeneous systematic samples) $\Rightarrow \mathrm{Var}(\bar{y}_{\mathrm{sy}})$ large;
- ρ small positive or negative (heterogeneous samples) $\Rightarrow \mathrm{Var}(\bar{y}_{\mathrm{sy}})$ small;
- $\rho \approx 0 \Rightarrow \mathrm{Var}(\bar{y}_{\mathrm{sy}}) \approx \mathrm{Var}(\bar{y}_{\mathrm{ran}})$. Indeed $\rho = -\frac{1}{N-1}$ reproduces the SRS variance exactly.

3.4 ANOVA form

Arrange the population as an $n \times k$ table whose columns are the systematic samples, with column means $\bar{y}_1, \ldots, \bar{y}_k$ and grand mean μ. Let

$$SS_T = \sum_i \sum_j (y_{ij}-\mu)^2, \quad SS_W = \sum_i \sum_j (y_{ij}-\bar{y}_i)^2, \quad SS_B = n\sum_i (\bar{y}_i - \mu)^2,$$

so $SS_T = SS_W + SS_B$ and $S^2 = SS_T/(N-1)$. Then

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{k}\sum_i (\bar{y}_i - \mu)^2 = \frac{SS_B}{N},$$

and equating with the ρ form, $\frac{SS_T}{nN}[1+(n-1)\rho] = \frac{SS_B}{N}$, gives

$$1+(n-1)\rho = \frac{n\,SS_B}{SS_T} \quad \Longrightarrow \quad \rho = 1 - \frac{n}{n-1}\frac{SS_W}{SS_T} \approx 1 - \frac{SS_W}{SS_T} = \frac{SS_B}{SS_T}.$$

A large between-sample sum of squares means a large ρ and a large variance.

Example 4.9: Two orderings of $1, \ldots, 9$

Here $\mu = 5$, $S^2 = 60/8 = 7.5$, $n = k = 3$, and the SRS variance is $\mathrm{Var}(\bar{y}_{\mathrm{ran}}) = \frac{7.5}{3} \cdot \frac{6}{9} = 1.667$.

- Order 1 2 3 | 4 5 6 | 7 8 9: the samples are $(1,4,7), (2,5,8), (3,6,9)$ with means $4, 5, 6$. Then $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{3}(1+0+1) = \frac{2}{3}$ and $\rho = -21/60$. Heterogeneous samples: systematic beats SRS.
- Order 1 4 7 | 2 5 8 | 3 6 9: the samples are $(1,2,3), (4,5,6), (7,8,9)$ with means $2, 5, 8$. Then $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{3}(9+0+9) = 6$ and $\rho = 51/60$. Homogeneous samples: systematic is much worse than SRS.

Check: $\frac{7.5}{3} \cdot \frac{8}{9}(1+2\rho) = \frac{20}{9}(1+2\rho)$ equals $\frac{2}{3}$ for $\rho = -0.35$ and 6 for $\rho = 0.85$.

3.5 Types of population

We want systematic samples with heterogeneous units (small ρ). Which populations produce them?

1. **Random order.** E.g. estimating the average weight of $N = 3000$ SNU freshmen from an alphabetical list. The units in a systematic sample are effectively in random order, $\rho \approx 0$, and $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) \approx \mathrm{Var}(\bar{y}_{\mathrm{ran}})$: the systematic sample may be treated as an SRS.

2. **Ordered population.** E.g. farms ordered by area when estimating corn yield. A systematic sample covers the whole range, is heterogeneous, and $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) < \mathrm{Var}(\bar{y}_{\mathrm{ran}})$. It avoids samples with too many large or too many small farms, so it tends to be more representative than an SRS.

3. **Periodic variation.** E.g. supermarket sales, high on Friday–Saturday and low on Monday–Tuesday. If k matches the period, the units are homogeneous, ρ is large, and $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) > \mathrm{Var}(\bar{y}_{\mathrm{ran}})$.

3.6 Estimating the variance

A single systematic sample contains no information about between-sample variation, so $\mathrm{Var}(\bar{y}_{\mathrm{sy}})$ cannot be estimated unbiasedly from it. Assuming a randomly ordered population ($\rho \approx 0$), we use the SRS formula:

$$\bar{y}_{\mathrm{sy}} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad \widehat{\mathrm{Var}}(\bar{y}_{\mathrm{sy}}) = \frac{s^2}{n}\cdot\frac{N-n}{N}.$$

This is conservative for an ordered population and anti-conservative under periodicity.

4 Population Total and Proportion

Assuming a randomly ordered population:

$$\hat{\tau} = N\bar{y}_{\mathrm{sy}}, \qquad \widehat{\mathrm{Var}}(\hat{\tau}) = N^2\widehat{\mathrm{Var}}(\bar{y}_{\mathrm{sy}}) = N^2\frac{s^2}{n}\frac{N-n}{N};$$

$$\hat{p}_{\mathrm{sy}} = \bar{y}_{\mathrm{sy}} = \frac{1}{n}\sum y_i \ (y_i \in \{0,1\}), \qquad \widehat{\mathrm{Var}}(\hat{p}_{\mathrm{sy}}) = \frac{\hat{p}_{\mathrm{sy}}\hat{q}_{\mathrm{sy}}}{n-1}\cdot\frac{N-n}{N}.$$

5 Selecting the Sample Size

Under the same assumption the SRS formulas apply: for μ with bound B,

$$n = \frac{N\sigma^2}{(N-1)D+\sigma^2} = \frac{NS^2}{ND+S^2}, \qquad D = \frac{B^2}{4},$$

and for p, $n = \dfrac{Npq}{(N-1)D+pq}$ with $D = B^2/4$.

6 Selected exercises

Motivation. The first item combines systematic selection with stratification; the second quantifies exactly how much a systematic sample gains in an ordered population, the case described qualitatively above; the third is a computation for a total under the random-order assumption.

Exercise 4.1 (Systematic sampling within strata)

A population of N elements is divided into L strata of sizes $N_\ell = n_\ell k$, $\ell = 1,\dots,L$, with the same k in every stratum. In each stratum a 1-in-k systematic sample with a random start is drawn (Method A), independently across strata. Let $\bar{y}_\ell$ be the mean of the sample from stratum ℓ and $\hat{\mu} = \sum_\ell w_\ell \bar{y}_\ell$, $w_\ell = N_\ell/N$. Find $\mathrm{E}(\hat{\mu})$ and $\mathrm{Var}(\hat{\mu})$.

Solution. Write $y_{\ell ij}$ for the jth element of the ith possible systematic sample in stratum ℓ, $\bar{y}_{\ell i} = \frac{1}{n_\ell}\sum_j y_{\ell ij}$, and μ_ℓ, S_ℓ^2, ρ_ℓ for the stratum mean, variance and intraclass correlation.

Step 1. (Unbiasedness.) In stratum ℓ the sample i is chosen with probability $1/k$, so by Case 1 of the unbiasedness subsection, $\mathrm{E}(\bar{y}_\ell) = \frac{1}{k}\sum_i \bar{y}_{\ell i} = \frac{1}{N_\ell}\sum_i\sum_j y_{\ell ij} = \mu_\ell$. Hence $\mathrm{E}(\hat{\mu}) = \sum_\ell w_\ell \mu_\ell = \frac{1}{N}\sum_\ell \tau_\ell = \mu$.

Step 2. (Variance.) The random starts are independent across strata, so the covariances vanish:

$$\mathrm{Var}(\hat{\mu}) = \sum_{\ell=1}^{L} w_\ell^2 \mathrm{Var}(\bar{y}_\ell), \qquad \mathrm{Var}(\bar{y}_\ell) = \frac{1}{k}\sum_{i=1}^{k}(\bar{y}_{\ell i} - \mu_\ell)^2.$$

Step 3. (Intraclass form.) Applying the intraclass-correlation theorem within stratum ℓ,

$$\mathrm{Var}(\hat{\mu}) = \sum_{\ell=1}^{L} w_\ell^2 \,\frac{S_\ell^2}{n_\ell}\cdot\frac{N_\ell - 1}{N_\ell}\big[1 + (n_\ell - 1)\rho_\ell\big].$$

Step 4. (Interpretation.) Since $n_\ell = N_\ell/k$, the design is automatically proportionally allocated with common sampling fraction $1/k$. If every stratum is in random order ($\rho_\ell \approx -1/(N_\ell - 1)$), the bracket times $\frac{N_\ell - 1}{N_\ell}$ is $1 - \frac{1}{k}$ and $\mathrm{Var}(\hat{\mu})$ reduces to that of proportional stratified sampling. A single systematic sample from each stratum still gives no unbiased variance estimator, so in practice $\mathrm{Var}(\bar{y}_\ell)$ is estimated by the SRS formula, stratum by stratum. ■

Exercise 4.2 (Population with a linear trend)

Let $y_t = a + bt$, $t = 1, \dots, N$, with $b \neq 0$ and $N = nk$, $n \geq 2$. Show that

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{b^2(k^2-1)}{12}, \qquad \mathrm{Var}(\bar{y}_{\mathrm{ran}}) = \frac{b^2(k-1)(N+1)}{12}, \qquad \frac{\mathrm{Var}(\bar{y}_{\mathrm{sy}})}{\mathrm{Var}(\bar{y}_{\mathrm{ran}})} = \frac{k+1}{N+1} < 1.$$

Solution. [Supplement]

Step 1. (A discrete uniform variance.) If I is uniform on $\{1, \dots, k\}$, then $\mathrm{E}I = \frac{k+1}{2}$, $\mathrm{E}I^2 = \frac{(k+1)(2k+1)}{6}$ and

$$\mathrm{Var}I = \frac{(k+1)(2k+1)}{6} - \frac{(k+1)^2}{4} = \frac{(k+1)\big[2(2k+1) - 3(k+1)\big]}{12} = \frac{k^2-1}{12}.$$

Step 2. (Systematic mean.) Sample i consists of $t = i, i+k, \dots, i+(n-1)k$, so $\bar{y}_i = a + b(i + \frac{(n-1)k}{2})$. The start I is uniform on $\{1, \dots, k\}$, so $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = b^2\mathrm{Var}I = \frac{b^2(k^2-1)}{12}$.

Step 3. (SRS.) Step 1 with k replaced by N gives $\sum_{t=1}^{N}\left(t - \frac{N+1}{2}\right)^2 = N\frac{N^2-1}{12}$, so $S^2 = \frac{b^2}{N-1}\cdot\frac{N(N^2-1)}{12} = \frac{b^2N(N+1)}{12}$. With $1 - f = 1 - \frac{1}{k}$ and $N = nk$,

$$\mathrm{Var}(\bar{y}_{\mathrm{ran}}) = \frac{k-1}{k}\cdot\frac{1}{n}\cdot\frac{b^2N(N+1)}{12} = \frac{b^2(k-1)(N+1)}{12}.$$

Step 4. (Ratio.) Dividing, $\frac{(k^2-1)}{(k-1)(N+1)} = \frac{k+1}{N+1}$, which is below 1 because $k < N$ when $n \geq 2$. For $N = 1000$ and $k = 10$ the systematic sample has about 1% of the SRS variance: it cannot pick only small or only large units, which is the "ordered population" case of the text made exact. ■

Exercise 4.3 (Families who rent)

A 1-in-50 systematic sample is drawn from the tax records of a city with $N = 15{,}200$ households to estimate the total number of families who rent. Of the $n = 304$ sampled families, $\sum y_i = 88$ rent ($y_i = 1$ if the family rents). Estimate the total and place a bound on the error, assuming the records are in random order with respect to renting.

Solution.

Step 1. $\hat{p}_{\text{sy}} = 88/304 = 0.2895$ and $\hat{\tau} = N\hat{p}_{\text{sy}} = 15{,}200 \times 0.2895 = 4400$.

Step 2. Under the random-order assumption the SRS formulas apply:

$$\widehat{\text{Var}}(\hat{p}_{\text{sy}}) = \frac{\hat{p}\hat{q}}{n-1} \cdot \frac{N-n}{N} = \frac{0.2895 \times 0.7105}{303} \cdot \frac{14{,}896}{15{,}200} = 0.000665.$$

Step 3. $\widehat{\text{Var}}(\hat{\tau}) = N^2 \widehat{\text{Var}}(\hat{p}_{\text{sy}}) = 15{,}200^2 \times 0.000665 = 153{,}700$ and $2\sqrt{153{,}700} = 784$. The estimate is 4400 ± 784 renting families. The bound is trustworthy only if the list has no periodicity of period 50 related to renting (e.g. apartment blocks listed in groups of 50). ■

Chapter 5. Ratio Estimation

1 Auxiliary Information

So far the data were modeled as $y_i = \mu + \varepsilon_i$ with $\mathrm{E}(\varepsilon_i) = 0$, so $\mathrm{E}(y_i) = \mu$ for every unit. If an auxiliary variable x related to y is available for each sampled unit, a better model is

$$y_i = f(x_i) + \varepsilon_i, \qquad \mathrm{E}(y_i) = f(x_i) \quad (\text{e.g. } f(x) = \beta_0 + \beta_1 x),$$

and knowledge of the population mean μ_x or total τ_x of x can be used to sharpen the estimate of μ_y or τ_y. This chapter uses a line through the origin (ratio estimation); Ch. 7 uses a general line (regression estimation).

Example 5.1: Sugar in a truckload of oranges

The wholesale price of oranges is based on the total sugar content τ_y of the load. The estimate $N\bar{y}$ needs the number of oranges N, which is unknown. But sugar content y is closely related to weight x, and the total weight τ_x of the truckload is easy to obtain. Since

$$\frac{\mu_y}{\mu_x} = \frac{N\mu_y}{N\mu_x} = \frac{\tau_y}{\tau_x} \quad \Longrightarrow \quad \tau_y = \frac{\mu_y}{\mu_x}\tau_x,$$

the **ratio estimator** of the total is

$$\hat{\tau}_y = \frac{\bar{y}}{\bar{x}}\tau_x = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} x_i}\tau_x,$$

which does not require N.

Other ratios of interest: total monthly food budget to total monthly income per family; number of children to number of residents; average amount of a new drug needed to evoke a response relative to that of a standard drug.

2 Ratio Estimation under SRS

Motivation. If an auxiliary variable x with known population total is strongly correlated with y, the ratio $\bar{y}/\bar{x}$ is more stable than $\bar{y}$ itself. Multiplying it by the known total of x gives a ratio estimator with much smaller variance.

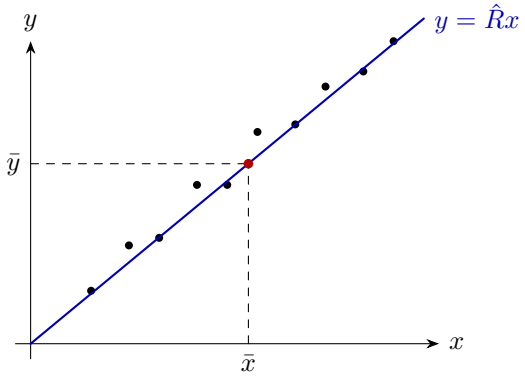


Figure. *Ratio estimation fits a line through the origin with slope $\hat{R} = \bar{y}/\bar{x}$. When y is roughly proportional to x, $\hat{\tau}_y = \hat{R}\,\tau_x$ uses the known total τ_x and has much smaller variance than $N\bar{y}$.*

Let (x_i, y_i), $i = 1, \dots, n$, be an SRS of pairs from a population of size N. The population ratio and its estimator are

$$R = \frac{\sum_{i=1}^{N} y_i}{\sum_{i=1}^{N} x_i} = \frac{\mu_y}{\mu_x}, \qquad r = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} x_i} = \frac{\bar{y}}{\bar{x}}.$$

Theorem 5.2: Ratio estimators

With $f = n/N$ and $s_r^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - r x_i)^2$,

ratio: $r = \bar{y}/\bar{x}$, $\quad \widehat{\mathrm{Var}}(r) = \dfrac{1-f}{n} \cdot \dfrac{1}{\mu_x^2}\, s_r^2$ (μ_x replaced by $\bar{x}$ if unknown);

total: $\hat{\tau}_y = r\tau_x$, $\quad \widehat{\mathrm{Var}}(\hat{\tau}_y) = \tau_x^2 \widehat{\mathrm{Var}}(r) = \tau_x^2 \dfrac{1-f}{n} \dfrac{s_r^2}{\bar{x}^2}$;

mean: $\hat{\mu}_y = r\mu_x$, $\quad \widehat{\mathrm{Var}}(\hat{\mu}_y) = \mu_x^2 \widehat{\mathrm{Var}}(r) = \dfrac{1-f}{n} s_r^2$.

The residuals $y_i - r x_i$ measure the departure from a line through the origin. If $y_i \approx r x_i$ for all units, s_r^2 is small and the ratio estimator is very precise.

Derivation: first-order variance of r

Write $d_i = y_i - R x_i$. For N population units, $\bar{D} = \frac{1}{N}\sum_{i=1}^{N}(y_i - R x_i) = \mu_y - R\mu_x = 0$. If $\bar{x} \approx \mu_x$,

$$r - R = \frac{\bar{y} - R\bar{x}}{\bar{x}} \approx \frac{\bar{y} - R\bar{x}}{\mu_x} = \frac{1}{\mu_x} \cdot \frac{1}{n}\sum_{i=1}^{n}(y_i - R x_i) = \frac{\bar{d}}{\mu_x}.$$

Thus $r - R$ is approximately the SRS mean of the d_i divided by μ_x, and by Ch. 2

$$\mathrm{Var}(r) \approx \mathrm{E}(r-R)^2 = \frac{1}{\mu_x^2}\mathrm{E}(\bar{d} - \bar{D})^2 = \frac{1}{\mu_x^2} \cdot \frac{N-n}{N} \cdot \frac{1}{n} S_r^2, \qquad S_r^2 = \frac{1}{N-1}\sum_{i=1}^{N}(y_i - R x_i)^2.$$

Expanding the square,

$$\sum_{i=1}^{N}(y_i - R x_i)^2 = \sum_{i=1}^{N}\left[(y_i - \mu_y) - R(x_i - \mu_x)\right]^2 = (N-1)(S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y),$$

using $\mu_y = R\mu_x$ and $\sum(x_i - \mu_x)(y_i - \mu_y) = (N-1)\rho S_x S_y$. Hence

$$\mathrm{Var}(r) \approx \frac{1}{\mu_x^2} \cdot \frac{1-f}{n}(S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y).$$

Replacing R by r and S_r^2 by s_r^2 gives $\widehat{\mathrm{Var}}(r)$. Note that $\mathrm{Var}(r)$ is smallest when $y_i = R x_i$ exactly.

2.1 Correlation form

Expanding s_r^2 gives the equivalent estimate

$$\widehat{\mathrm{Var}}(r) = \frac{1-f}{n} \cdot \frac{1}{\bar{x}^2}(s_y^2 + r^2 s_x^2 - 2r\hat{\rho} s_x s_y), \qquad \hat{\rho} = \frac{s_{xy}}{s_x s_y},$$

with $s_{xy} = \frac{1}{n-1}\sum(x_i - \bar{x})(y_i - \bar{y})$. A larger positive correlation $\hat{\rho}$ means a smaller variance.

Example 5.3: Assessed values of homes

To estimate the relative change in the assessed value of homes over two years (x: earlier value, y: current value), $n = 20$ homes give

$$r = \frac{\sum_{i=1}^{20} y_i}{\sum_{i=1}^{20} x_i} = 1.066, \qquad \sqrt{\widehat{\mathrm{Var}}(r)} = 0.00745, \qquad B = 2\sqrt{\widehat{\mathrm{Var}}(r)} = 0.0149,$$

so an approximate 95% interval for R is $[1.0511, 1.0809]$.

Example 5.4: Oranges, continued

Ten oranges give $\sum y_i = 0.246$ (sugar) and $\sum x_i = 4.350$ (weight); the load weighs $\tau_x = 1800$ pounds:

$$\hat{\tau}_y = \frac{0.246}{4.350}(1800) = 101.79, \qquad 2\sqrt{\widehat{\mathrm{Var}}(\hat{\tau}_y)} = 6.3.$$

3 Bias of the Ratio Estimator

Motivation. A ratio of unbiased estimators is not unbiased. The bias is of order $1/n$, smaller than the standard error, so in moderate samples it is negligible.

r is a ratio of random variables and is generally biased for R. Write $\delta_y = (\bar{y} - \mu_y)/\mu_y$ and $\delta_x = (\bar{x} - \mu_x)/\mu_x$, so

$$r = R\,(1 + \delta_y)(1 + \delta_x)^{-1} = R(1 + \delta_y)\,(1 - \delta_x + \delta_x^2 - \cdots) \approx R[1 + \delta_y - \delta_x + \delta_x^2 - \delta_x\delta_y].$$

- **Case 1** ($\bar{x} \approx \mu_x$, keep only δ_y): $r \approx R + R\delta_y$, so $\mathrm{E}(r) \approx R$.
- **Case 2** (first order): $\mathrm{E}(r) \approx R + R[\mathrm{E}\delta_y - \mathrm{E}\delta_x] = R$.
- **Case 3** (second order):
$$\mathrm{E}(r) - R \approx R\,\mathrm{E}(\delta_x^2) - R\,\mathrm{E}(\delta_x\delta_y).$$

Theorem 5.5: Approximate bias

$$\mathrm{E}(r) - R \approx \frac{1}{\mu_x^2} \cdot \frac{1 - f}{n}(RS_x^2 - \rho S_x S_y).$$

The bias vanishes (to this order) when $RS_x = \rho S_y$, i.e. when $R = \rho S_y/S_x = S_{xy}/S_x^2$ is the least-squares slope.

Proof. Under SRS, $\mathrm{E}[(\bar{x} - \mu_x)^2] = \frac{1-f}{n}S_x^2$ and $\mathrm{E}[(\bar{x} - \mu_x)(\bar{y} - \mu_y)] = \frac{1-f}{n}S_{xy} = \frac{1-f}{n}\rho S_x S_y$. Therefore

$$\mathrm{E}(r) - R \approx \frac{R}{\mu_x^2}\frac{1 - f}{n}S_x^2 - \frac{R}{\mu_x\mu_y}\frac{1 - f}{n}\rho S_x S_y,$$

and $R/(\mu_x\mu_y) = 1/\mu_x^2$. ∎

Remark 5.6: When is r unbiased?

If $y_i = \beta x_i$ (a line through the origin), then $R = \sum y_i / \sum x_i = \beta$ and also

$$\rho \frac{S_y}{S_x} = \frac{S_{xy}}{S_x^2} = \frac{\frac{1}{N-1}\sum (x_i - \mu_x)\,\beta(x_i - \mu_x)}{\frac{1}{N-1}\sum (x_i - \mu_x)^2} = \beta,$$

so the bias term is zero. The bias is negligible whenever the relation between y and x is a straight line through the origin.

Theorem 5.7: An exact bias bound

$$\mathrm{E}(r) - R = -\frac{\mathrm{Cov}(r, \bar{x})}{\mu_x}, \qquad \frac{|\mathrm{E}(r) - R|}{\sqrt{\mathrm{Var}(r)}} \le \frac{\sqrt{\mathrm{Var}(\bar{x})}}{\mu_x} = CV(\bar{x}).$$

Proof. Since $r\bar{x} = \bar{y}$,

$$\mathrm{Cov}(r, \bar{x}) = \mathrm{E}(r\bar{x}) - \mathrm{E}(r)\mathrm{E}(\bar{x}) = \mu_y - \mathrm{E}(r)\mu_x = \mu_x(R - \mathrm{E}(r)).$$

By $|\mathrm{Cov}(r, \bar{x})| \le \sqrt{\mathrm{Var}(r)}\sqrt{\mathrm{Var}(\bar{x})}$, $|\mathrm{E}(r) - R| \le \frac{1}{\mu_x}\sqrt{\mathrm{Var}(r)}\sqrt{\mathrm{Var}(\bar{x})}$. ■

The bias is small relative to the standard error whenever the coefficient of variation of $\bar{x}$ is small, which holds for moderately large n.

4 Ratio Estimation in Stratified Random Sampling

Motivation. Ratio estimation and stratification can be combined, either one ratio per stratum (separate) or one ratio for the whole sample (combined). The choice depends on stratum sample sizes and on whether the ratios differ across strata.

With strata $h = 1, \ldots, L$ (weights $w_h = N_h/N$), there are two ways to build a ratio estimator of μ_y.

Definition 5.8

1. **Separate ratio estimator**: estimate the ratio within each stratum and combine,

$$\hat{\mu}_{yRS} = \sum_{h=1}^{L} w_h\, r_h\, \mu_{xh}, \qquad r_h = \frac{\bar{y}_h}{\bar{x}_h}.$$

   For two strata A and B: $\hat{\mu}_{yRS} = \frac{N_A}{N}\frac{\bar{y}_A}{\bar{x}_A}\mu_{xA} + \frac{N_B}{N}\frac{\bar{y}_B}{\bar{x}_B}\mu_{xB}$.

2. **Combined ratio estimator**: estimate μ_y and μ_x by the stratified means and take one ratio,

$$\hat{\mu}_{yRC} = r_C\, \mu_x, \qquad r_C = \frac{\bar{y}_{\mathrm{st}}}{\bar{x}_{\mathrm{st}}}, \quad \bar{y}_{\mathrm{st}} = \sum_h w_h \bar{y}_h,\ \bar{x}_{\mathrm{st}} = \sum_h w_h \bar{x}_h.$$

Theorem 5.9: Estimated variances

$$\widehat{\mathrm{Var}}(\hat{\mu}_{yRS}) = \sum_{h=1}^{L} w_h^2 \frac{N_h - n_h}{N_h n_h} \cdot \frac{\sum_{i=1}^{n_h} (y_{hi} - r_h x_{hi})^2}{n_h - 1},$$

$$\widehat{\mathrm{Var}}(\hat{\mu}_{yRC}) = \sum_{h=1}^{L} w_h^2 \frac{N_h - n_h}{N_h n_h} s_{rh}^2,$$

where s_{rh}^2 is the sample variance of the terms $y_{hi} - r_C x_{hi}$ in stratum h.

Derivation: which one is better?

The population versions are $\mathrm{Var}(\hat{\mu}_{yRS}) \approx \sum_h w_h^2 \frac{1-f_h}{n_h} S_{h,S}^2$ and $\mathrm{Var}(\hat{\mu}_{yRC}) \approx \sum_h w_h^2 \frac{1-f_h}{n_h} S_{h,C}^2$, with $R_h = \mu_{yh}/\mu_{xh}$, $R_C = \mu_y/\mu_x$ and

$$\begin{aligned}(N_h - 1)S_{h,S}^2 &= \sum_i (y_{hi} - R_h x_{hi})^2 = (N_h - 1)(S_{yh}^2 + R_h^2 S_{xh}^2 - 2R_h \rho_h S_{xh} S_{yh}),\\ (N_h - 1)S_{h,C}^2 &= \sum_i \left[(y_{hi} - \mu_{yh}) - R_C(x_{hi} - \mu_{xh})\right]^2\\ &= (N_h - 1)(S_{yh}^2 + R_C^2 S_{xh}^2 - 2R_C \rho_h S_{xh} S_{yh}).\end{aligned}$$

(The combined deviation $\hat{\mu}_{yRC} - \mu_y \approx \bar{y}_{\mathrm{st}} - R_C \bar{x}_{\mathrm{st}} = \bar{d}_{\mathrm{st}}$ is a stratified mean of $d_{hi} = y_{hi} - R_C x_{hi}$, whose stratum means are $\mu_{yh} - R_C \mu_{xh}$.) Subtracting,

$$\begin{aligned}S_{h,C}^2 - S_{h,S}^2 &= (R_C^2 - R_h^2)S_{xh}^2 - 2(R_C - R_h)\rho_h S_{xh} S_{yh}\\ &= (R_C - R_h)^2 S_{xh}^2 + 2(R_C - R_h)(R_h S_{xh}^2 - \rho_h S_{xh} S_{yh}).\end{aligned}$$

The last bracket is zero when each stratum ratio is (approximately) unbiased, $R_h S_{xh} = \rho_h S_{yh}$. Then

$$\mathrm{Var}(\hat{\mu}_{yRC}) - \mathrm{Var}(\hat{\mu}_{yRS}) \approx \sum_h w_h^2 \frac{1 - f_h}{n_h} (R_C - R_h)^2 S_{xh}^2 \ \geq 0.$$

Remark 5.10: Practical guidance

- When the stratum ratios R_h differ greatly, the separate estimator is more efficient.
- But each r_h carries its own ratio bias, and these add up across strata. When the n_h are small the separate estimator can be badly biased; when the n_h are large (over about 50) the bias is negligible. With small strata the combined estimator is safer.

5 Ratio Estimator versus the Sample Mean

Motivation. The ratio estimator beats $\bar{y}$ when the correlation between x and y exceeds half the ratio of their coefficients of variation; this condition tells us when the auxiliary information is worth using.

Compare $\bar{y}$ and $\hat{\mu}_y = r\mu_x$ by relative efficiency $\mathrm{RE}(\hat{\theta}_1, \hat{\theta}_2) = \mathrm{Var}(\hat{\theta}_1)/\mathrm{Var}(\hat{\theta}_2)$:

$$\mathrm{Var}(\bar{y}) = \frac{1-f}{n} S_y^2, \qquad \mathrm{Var}(\hat{\mu}_y) \approx \frac{1-f}{n}(S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y),$$

$$\mathrm{RE}(\bar{y}, \hat{\mu}_y) = \frac{\mathrm{Var}(\bar{y})}{\mathrm{Var}(\hat{\mu}_y)} = \frac{S_y^2}{S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y}.$$

Theorem 5.11: When the ratio estimator wins

$\mathrm{RE}(\bar{y}, \hat{\mu}_y) > 1$ if and only if

$$\rho > \frac{1}{2} \cdot \frac{S_x/\mu_x}{S_y/\mu_y} = \frac{1}{2} \cdot \frac{CV_x}{CV_y}.$$

In particular, if x and y have equal coefficients of variation, the ratio estimator is better as soon as $\rho > 1/2$.

Proof. $\mathrm{RE} > 1 \iff R^2 S_x^2 < 2R\rho S_x S_y \iff RS_x < 2\rho S_y \iff \rho > \frac{RS_x}{2S_y}$, and $R = \mu_y/\mu_x$. The sample version replaces everything by estimates: $\hat{\rho} > \frac{1}{2}\frac{s_x/\bar{x}}{s_y/\bar{y}}$. ■

The ratio estimator is more efficient than $\bar{y}$ when x and y are highly positively correlated; the estimated relative efficiency is meaningful only for large samples.

6 Selected exercises

Motivation. The ratio estimator was justified above by design-based approximations. The first item shows that it is also the exact optimal estimator under a natural model, and the second is a complete computation for a total.

Exercise 5.1 (The ratio estimator as a weighted least-squares fit)

Consider the model $y_i = \beta x_i + \varepsilon_i$, $i = 1, \dots, n$, with fixed $x_i > 0$, $\mathrm{E}(\varepsilon_i) = 0$, $\mathrm{Var}(\varepsilon_i) = \sigma^2 x_i$ and uncorrelated errors.
(a) Show that $r = \sum y_i / \sum x_i$ minimizes the weighted sum of squares $\sum_i (y_i - \beta x_i)^2 / x_i$.
(b) Show that r is unbiased for β with $\mathrm{Var}(r) = \sigma^2 / \sum_i x_i$, and that it has the smallest variance among all linear unbiased estimators $\sum_i a_i y_i$ of β.

Solution. [Supplement]

(a) $Q(\beta) = \sum_i (y_i - \beta x_i)^2 / x_i$ is a quadratic in β with $Q'(\beta) = -2\sum_i (y_i - \beta x_i) = -2(\sum_i y_i - \beta \sum_i x_i)$ and $Q'' = 2\sum_i x_i > 0$, so the unique minimizer is $\beta = \sum y_i / \sum x_i = r$.

(b)

Step 1. (Unbiasedness and variance.) $r = \sum_i a_i^0 y_i$ with $a_i^0 = 1/\sum_j x_j$. Then $\mathrm{E}(r) = \beta \sum_i a_i^0 x_i = \beta$ and $\mathrm{Var}(r) = \sigma^2 \sum_i (a_i^0)^2 x_i = \sigma^2 / \sum_i x_i$.

Step 2. (Unbiasedness constraint.) A linear estimator $\sum_i a_i y_i$ has mean $\beta \sum_i a_i x_i$, so it is unbiased for every β iff $\sum_i a_i x_i = 1$. Its variance is $\sigma^2 \sum_i a_i^2 x_i$.

Step 3. (Cauchy–Schwarz.) Writing $a_i x_i = (a_i \sqrt{x_i})\sqrt{x_i}$,

$$1 = \Big(\sum_i a_i x_i\Big)^2 \le \Big(\sum_i a_i^2 x_i\Big)\Big(\sum_i x_i\Big),$$

so $\sigma^2 \sum_i a_i^2 x_i \ge \sigma^2 / \sum_i x_i = \mathrm{Var}(r)$, with equality iff $a_i \sqrt{x_i} \propto \sqrt{x_i}$, i.e. a_i constant, i.e. $a_i = a_i^0$.

Thus the ratio estimator is the best linear unbiased estimator when y is proportional to x with error variance proportional to x, the situation of the oranges example. ■

Exercise 5.2 (Sales of television sets)

A corporation wants the total earnings τ_y from sales of television sets in the current three-month period. An SRS of $n = 13$ of its $N = 123$ district offices gives the previous-year figures

x_i and current figures y_i below, and the previous-year total for all offices is $\tau_x = 128{,}200$.

x_i	550	720	1500	1020	620	980	928	1200	1350	1750	670	729	1530
y_i	610	780	1600	1030	600	1050	977	1440	1570	2210	980	865	1710

Estimate τ_y by the ratio estimator and place a bound on the error.

Solution.

Step 1. $\sum x_i = 13{,}547$, $\sum y_i = 15{,}422$, so $r = 15{,}422/13{,}547 = 1.1384$ and $\hat{\tau}_y = r\tau_x = 1.1384 \times 128{,}200 = 145{,}944$.

Step 2. The residual variance is $s_r^2 = \frac{1}{12}\sum_i (y_i - r x_i)^2 = 12{,}990$, and $\mu_x = \tau_x/N = 1042.3$ is known.

Step 3. By the ratio-estimator theorem,

$$\widehat{\mathrm{Var}}(\hat{\tau}_y) = \tau_x^2 \, \frac{1-f}{n} \, \frac{s_r^2}{\mu_x^2} = N^2 \, \frac{1-f}{n} \, s_r^2 = 123^2 \cdot \frac{1 - 13/123}{13} \cdot 12{,}990 = 1.352 \times 10^7,$$

so the bound is $2\sqrt{1.352 \times 10^7} = 7354$ and τ_y is estimated by $145{,}944 \pm 7354$.

Step 4. For comparison, $N\bar{y} = 123 \times 1186.3 = 145{,}916$ has $s_y^2 = 231{,}543$ and bound

$$2N\sqrt{(1-f)s_y^2/n} \approx 31{,}050 :$$

the ratio estimator is about four times as precise, since current and previous sales are almost proportional. ■

Chapter 6. Simple Linear Regression

This chapter reviews simple linear regression, the tool behind the regression estimator of Ch. 7. Regression compresses n-dimensional data into a 2-dimensional summary $(\hat\beta_0, \hat\beta_1)$: data = model structure + error.

1 The Model

Definition 6.1

The simple linear regression model with a single regressor x is

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \dots, n,$$

where the errors are uncorrelated with mean 0 and unknown variance σ^2. The regressor x is treated as fixed; y is random. β_0 (intercept) and β_1 (slope) are the **regression coefficients**.

Then

$$\mathrm{E}(y \mid x) = \beta_0 + \beta_1 x = f(x), \qquad \mathrm{Var}(y \mid x) = \mathrm{Var}(\varepsilon) = \sigma^2,$$

so the true regression line is the line of mean responses. The goals are to estimate β_0, β_1 and to test $H_0 : \beta_1 = 0$.

2 Least-Squares Estimation

Motivation. Ratio estimation assumes y is proportional to x. When the relation has an intercept, a regression line is the natural tool; this section reviews least squares as preparation for the regression estimator.

The least-squares (LS) estimators minimize

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2 = \sum_{i=1}^{n} \varepsilon_i^2.$$

Setting $\partial S/\partial \beta_0 = \partial S/\partial \beta_1 = 0$ gives the **normal equations**

$$n\hat\beta_0 + \hat\beta_1 \sum x_i = \sum y_i, \qquad \hat\beta_0 \sum x_i + \hat\beta_1 \sum x_i^2 = \sum x_i y_i.$$

Theorem 6.2: LS estimators

$$\hat\beta_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum (x_i - \bar x)(y_i - \bar y)}{\sum (x_i - \bar x)^2} = \frac{\sum (x_i - \bar x) y_i}{\sum (x_i - \bar x)^2}, \qquad \hat\beta_0 = \bar y - \hat\beta_1 \bar x.$$

The **fitted values** are $\hat y_i = \hat\beta_0 + \hat\beta_1 x_i = \widehat{\mathrm{E}}(y_i \mid x_i)$ and the **residuals** are $e_i = y_i - \hat y_i$, so $y_i = \hat\beta_0 + \hat\beta_1 x_i + e_i$; residuals are the basis of model-adequacy checks (residual analysis).

Example 6.3: Rocket propellant data, $n = 20$

y = shear strength, x = age of propellant. From $\bar x = 13.3625$, $\bar y = 2131.358$, $S_{xx} = 1106.559$, $S_{xy} = -41112.65$:

$$\hat\beta_1 = \frac{-41112.65}{1106.559} = -37.15, \qquad \hat\beta_0 = 2131.358 + 37.15 \times 13.3625 = 2627.82,$$

so $\hat{y} = 2627.82 - 37.15x$. R output: intercept 2627.822 (SE 44.184), slope -37.154 (SE 2.889, $t = -12.86$, $p = 1.64 \times 10^{-10}$).

2.1 Properties of the estimators

Theorem 6.4

$\mathrm{E}(\hat{\beta}_1) = \beta_1$, $\mathrm{E}(\hat{\beta}_0) = \beta_0$, and

$$\mathrm{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}, \qquad \mathrm{Var}(\hat{\beta}_0) = \sigma^2\Big(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\Big), \qquad \mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\bar{x}\frac{\sigma^2}{S_{xx}}.$$

Derivation: proof

Write $\hat{\beta}_1 = \sum c_i y_i$ with $c_i = (x_i - \bar{x})/S_{xx}$. Then $\sum c_i = 0$ and $\sum c_i x_i = \sum (x_i - \bar{x})x_i / S_{xx} = 1$, so

$$\mathrm{E}(\hat{\beta}_1) = \sum c_i(\beta_0 + \beta_1 x_i) = \beta_0 \sum c_i + \beta_1 \sum c_i x_i = \beta_1,$$

$$\mathrm{Var}(\hat{\beta}_1) = \sigma^2 \sum c_i^2 = \sigma^2 \frac{S_{xx}}{S_{xx}^2} = \frac{\sigma^2}{S_{xx}}.$$

Since $\mathrm{E}(\bar{y}) = \beta_0 + \beta_1 \bar{x}$, $\mathrm{E}(\hat{\beta}_0) = \beta_0 + \beta_1 \bar{x} - \bar{x}\beta_1 = \beta_0$. With $\bar{y} = \sum d_i y_i$, $d_i = 1/n$,

$$\mathrm{Cov}(\bar{y}, \hat{\beta}_1) = \sigma^2 \sum d_i c_i = \frac{\sigma^2}{n} \sum c_i = 0,$$

hence

$$\mathrm{Var}(\hat{\beta}_0) = \mathrm{Var}(\bar{y}) + \bar{x}^2 \mathrm{Var}(\hat{\beta}_1) - 2\bar{x}\mathrm{Cov}(\bar{y}, \hat{\beta}_1) = \sigma^2\Big(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\Big),$$

$$\mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1) = \mathrm{Cov}(\bar{y}, \hat{\beta}_1) - \bar{x}\mathrm{Var}(\hat{\beta}_1) = -\bar{x}\sigma^2 / S_{xx}.$$

2.2 Properties of the fit

Theorem 6.5

(1) $\sum e_i = 0$, equivalently $\sum y_i = \sum \hat{y}_i$;

(2) the fitted line passes through $(\bar{x}, \bar{y})$;

(3) $\sum x_i e_i = 0$;

(4) $\sum \hat{y}_i e_i = 0$.

Proof. (1) and (3) are the two normal equations: $\sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$ and $\sum x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$. (2): at $x = \bar{x}$, $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x} = \bar{y}$. (4): $\sum \hat{y}_i e_i = \hat{\beta}_0 \sum e_i + \hat{\beta}_1 \sum x_i e_i = 0$. ■

2.3 Centered model

Rewriting

$$y_i = (\beta_0 + \beta_1 \bar{x}) + \beta_1 (x_i - \bar{x}) + \varepsilon_i = \beta_0' + \beta_1 (x_i - \bar{x}) + \varepsilon_i$$

gives $\hat{\beta}_0' = \bar{y}$ and $\hat{\beta}_1 = S_{xy}/S_{xx}$, and now $\mathrm{Cov}(\hat{\beta}_0', \hat{\beta}_1) = \mathrm{Cov}(\bar{y}, \hat{\beta}_1) = 0$: centering makes the estimators uncorrelated.

3 Estimation of σ^2

Motivation. To make inference about the slope we need an estimate of the error variance, the residual mean square.

Theorem 6.6: Sum-of-squares decomposition

$$\underbrace{\sum (y_i - \bar{y})^2}_{SS_T} = \underbrace{\sum (y_i - \hat{y}_i)^2}_{SS_E} + \underbrace{\sum (\hat{y}_i - \bar{y})^2}_{SS_R}, \qquad SS_E = SS_T - \hat{\beta}_1 S_{xy}, \quad SS_R = \hat{\beta}_1 S_{xy}.$$

Proof. Expanding $\sum (y_i - \hat{y}_i + \hat{y}_i - \bar{y})^2$ leaves the cross term $2\sum e_i(\hat{y}_i - \bar{y}) = 2(\sum e_i \hat{y}_i - \bar{y}\sum e_i) = 0$. Since $e_i = (y_i - \bar{y}) - \hat{\beta}_1(x_i - \bar{x})$,

$$SS_E = SS_T + \hat{\beta}_1^2 S_{xx} - 2\hat{\beta}_1 S_{xy} = SS_T - \hat{\beta}_1 S_{xy}.$$

■

SS_E has $n-2$ degrees of freedom, and the unbiased estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{SS_E}{n-2} = MS_E \quad \text{(the mean square error)}.$$

$\hat{\sigma}$ is the standard error of regression; it is a model-dependent estimate of σ.

Example 6.7: Rocket propellant

$SS_T = 1{,}693{,}737.6$ and $SS_E = 1{,}693{,}737.6 - (-37.15)(-41112.65) = 166{,}402.65$, so $\hat{\sigma}^2 = 166402.65/18 = 9244.59$.

4 Hypothesis Testing on the Slope

Motivation. If the slope is not significantly different from zero, the auxiliary variable is of little help for estimation.

Now assume $\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$. Then $\hat{\beta}_1 \sim N(\beta_1, \sigma^2/S_{xx})$.

Theorem 6.8: t-test for significance of regression

For $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$,

$$t_0 = \frac{\hat{\beta}_1}{\text{se}(\hat{\beta}_1)}, \qquad \text{se}(\hat{\beta}_1) = \sqrt{MS_E/S_{xx}},$$

which is t_{n-2} under H_0. Reject H_0 if $|t_0| > t_{\alpha/2, n-2}$ (equivalently, if the P-value $\leq \alpha$). If σ^2 were known, $Z_0 = \hat{\beta}_1/\sqrt{\sigma^2/S_{xx}} \sim N(0,1)$ would be used instead.

Failing to reject H_0 means there is no evidence of a *linear* relationship between x and y.

Example 6.9: Rocket propellant

$t_0 = -37.15/\sqrt{9244.59/1106.56} = -12.85$ and $t_{0.025,18} = 2.101$, so H_0 is rejected; the P-value is 1.64×10^{-10}.

4.1 Analysis of variance

Theorem 6.10: F-test

With degrees of freedom $(n-1) = 1 + (n-2)$,

$$F_0 = \frac{SS_R/1}{SS_E/(n-2)} = \frac{MS_R}{MS_E} \sim F_{1,n-2} \quad \text{under } H_0,$$

and H_0 is rejected if $F_0 > F_{\alpha,1,n-2}$. Since $\mathrm{E}(MS_R) = \sigma^2 + \beta_1^2 S_{xx}$ and $\mathrm{E}(MS_E) = \sigma^2$, F_0 is near 1 under H_0 and large otherwise. In simple regression $F_0 = t_0^2$.

ANOVA gives the same answer as the t-test here, but it is the natural tool for multiple regression.

Example 6.11: Rocket propellant

$SS_R = (-37.15)(-41112.65) = 1{,}527{,}334.95$, $SS_E = 166{,}402.65$,

$$F_0 = \frac{1527334.95/1}{166402.65/18} = 165.21 > F_{0.05,1,18} = 4.41,$$

so H_0 is rejected (R: $F = 165.4$ on 1 and 18 df, $p = 1.64 \times 10^{-10}$).

5 Interval Estimation

5.1 Coefficients

Under normal errors, $(\hat\beta_j - \beta_j)/\mathrm{se}(\hat\beta_j) \sim t_{n-2}$, so

$$\hat\beta_1 \pm t_{\alpha/2,n-2}\,\mathrm{se}(\hat\beta_1), \qquad \hat\beta_0 \pm t_{\alpha/2,n-2}\,\mathrm{se}(\hat\beta_0),$$

$$\mathrm{se}(\hat\beta_1) = \sqrt{\frac{MS_E}{S_{xx}}}, \qquad \mathrm{se}(\hat\beta_0) = \sqrt{MS_E\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big)}.$$

Example 6.12

For the rocket data with $t_{0.025,18} = 2.101$:

$$2534.99 \le \beta_0 \le 2720.65, \qquad -43.22 \le \beta_1 \le -31.08.$$

5.2 Error variance

Since $(n-2)MS_E/\sigma^2 \sim \chi^2_{n-2}$,

$$\frac{(n-2)MS_E}{\chi^2_{\alpha/2,n-2}} \le \sigma^2 \le \frac{(n-2)MS_E}{\chi^2_{1-\alpha/2,n-2}}.$$

5.3 Mean response

For x_0 within the range of the data, the estimate is $\hat{\mu}_{y|x_0} = \hat{\beta}_0 + \hat{\beta}_1 x_0$, with $\mathrm{E}(\hat{\mu}_{y|x_0}) = \beta_0 + \beta_1 x_0$ and

$$\begin{aligned}\mathrm{Var}(\hat{\mu}_{y|x_0}) &= \mathrm{Var}(\hat{\beta}_0) + x_0^2 \mathrm{Var}(\hat{\beta}_1) + 2x_0 \mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1) \\ &= \sigma^2\Big(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\Big) + x_0^2 \frac{\sigma^2}{S_{xx}} - 2x_0 \bar{x} \frac{\sigma^2}{S_{xx}} = \sigma^2\Big(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\Big).\end{aligned}$$

Hence

$$\hat{\mu}_{y|x_0} \pm t_{\alpha/2, n-2} \sqrt{MS_E\Big(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\Big)}.$$

At $x_0 = \bar{x} = 13.3625$ in the rocket data the 95% interval is $2086.2 \le \mathrm{E}(y \mid x_0) \le 2176.6$. The interval is narrowest at $\bar{x}$ and widens away from it.

5.4 Prediction of a new observation

The point prediction of y_0 at x_0 is $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$. Since the new y_0 is independent of the data,

$$\mathrm{E}(\hat{y}_0 - y_0)^2 = \mathrm{E}[\hat{y}_0 - \mathrm{E}(y_0)]^2 + \mathrm{E}[y_0 - \mathrm{E}(y_0)]^2 = \sigma^2\Big(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\Big),$$

and the $100(1-\alpha)\%$ prediction interval is

$$\hat{y}_0 \pm t_{\alpha/2, n-2} \sqrt{MS_E\Big(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\Big)}.$$

For the rocket data at $x_0 = 10$: $[2048.32,\ 2464.32]$.

6 Coefficient of Determination

Definition 6.13

$R^2 = \dfrac{SS_R}{SS_T} = 1 - \dfrac{SS_E}{SS_T}$, $0 \le R^2 \le 1$: the proportion of variation in y explained by x.

In simple regression,

$$R^2 = \frac{\hat{\beta}_1 S_{xy}}{S_{yy}} = \frac{S_{xy}^2}{S_{xx} S_{yy}} = r_{xy}^2, \qquad F_0 = \frac{SS_R/1}{SS_E/(n-2)} = \frac{R^2(n-2)}{1 - R^2}.$$

For the rocket data $R^2 = 1527334.95/1693737.6 = 0.9018 = (-0.9497)^2$: 90.18% of the variability in strength is explained by age.

7 Selected exercises

Motivation. Several facts above were stated without proof: the unbiasedness of MS_E, the expected regression mean square, and the optimality of least squares. The items below prove them and treat the no-intercept model, which links regression to the ratio estimator of Ch. 5.

Exercise 6.1 (Expected mean squares)

In the simple linear regression model show that
(a) $\mathrm{E}(SS_R) = \sigma^2 + \beta_1^2 S_{xx}$;
(b) $\mathrm{E}(SS_E) = (n-2)\sigma^2$, so MS_E is unbiased for σ^2;
(c) $F_0 = t_0^2$.

Solution. [Supplement]

(a) Since $SS_R = \hat\beta_1 S_{xy}$ and $S_{xy} = \hat\beta_1 S_{xx}$, $SS_R = \hat\beta_1^2 S_{xx}$. Hence, by the theorem on the properties of the estimators,

$$\mathrm{E}(SS_R) = S_{xx}\big[\mathrm{Var}(\hat\beta_1) + (\mathrm{E}\hat\beta_1)^2\big] = S_{xx}\Big(\frac{\sigma^2}{S_{xx}} + \beta_1^2\Big) = \sigma^2 + \beta_1^2 S_{xx}.$$

(b)

Step 1. Averaging the model, $\bar y = \beta_0 + \beta_1 \bar x + \bar\varepsilon$, so $y_i - \bar y = \beta_1(x_i - \bar x) + (\varepsilon_i - \bar\varepsilon)$.

Step 2. Squaring and summing, the cross term $2\beta_1 \sum (x_i - \bar x)(\varepsilon_i - \bar\varepsilon)$ has mean zero, and

$$\mathrm{E}\sum(\varepsilon_i - \bar\varepsilon)^2 = \mathrm{E}\Big[\sum \varepsilon_i^2 - n\bar\varepsilon^2\Big] = n\sigma^2 - n\cdot\frac{\sigma^2}{n} = (n-1)\sigma^2,$$

using that the errors are uncorrelated. Hence $\mathrm{E}(SS_T) = \beta_1^2 S_{xx} + (n-1)\sigma^2$.

Step 3. By the sum-of-squares decomposition and (a), $\mathrm{E}(SS_E) = \mathrm{E}(SS_T) - \mathrm{E}(SS_R) = (n-1)\sigma^2 - \sigma^2 = (n-2)\sigma^2$.

(c) $F_0 = \dfrac{\hat\beta_1^2 S_{xx}}{MS_E} = \Big(\dfrac{\hat\beta_1}{\sqrt{MS_E/S_{xx}}}\Big)^2 = t_0^2$. Under H_0, (a) and (b) give $\mathrm{E}(MS_R) = \mathrm{E}(MS_E) = \sigma^2$, which is why F_0 is near 1. ■

Exercise 6.2 (Least squares is best among linear unbiased estimators)

Show that $\hat\beta_1$ has the smallest variance among all estimators of the form $\sum_i a_i y_i$ that are unbiased for β_1 whatever β_0, β_1 are.

Solution. [Supplement]

Step 1. (Constraints.) $\mathrm{E}\sum a_i y_i = \beta_0 \sum a_i + \beta_1 \sum a_i x_i$ equals β_1 for all (β_0, β_1) iff $\sum a_i = 0$ and $\sum a_i x_i = 1$. Its variance is $\sigma^2 \sum a_i^2$.

Step 2. (Decomposition.) Let $c_i = (x_i - \bar x)/S_{xx}$, so $\hat\beta_1 = \sum c_i y_i$ and the c_i satisfy both constraints. Put $d_i = a_i - c_i$; then $\sum d_i = 0$ and $\sum d_i x_i = 0$.

Step 3. (Orthogonality.) $\sum c_i d_i = \frac{1}{S_{xx}}\big(\sum d_i x_i - \bar x \sum d_i\big) = 0$, so

$$\mathrm{Var}\Big(\sum a_i y_i\Big) = \sigma^2 \sum (c_i + d_i)^2 = \sigma^2 \sum c_i^2 + \sigma^2 \sum d_i^2 \;\geq\; \sigma^2 \sum c_i^2 = \mathrm{Var}(\hat\beta_1),$$

with equality iff all $d_i = 0$, i.e. the estimator is $\hat\beta_1$ itself. This is the Gauss–Markov theorem for the slope; normality is not needed. ■

Exercise 6.3 (Regression through the origin)

Let $y_i = \beta x_i + \varepsilon_i$, $i = 1, \ldots, n$, with fixed x_i not all zero and uncorrelated errors of mean 0 and variance σ^2.
(a) Find the LS estimator $\hat\beta$, its mean and variance.

(b) With $e_i = y_i - \hat{\beta} x_i$, show that $\sum x_i e_i = 0$ but in general $\sum e_i \neq 0$.
(c) Show that $\mathrm{E} \sum e_i^2 = (n-1)\sigma^2$, so $\hat{\sigma}^2 = \sum e_i^2/(n-1)$ is unbiased.

Solution. [Supplement]

(a) $\frac{d}{d\beta} \sum (y_i - \beta x_i)^2 = -2 \sum x_i (y_i - \beta x_i) = 0$ gives $\hat{\beta} = \sum x_i y_i / \sum x_i^2$, the unique minimizer since the second derivative $2 \sum x_i^2$ is positive. Then $\mathrm{E}\hat{\beta} = \beta \sum x_i^2 / \sum x_i^2 = \beta$ and $\mathrm{Var}\hat{\beta} = \sigma^2 \sum x_i^2 / (\sum x_i^2)^2 = \sigma^2 / \sum x_i^2$.

(b) The normal equation of (a) is $\sum x_i e_i = 0$. There is no intercept, hence no equation $\sum e_i = 0$: for the data $(x, y) = (1, 1), (2, 1)$, $\hat{\beta} = 3/5$ and $e_1 + e_2 = 0.4 - 0.2 = 0.2 \neq 0$.

(c)

Step 1. By (b), $\sum e_i^2 = \sum e_i y_i - \hat{\beta} \sum e_i x_i = \sum e_i y_i = \sum y_i^2 - \hat{\beta} \sum x_i y_i = \sum y_i^2 - \hat{\beta}^2 \sum x_i^2$.

Step 2. $\mathrm{E} \sum y_i^2 = \sum (\sigma^2 + \beta^2 x_i^2) = n\sigma^2 + \beta^2 \sum x_i^2$, and by (a) $\mathrm{E}(\hat{\beta}^2) \sum x_i^2 = (\sigma^2 / \sum x_i^2 + \beta^2) \sum x_i^2 = \sigma^2 + \beta^2 \sum x_i^2$.

Step 3. Subtracting, $\mathrm{E} \sum e_i^2 = (n-1)\sigma^2$: only one degree of freedom is lost, for the single coefficient.

With constant error variance the no-intercept LS slope is $\sum x_i y_i / \sum x_i^2$; with variance proportional to x_i it becomes the ratio $\sum y_i / \sum x_i$ (Exercise 5.1). The ratio estimator is thus regression through the origin with a particular error model. ■

Chapter 7. Regression Estimation

1 The Regression Estimator

The ratio estimator works best when y is proportional to x (a line through the origin). If the relation is linear with a nonzero intercept, use the regression line instead. As in Ch. 6 the x's are treated as fixed (already observed) and the y's as random:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i = \beta_0' + \beta_1(x_i - \bar{x}) + \varepsilon_i, \qquad \hat{\beta}_0' = \bar{y}, \quad \hat{\beta}_1 = b = \frac{\sum(y_i - \bar{y})(x_i - \bar{x})}{\sum(x_i - \bar{x})^2},$$

so the fitted mean response is $\widehat{\mathrm{E}}(y \mid x) = \bar{y} + b(x - \bar{x})$. Evaluating it at the known population mean μ_x gives the estimator.

Definition 7.1

The **regression estimator** of μ_y is

$$\hat{\mu}_{yL} = \widehat{\mathrm{E}}(y \mid \mu_x) = \bar{y} + b(\mu_x - \bar{x}) = \frac{1}{n}\sum_{i=1}^{n}\left[y_i - b(x_i - \mu_x)\right],$$

i.e. the sample mean of the adjusted values $y_i - b(x_i - \mu_x)$.

If the sample happens to have $\bar{x} < \mu_x$ and $b > 0$, the sample mean $\bar{y}$ is probably too small as well, and $\hat{\mu}_{yL}$ corrects it upward by $b(\mu_x - \bar{x})$.

Theorem 7.2: Model unbiasedness

Under the model $\mathrm{E}(y_i) = \beta_0 + \beta_1 x_i$ with fixed x's,

$$\mathrm{E}(\hat{\mu}_{yL}) = \beta_0 + \beta_1 \mu_x = \mathrm{E}(y \mid \mu_x).$$

Proof. $\mathrm{E}(\bar{y}) = \beta_0 + \beta_1 \bar{x}$ and $\mathrm{E}(b) = \beta_1$ (Ch. 6), so $\mathrm{E}(\hat{\mu}_{yL}) = \beta_0 + \beta_1 \bar{x} + \beta_1(\mu_x - \bar{x}) = \beta_0 + \beta_1 \mu_x$. ■

2 Variance of the Regression Estimator

2.1 Fixed slope

First take a fixed constant β and $\bar{y}_{\mathrm{reg}} = \bar{y} + \beta(\mu_x - \bar{x})$. This is the SRS mean of $u_i = y_i - \beta(x_i - \mu_x)$, whose population mean is μ_y, so it is exactly unbiased.

Theorem 7.3

$$\mathrm{Var}(\bar{y}_{\mathrm{reg}}) = \frac{1-f}{n}(S_y^2 - 2\beta S_{xy} + \beta^2 S_x^2),$$

which is minimized at $\beta = S_{xy}/S_x^2$ (the population LS slope), where

$$\mathrm{Var}(\bar{y}_{\mathrm{reg}}) = \frac{1-f}{n}\left(S_y^2 - \frac{S_{xy}^2}{S_x^2}\right) = \frac{1-f}{n} S_y^2(1 - \rho^2).$$

Proof. By Ch. 2, $\mathrm{Var}(\bar{y}_{\mathrm{reg}}) = \frac{1-f}{n} S_u^2$ with

$$S_u^2 = \frac{1}{N-1}\sum_{i=1}^{N}\big[(y_i-\mu_y)-\beta(x_i-\mu_x)\big]^2$$
$$= \frac{1}{N-1}\Big[\sum(y_i-\mu_y)^2 - 2\beta\sum(x_i-\mu_x)(y_i-\mu_y) + \beta^2\sum(x_i-\mu_x)^2\Big] = S_y^2 - 2\beta S_{xy} + \beta^2 S_x^2.$$

This quadratic in β is minimized at $\beta = S_{xy}/S_x^2$, giving $S_y^2 - S_{xy}^2/S_x^2 = S_y^2(1-\rho^2)$. ∎

2.2 Estimated slope

With b estimated, write $\xi_i = (y_i-\mu_y) - \beta(x_i-\mu_x)$ for $\beta = S_{xy}/S_x^2$, so $y_i = \mu_y + \beta(x_i-\mu_x) + \xi_i$. Then

$$b = \frac{\sum(x_i-\bar{x})y_i}{\sum(x_i-\bar{x})^2} = \beta\frac{\sum(x_i-\bar{x})(x_i-\mu_x)}{\sum(x_i-\bar{x})^2} + \frac{\sum(x_i-\bar{x})\xi_i}{\sum(x_i-\bar{x})^2} = \beta + \frac{\sum(x_i-\bar{x})\xi_i}{\sum(x_i-\bar{x})^2},$$

and the last term is of smaller order since the ξ_i are uncorrelated with x in the population. Hence $b \approx \beta$ in large samples and

$$\mathrm{Var}(\hat{\mu}_{yL}) \approx \mathrm{Var}(\bar{y}_{\mathrm{reg}}) = \frac{1-f}{n}S^2, \qquad S^2 = \frac{1}{N-1}\sum_{i=1}^{N}\big[(y_i-\mu_y)-\beta(x_i-\mu_x)\big]^2.$$

Theorem 7.4: Estimated variance

$$\widehat{\mathrm{Var}}(\hat{\mu}_{yL}) = \frac{N-n}{Nn}\,MS_E, \qquad MS_E = \frac{1}{n-2}\sum_{i=1}^{n}\big[(y_i-\bar{y})-b(x_i-\bar{x})\big]^2,$$

the mean square error from the simple linear regression of y on x.

The sample analogue of S^2 is $\frac{1}{n-1}\sum e_i^2$ with residuals $e_i = (y_i-\bar{y}) - b(x_i-\bar{x})$; dividing by $n-2$ instead (two estimated coefficients) gives MS_E, and the difference is negligible for moderate n.

Example 7.5: Calculus marks

A mathematics achievement test was taken by all $N = 486$ entering students, with mean $\mu_x = 52$. For $n = 10$ sampled students:

Student	1	2	3	4	5	6	7	8	9	10
Test (x)	39	43	21	64	57	47	28	75	34	52
Final (y)	65	78	52	82	92	89	73	98	56	75

Here $\bar{x} = 46$, $\bar{y} = 76$, and the R fit gives $\hat{y} = 40.78 + 0.7656x$ with residual standard error 8.704 on 8 df ($MS_E = 75.76$, $R^2 = 0.705$). Then

$$\hat{\mu}_{yL} = 76 + 0.7656(52-46) = 80.6,$$

$$\widehat{\mathrm{Var}}(\hat{\mu}_{yL}) = \frac{486-10}{486\times 10}(75.76) = 7.42, \qquad B = 2\sqrt{7.42} = 5.45.$$

The fitted intercept (40.8) is far from zero, so a ratio estimator would be inappropriate here: $r\mu_x = (760/460)(52) = 85.9$ overshoots.

3 Relative Efficiency: Mean, Ratio and Regression

For two estimators E_1, E_2 of μ_y, $\mathrm{RE}(E_1/E_2) = \mathrm{Var}(E_2)/\mathrm{Var}(E_1)$; if it exceeds 1, E_1 is preferred. The estimated version (with estimated variances) is meaningful for large samples. The three estimated variances share the factor $\frac{1-f}{n}$:

$$\widehat{\mathrm{Var}}(\bar{y}) = \frac{1-f}{n} s_y^2, \qquad \widehat{\mathrm{Var}}(\hat{\mu}_y) = \frac{1-f}{n}(s_y^2 + r^2 s_x^2 - 2r\hat{\rho} s_x s_y), \qquad \widehat{\mathrm{Var}}(\hat{\mu}_{yL}) = \frac{1-f}{n} s_y^2(1-\hat{\rho}^2).$$

Derivation: the regression variance in correlation form

$$\begin{aligned}\frac{1}{n-1}\sum \left[(y_i - \bar{y}) - b(x_i - \bar{x})\right]^2 &= \frac{1}{n-1}\left[S_{yy} - 2b\, S_{xy} + b^2 S_{xx}\right] \\ &= s_y^2 - 2b\, s_{xy} + b^2 s_x^2 = s_y^2 - b^2 s_x^2,\end{aligned}$$

since $s_{xy} = b s_x^2$. Moreover $\hat{\rho} = \frac{s_{xy}}{s_x s_y} = \frac{b s_x^2}{s_x s_y}$ gives $\hat{\rho} s_y = b s_x$, so $s_y^2 - b^2 s_x^2 = s_y^2(1-\hat{\rho}^2)$.

3.1 Mean versus ratio

From Ch. 5,

$$\widehat{\mathrm{RE}}(\hat{\mu}_y/\bar{y}) = \frac{s_y^2}{s_y^2 + r^2 s_x^2 - 2r\hat{\rho} s_x s_y},$$

and the ratio estimator is more efficient iff $\hat{\rho} > \frac{1}{2} \cdot \frac{s_x/\bar{x}}{s_y/\bar{y}}$. Recall that r is generally biased for R; the bias is negligible when the relation is a line through the origin.

3.2 Mean versus regression

$$\widehat{\mathrm{RE}}(\hat{\mu}_{yL}/\bar{y}) = \frac{s_y^2}{s_y^2(1-\hat{\rho}^2)} = \frac{1}{1-\hat{\rho}^2} > 1 \quad \text{unless } \hat{\rho} = 0.$$

The regression estimator is (asymptotically) never worse than $\bar{y}$.

3.3 Ratio versus regression

Theorem 7.6

$$\widehat{\mathrm{RE}}(\hat{\mu}_{yL}/\hat{\mu}_y) = \frac{s_y^2 + r^2 s_x^2 - 2r\hat{\rho} s_x s_y}{s_y^2(1-\hat{\rho}^2)} \geq 1,$$

with equality iff $r = b$.

Proof. The numerator minus the denominator is

$$r^2 s_x^2 - 2r\hat{\rho} s_x s_y + \hat{\rho}^2 s_y^2 = (r s_x - \hat{\rho} s_y)^2 = (r s_x - b s_x)^2 = s_x^2(r-b)^2 \geq 0.$$

■

Both r and b are linear in the y's: $r = \sum y_i / \sum x_i = \sum d_i y_i$ with equal weights $d_i = 1/\sum x_j$, while $b = \sum w_i y_i$ with $w_i = (x_i - \bar{x})/\sum (x_j - \bar{x})^2$. They coincide only when the LS line passes through the origin, which is exactly when the ratio estimator is as good as the regression estimator.

4 Selected exercises

Motivation. The first two items carry regression estimation into stratified sampling, in parallel with the separate and combined ratio estimators of Ch. 5; the last two give the exact design bias of $\hat{\mu}_{yL}$ and its interpretation as a reweighting of the sample.

Exercise 7.1 (Separate regression estimator)

In stratified random sampling with strata $h = 1, \dots, L$, weights w_h and known stratum means μ_{xh}, the separate regression estimator is

$$\hat{\mu}_{yLS} = \sum_{h=1}^{L} w_h \left[\bar{y}_h + b_h(\mu_{xh} - \bar{x}_h)\right],$$

with b_h the LS slope in stratum h. Find its approximate variance and an estimator of it.

Solution.

Step 1. (Independence.) The term in brackets, $\hat{\mu}_{yL,h}$, is the regression estimator computed from the SRS in stratum h alone, and the strata are sampled independently, so

$$\mathrm{Var}(\hat{\mu}_{yLS}) = \sum_h w_h^2 \mathrm{Var}(\hat{\mu}_{yL,h}).$$

Step 2. (Each stratum.) By the estimated-slope argument of this chapter applied in stratum h, $b_h \approx \beta_h = S_{xyh}/S_{xh}^2$ for large n_h, and the fixed-slope theorem gives

$$\mathrm{Var}(\hat{\mu}_{yL,h}) \approx \frac{1 - f_h}{n_h} S_{yh}^2 (1 - \rho_h^2), \qquad \rho_h = \frac{S_{xyh}}{S_{xh} S_{yh}}.$$

Step 3. Hence

$$\mathrm{Var}(\hat{\mu}_{yLS}) \approx \sum_{h=1}^{L} w_h^2 \frac{1 - f_h}{n_h} S_{yh}^2 (1 - \rho_h^2), \qquad \widehat{\mathrm{Var}}(\hat{\mu}_{yLS}) = \sum_{h=1}^{L} w_h^2 \frac{1 - f_h}{n_h} MS_{E,h},$$

where $MS_{E,h}$ is the residual mean square of the regression of y on x in stratum h.

Step 4. As for the separate ratio estimator, the approximation of Step 2 needs every n_h to be large; with small strata the slope errors (and the resulting biases) accumulate over the L strata. ■

Exercise 7.2 (Combined regression estimator with a given slope)

With a slope β fixed in advance (the same for all strata), let $\hat{\mu}_{yLC} = \bar{y}_{\mathrm{st}} + \beta(\mu_x - \bar{x}_{\mathrm{st}})$.
(a) Show that $\hat{\mu}_{yLC}$ is unbiased and find $\mathrm{Var}(\hat{\mu}_{yLC})$.
(b) Find the β minimizing the variance and compare $\hat{\mu}_{yLC}$ with the combined ratio estimator $\hat{\mu}_{yRC}$.

Solution. Write $a_h = w_h^2(1 - f_h)/n_h$ and S_{xyh} for the covariance of x and y in stratum h.

(a)

Step 1. $\hat{\mu}_{yLC} = \sum_h w_h \bar{u}_h + \beta\mu_x$ is the stratified mean of $u_{hi} = y_{hi} - \beta x_{hi}$ plus a constant. Since $\mathrm{E}(\bar{y}_{\mathrm{st}}) = \mu_y$ and $\mathrm{E}(\bar{x}_{\mathrm{st}}) = \mu_x$, $\mathrm{E}(\hat{\mu}_{yLC}) = \mu_y$ exactly.

Step 2. By the stratified variance theorem applied to u, whose stratum variance is $S_{yh}^2 - 2\beta S_{xyh} + \beta^2 S_{xh}^2$,

$$\mathrm{Var}(\hat{\mu}_{yLC}) = Q(\beta) := \sum_h a_h (S_{yh}^2 - 2\beta S_{xyh} + \beta^2 S_{xh}^2) = A - 2\beta B + \beta^2 C,$$

with $A = \sum_h a_h S_{yh}^2$, $B = \sum_h a_h S_{xyh}$, $C = \sum_h a_h S_{xh}^2 > 0$.

(b)

Step 1. Q is minimized at $\beta_{\text{opt}} = B/C = \sum_h a_h S_{xyh} / \sum_h a_h S_{xh}^2$, a weighted average of the within-stratum covariances over variances, and completing the square gives

$$Q(\beta) = Q(\beta_{\text{opt}}) + C(\beta - \beta_{\text{opt}})^2.$$

Step 2. By the classbox on separate versus combined ratio estimators in Ch. 5, $\text{Var}(\hat{\mu}_{yRC}) \approx \sum_h a_h S_{h,C}^2$ with $S_{h,C}^2 = S_{yh}^2 - 2R_C \rho_h S_{xh} S_{yh} + R_C^2 S_{xh}^2$, i.e. $\text{Var}(\hat{\mu}_{yRC}) \approx Q(R_C)$: to first order the combined ratio estimator is the combined regression estimator with slope $R_C = \mu_y/\mu_x$.

Step 3. Therefore

$$\text{Var}(\hat{\mu}_{yRC}) - \text{Var}(\hat{\mu}_{yLC})\big|_{\beta=\beta_{\text{opt}}} \approx C\,(R_C - \beta_{\text{opt}})^2 \;\geq 0,$$

with equality iff $R_C = \beta_{\text{opt}}$, i.e. when the pooled within-stratum regression line passes through the origin. In practice β_{opt} is estimated by the pooled sample slope, which changes the variance only by a term of smaller order. ■

Exercise 7.3 (Exact bias of the regression estimator)

Under SRS, assume the sample x's are never all equal, so b is defined. Show that

$$\text{E}(\hat{\mu}_{yL}) - \mu_y = -\text{Cov}(b, \bar{x}), \qquad |\text{E}(\hat{\mu}_{yL}) - \mu_y| \leq \sqrt{\text{Var}(b)}\,\sqrt{\frac{1-f}{n}}\,S_x,$$

and explain why the bias is negligible relative to the standard error for large n.

Solution. [Supplement]

Step 1. $\text{E}(\bar{y}) = \mu_y$ and $\text{E}(\bar{x} - \mu_x) = 0$ under SRS, so

$$\text{E}(\hat{\mu}_{yL}) - \mu_y = -\text{E}[b(\bar{x} - \mu_x)] = -\{\text{E}[b(\bar{x} - \mu_x)] - \text{E}(b)\,\text{E}(\bar{x} - \mu_x)\} = -\text{Cov}(b, \bar{x}).$$

Step 2. By the Cauchy–Schwarz inequality $|\text{Cov}(b, \bar{x})| \leq \sqrt{\text{Var}(b)\text{Var}(\bar{x})}$, and $\text{Var}(\bar{x}) = \frac{1-f}{n} S_x^2$.

Step 3. Both $\sqrt{\text{Var}(b)}$ and $\sqrt{\text{Var}(\bar{x})}$ are of order $n^{-1/2}$, so the bias is of order $1/n$, while the standard error of $\hat{\mu}_{yL}$ is of order $n^{-1/2}$. This is the exact analogue of the bias bound $|\text{E}(r) - R| \leq CV(\bar{x})\sqrt{\text{Var}(r)}$ for the ratio estimator. The bias vanishes when b is uncorrelated with $\bar{x}$, e.g. under the linear model with fixed x's (the model-unbiasedness theorem). ■

Exercise 7.4 (The regression estimator as a calibrated weighting)

Let $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 > 0$ and $g_i = 1 + \dfrac{n(\mu_x - \bar{x})(x_i - \bar{x})}{S_{xx}}$.

(a) Show that $\hat{\mu}_{yL} = \frac{1}{n}\sum_i g_i y_i$.

(b) Show that $\frac{1}{n}\sum_i g_i = 1$ and $\frac{1}{n}\sum_i g_i x_i = \mu_x$: the weights reproduce the known mean of x exactly.

(c) Show that, among all weights (g_i') satisfying the two equations in (b), the g_i minimize $\sum_i (g_i' - 1)^2$.

Solution. [Supplement]

(a) $b = \sum_i (x_i - \bar{x}) y_i / S_{xx}$, so $\bar{y} + b(\mu_x - \bar{x}) = \frac{1}{n}\sum_i y_i [1 + n(\mu_x - \bar{x})(x_i - \bar{x})/S_{xx}] = \frac{1}{n}\sum_i g_i y_i$.

(b) Since $\sum_i (x_i - \bar{x}) = 0$, $\sum_i g_i = n$. Since $\sum_i (x_i - \bar{x}) x_i = S_{xx}$, $\sum_i g_i x_i = n\bar{x} + n(\mu_x - \bar{x}) = n\mu_x$.

(c)

Step 1. Put $h_i = g_i - 1 = \lambda(x_i - \bar{x})$ with $\lambda = n(\mu_x - \bar{x})/S_{xx}$, and let g_i' be any feasible weights, $d_i = g_i' - g_i$. Subtracting the constraints for g' and g, $\sum_i d_i = 0$ and $\sum_i d_i x_i = 0$.

Step 2. Then $\sum_i h_i d_i = \lambda\big(\sum_i d_i x_i - \bar{x}\sum_i d_i\big) = 0$, so

$$\sum_i (g_i' - 1)^2 = \sum_i (h_i + d_i)^2 = \sum_i h_i^2 + \sum_i d_i^2 \geq \sum_i h_i^2,$$

with equality iff $d = 0$.

Thus $\hat{\mu}_{yL}$ is the weighted mean whose weights stay as close as possible to the SRS weights 1 while being calibrated to the known μ_x. If $\bar{x} < \mu_x$, units with large x receive weights above 1, which is the upward correction described at the start of the chapter. ■

Chapter 8. Cluster Sampling

1 Introduction

Suppose we want the weekly food expenditure of families in Seoul. SRS, stratified or systematic sampling of families requires a list of all families (rarely available), and the sampled families would be scattered all over the city, which makes interviewing expensive. A list of city blocks, however, is easy to obtain, and interviewing every family in a few blocks is cheap.

Definition 8.1

A **cluster sample** is a simple random sample in which each sampling unit is a collection (cluster) of elements; every element of a selected cluster is observed.

Cluster sampling is effective when

- a good frame of elements is unavailable or costly, while a frame of clusters is easy to obtain;
- the cost of observation increases with the distance between elements.

Since it is SRS of clusters, the estimators parallel those of Ch. 2, with cluster totals playing the role of observations.

Notation.

N	number of clusters in the population
n	number of clusters selected by SRS
m_i	number of elements in cluster i, $i = 1, \dots, N$
$\bar{m} = \frac{1}{n}\sum_{i=1}^{n} m_i$	average cluster size in the sample
$M = \sum_{i=1}^{N} m_i$	number of elements in the population
$\bar{M} = M/N$	average cluster size in the population
y_{ij}	value of the jth element in cluster i
$y_i = \sum_{j=1}^{m_i} y_{ij}$	total of cluster i

2 Estimation of the Population Mean

The population mean per element is

$$\mu = \frac{\sum_{i=1}^{N}\sum_{j=1}^{m_i} y_{ij}}{\sum_{i=1}^{N} m_i} = \frac{\sum_{i=1}^{N} y_i}{M},$$

a ratio of the cluster totals y_i to the cluster sizes m_i. Its natural estimator is therefore a ratio estimator (Ch. 5) with $x_i = m_i$.

Theorem 8.2: Cluster estimator of μ

$$\bar{y}_{\text{cl}} = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} m_i}, \qquad \widehat{\text{Var}}(\bar{y}_{\text{cl}}) = \frac{N-n}{Nn\bar{M}^2}\, s_c^2, \qquad s_c^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y}_{\text{cl}} m_i)^2,$$

where $\bar{M}$ is replaced by $\bar{m}$ if M is unknown. An approximate 95% interval is $\bar{y}_{\text{cl}} \pm 2\sqrt{\widehat{\text{Var}}(\bar{y}_{\text{cl}})}$.

Derivation: from the ratio estimator

With $x_i = m_i$, $R = \mu$ and $\mu_x = \bar{M}$, Ch. 5 gives

$$\mathrm{Var}(\bar{y}_{\mathrm{cl}}) \approx \frac{1}{\bar{M}^2} \cdot \frac{N-n}{N} \cdot \frac{1}{n} \cdot \frac{1}{N-1} \sum_{i=1}^{N} (y_i - \mu m_i)^2 = \frac{1}{\bar{M}^2} \frac{N-n}{Nn} (S_y^2 + \mu^2 S_m^2 - 2\mu\rho S_m S_y),$$

and replacing population quantities by sample ones yields $\widehat{\mathrm{Var}}(\bar{y}_{\mathrm{cl}})$ and the alternative form

$$\widehat{\mathrm{Var}}(\bar{y}_{\mathrm{cl}}) = \frac{N-n}{Nn\bar{M}^2} (s_y^2 + \bar{y}_{\mathrm{cl}}^2 s_m^2 - 2\bar{y}_{\mathrm{cl}}\hat{\rho} s_m s_y), \qquad s_m^2 = \frac{1}{n-1} \sum_{i=1}^{n} (m_i - \bar{m})^2,$$

where s_y^2 is the sample variance of the cluster totals. Cluster totals nearly proportional to cluster sizes ($y_i \approx \mu m_i$) give a small variance.

3 Estimation of the Population Total

3.1 M known

Since $\tau = M\mu$,

$$\hat{\tau} = M\bar{y}_{\mathrm{cl}} = M \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} m_i}, \qquad \widehat{\mathrm{Var}}(\hat{\tau}) = M^2 \widehat{\mathrm{Var}}(\bar{y}_{\mathrm{cl}}) = N^2 \frac{N-n}{Nn} s_c^2.$$

3.2 M unknown

Often the number of elements is unknown. Then use the mean of the cluster totals,

$$\bar{y}_t = \frac{1}{n} \sum_{i=1}^{n} y_i, \qquad \mathrm{E}(\bar{y}_t) = \frac{1}{N} \sum_{i=1}^{N} y_i = \frac{\tau}{N},$$

which is unbiased for the average cluster total, so $N\bar{y}_t$ is unbiased for τ:

$$\hat{\tau} = N\bar{y}_t = \frac{N}{n} \sum_{i=1}^{n} y_i, \qquad \widehat{\mathrm{Var}}(\hat{\tau}) = N^2 \frac{N-n}{Nn} s_t^2, \qquad s_t^2 = \frac{1}{n-1} \sum_{i=1}^{n} (y_i - \bar{y}_t)^2.$$

This is SRS of the cluster totals and ignores the sizes m_i; it is less precise than $M\bar{y}_{\mathrm{cl}}$ when the totals vary mainly because the sizes vary.

If all $m_i = m$, then $M = Nm$, $\bar{y}_{\mathrm{cl}} = \bar{y}_t/m$, and the two estimators of τ coincide.

4 Comparison with Simple Random Sampling

Assume equal cluster sizes $m_i = m$, so $M = Nm$ and

$$\bar{y}_{\mathrm{cl}} = \frac{\sum_{i=1}^{n} y_i}{nm} = \frac{1}{nm} \sum_{i=1}^{n} \sum_{j=1}^{m} y_{ij} = \frac{\bar{y}_t}{m}.$$

Let $\bar{y}_i = y_i/m$ be the mean of cluster i. Then

$$\widehat{\mathrm{Var}}(\bar{y}_{\mathrm{cl}}) = \frac{1}{m^2} \widehat{\mathrm{Var}}(\bar{y}_t) = \frac{1}{m^2} \frac{N-n}{Nn} \frac{1}{n-1} \sum_{i} (m\bar{y}_i - m\bar{y}_{\mathrm{cl}})^2 = \frac{N-n}{Nn} \frac{1}{n-1} \sum_{i=1}^{n} (\bar{y}_i - \bar{y}_{\mathrm{cl}})^2.$$

Treat the nm observed elements as an $m \times n$ table with column means $\bar{y}_i$ and grand mean $\bar{y}_{\text{cl}}$:

$$\underbrace{\sum_{i=1}^{n}\sum_{j=1}^{m}(y_{ij}-\bar{y}_{\text{cl}})^2}_{SST} = \underbrace{\sum_{i}\sum_{j}(y_{ij}-\bar{y}_i)^2}_{SSW} + \underbrace{m\sum_{i}(\bar{y}_i-\bar{y}_{\text{cl}})^2}_{SSB},$$

with $MSB = SSB/(n-1)$ and $MSW = SSW/[n(m-1)]$. Then

$$\widehat{\text{Var}}(\bar{y}_{\text{cl}}) = \frac{N-n}{Nnm}\,MSB.$$

If the same nm elements had been drawn by SRS from the Nm elements, with sample variance s^2,

$$\widehat{\text{Var}}(\bar{y}) = \frac{Nm-nm}{Nm}\cdot\frac{s^2}{nm} = \frac{N-n}{Nnm}\,s^2.$$

Remark 8.3: Efficiency of cluster sampling

The relative efficiency of cluster sampling versus SRS is measured by comparing MSB with s^2. Cluster sampling is more efficient when there is little variation between clusters, i.e. when each cluster is a miniature of the whole population (heterogeneous within). In most real populations, neighboring elements are alike, $MSB > s^2$, and cluster sampling is less efficient per element than SRS; it is chosen for its lower cost per element.

5 Selecting the Sample Size

Setting $B = 2\sqrt{\frac{N-n}{NnM^2}s_c^2}$ and solving as in Ch. 2:

- for μ with $\bar{y}_{\text{cl}}$: $n = \dfrac{Ns_c^2}{ND+s_c^2}$, $D = \dfrac{B^2\bar{M}^2}{4}$;
- for τ with $M\bar{y}_{\text{cl}}$: $n = \dfrac{Ns_c^2}{ND+s_c^2}$, $D = \dfrac{B^2}{4N^2}$;
- for τ with $N\bar{y}_t$: $n = \dfrac{Ns_t^2}{ND+s_t^2}$, $D = \dfrac{B^2}{4N^2}$,

with s_c^2, s_t^2 taken from a previous survey or a pilot sample.

6 Estimation of a Population Proportion

Let a_i be the number of elements in cluster i having the characteristic. Then p is the ratio $\sum a_i / \sum m_i$ over the population, and

$$\hat{p}_{\text{cl}} = \frac{\sum_{i=1}^{n} a_i}{\sum_{i=1}^{n} m_i}, \qquad \widehat{\text{Var}}(\hat{p}_{\text{cl}}) = \frac{N-n}{Nn\bar{M}^2}s_p^2, \qquad s_p^2 = \frac{1}{n-1}\sum_{i=1}^{n}(a_i-\hat{p}_{\text{cl}}m_i)^2,$$

with $\bar{M} \approx \bar{m}$ if M is unknown. The sample size for a bound B is $n = Ns_p^2/(ND+s_p^2)$ with $D = B^2\bar{M}^2/4$.

7 Cluster Sampling Combined with Stratification

To survey the number of books read by 4th and 5th graders, divide the population into two strata (grades) and take a cluster sample (of classes) in each. In stratum ℓ let N_ℓ be the number of clusters, $\bar{y}_{t\ell}$ the mean cluster total and $\bar{m}_\ell$ the mean cluster size in the sample. Then

$$\text{average cluster total: } \frac{1}{N}(N_1\bar{y}_{t1} + N_2\bar{y}_{t2}), \qquad \text{average cluster size: } \frac{1}{N}(N_1\bar{m}_1 + N_2\bar{m}_2),$$

$$\bar{y}_c = \frac{N_1\bar{y}_{t1} + N_2\bar{y}_{t2}}{N_1\bar{m}_1 + N_2\bar{m}_2} = \frac{\sum_\ell N_\ell \bar{y}_{t\ell}}{\sum_\ell N_\ell \bar{m}_\ell},$$

a combined ratio estimator. Its estimated variance is

$$\widehat{\mathrm{Var}}(\bar{y}_c) = \frac{1}{\left(\sum_\ell N_\ell \bar{m}_\ell\right)^2} \sum_{\ell=1}^{L} N_\ell(N_\ell - n_\ell)\frac{s_{t\ell}^2}{n_\ell},$$

where $s_{t\ell}^2 = \frac{1}{n_\ell - 1}\sum_{i=1}^{n_\ell}(y_i - \bar{y}_{t\ell})^2$ is the sample variance of the cluster totals in stratum ℓ. The numerator is $\sum_\ell N_\ell^2 \widehat{\mathrm{Var}}(\bar{y}_{t\ell})$, the estimated variance of the stratified estimate of the population total, and the denominator is the squared estimated number of elements.

8 Cluster Sampling with Probabilities Proportional to Size

8.1 The PPS estimator

Let the population consist of units with values $y_1, \ldots, y_N$ and suppose sampling is with replacement, with z_i the probability that unit i is selected on any one draw ($\sum z_i = 1$).

Theorem 8.4: Hansen–Hurwitz estimator

$$\hat{\tau}_{\text{pps}} = \frac{1}{n}\sum_{i=1}^{n}\frac{y_i}{z_i}$$

is unbiased for τ for any choice of $z_i > 0$, with $\widehat{\mathrm{Var}}(\hat{\tau}_{\text{pps}}) = \frac{1}{n(n-1)}\sum_{i=1}^{n}\left(\frac{y_i}{z_i} - \hat{\tau}_{\text{pps}}\right)^2$.

Proof. Each draw gives an iid value $t = y_I/z_I$ with $P(I = k) = z_k$, so $\mathrm{E}(t) = \sum_k z_k \frac{y_k}{z_k} = \tau$. The average of n iid unbiased values is unbiased, and its variance is estimated by s_t^2/n. ■

With $z_i = 1/N$ (equal probability), $\hat{\tau}_{\text{pps}} = \frac{N}{n}\sum y_i = N\bar{y}$, the usual estimator. The variance is small when $z_i \propto y_i$, so we want z_i roughly proportional to y_i.

8.2 Application to clusters

Cluster totals tend to be proportional to cluster sizes, so choose $z_i = m_i/M$. With y_i the cluster total and $\bar{y}_i = y_i/m_i$ the cluster mean,

$$\hat{\tau}_{\text{pps}} = \frac{1}{n}\sum_{i=1}^{n}\frac{y_i}{m_i/M} = \frac{M}{n}\sum_{i=1}^{n}\bar{y}_i, \qquad \hat{\mu}_{\text{pps}} = \frac{\hat{\tau}_{\text{pps}}}{M} = \frac{1}{n}\sum_{i=1}^{n}\bar{y}_i :$$

the estimator of μ is the plain average of the sampled cluster means.

Theorem 8.5

$$\widehat{\mathrm{Var}}(\hat{\tau}_{\mathrm{pps}}) = \frac{M^2}{n(n-1)}\sum_{i=1}^{n}(\bar{y}_i - \hat{\mu}_{\mathrm{pps}})^2, \qquad \widehat{\mathrm{Var}}(\hat{\mu}_{\mathrm{pps}}) = \frac{1}{n(n-1)}\sum_{i=1}^{n}(\bar{y}_i - \hat{\mu}_{\mathrm{pps}})^2.$$

Proof. The $\bar{y}_i$ are iid draws (with replacement), so $\widehat{\mathrm{Var}}(\frac{1}{n}\sum \bar{y}_i) = \frac{1}{n}\cdot\frac{1}{n-1}\sum(\bar{y}_i - \hat{\mu}_{\mathrm{pps}})^2$; multiply by M^2 for the total. ■

Remark 8.6: Equal sizes

If $m_i = m$, then $\hat{\mu}_{\mathrm{pps}} = \frac{1}{n}\sum_i \frac{1}{m}\sum_j y_{ij} = \bar{y}_{\mathrm{cl}} = \bar{y}_t/m$, and since $s_t^2 = \frac{m^2}{n-1}\sum(\bar{y}_i - \bar{y}_{\mathrm{cl}})^2$,

$$\widehat{\mathrm{Var}}(\bar{y}_{\mathrm{cl}}) = \frac{1}{m^2}\frac{N-n}{Nn}s_t^2 = \frac{N-n}{N}\cdot\frac{1}{n(n-1)}\sum(\bar{y}_i - \bar{y}_{\mathrm{cl}})^2,$$

which is $\widehat{\mathrm{Var}}(\hat{\mu}_{\mathrm{pps}})$ up to the fpc (PPS draws are with replacement).

9 Selected exercises

Motivation. The comparison with SRS above was made through estimated variances. The first item gives the exact population version, the design effect $1+(m-1)\rho$; the second derives the exact variance of the PPS estimator and shows why selecting clusters proportional to size removes the variation of cluster sizes from the error.

Exercise 8.1 (Design effect of cluster sampling)

Let all N clusters have m elements ($M = Nm$), let μ and $S^2 = \frac{1}{Nm-1}\sum_{i=1}^{N}\sum_{j=1}^{m}(y_{ij}-\mu)^2$ be the element mean and variance, and define the intracluster correlation

$$\rho = \frac{\sum_{i=1}^{N}\sum_{j\neq j'}(y_{ij}-\mu)(y_{ij'}-\mu)}{(m-1)(Nm-1)S^2}.$$

For an SRS of n clusters, $f = n/N$, show that

$$\mathrm{Var}(\bar{y}_{\mathrm{cl}}) = \frac{1-f}{nm}S^2\cdot\frac{Nm-1}{m(N-1)}[1+(m-1)\rho],$$

and that the ratio of $\mathrm{Var}(\bar{y}_{\mathrm{cl}})$ to the variance of the mean of an SRS of nm elements is approximately $1+(m-1)\rho$ when N is large.

Solution. [Supplement]

Step 1. With equal sizes, $\bar{y}_{\mathrm{cl}} = \frac{1}{n}\sum_{i\le n}\bar{y}_i$ is the SRS mean of the N cluster means $\bar{y}_i = \frac{1}{m}\sum_j y_{ij}$, whose population mean is μ. By Ch. 2,

$$\mathrm{Var}(\bar{y}_{\mathrm{cl}}) = \frac{1-f}{n}S_b^2, \qquad S_b^2 = \frac{1}{N-1}\sum_{i=1}^{N}(\bar{y}_i-\mu)^2.$$

Step 2. Expanding a square of a sum,

$$\sum_{i=1}^{N}(\bar{y}_i-\mu)^2 = \frac{1}{m^2}\sum_i\Big(\sum_j(y_{ij}-\mu)\Big)^2 = \frac{1}{m^2}\Big[\sum_i\sum_j(y_{ij}-\mu)^2 + \sum_i\sum_{j\neq j'}(y_{ij}-\mu)(y_{ij'}-\mu)\Big]$$

$$= \frac{1}{m^2}\Big[(Nm-1)S^2 + (m-1)(Nm-1)S^2\rho\Big] = \frac{(Nm-1)S^2}{m^2}\big[1+(m-1)\rho\big].$$

Step 3. Substituting into Step 1 gives the stated formula.

Step 4. (Design effect.) An SRS of nm of the Nm elements has sampling fraction $nm/(Nm) = f$ and variance $\frac{1-f}{nm}S^2$. The ratio is $\frac{Nm-1}{m(N-1)}[1+(m-1)\rho]$, and $\frac{Nm-1}{m(N-1)} \to 1$ as $N \to \infty$.

When elements of a cluster are alike ($\rho > 0$), a cluster sample of nm elements is worth only about $nm/[1+(m-1)\rho]$ independently drawn elements; even $\rho = 0.05$ doubles the variance for clusters of size $m = 21$. Cluster sampling is efficient per element only if $\rho \le 0$, matching the remark on efficiency above. ■

Exercise 8.2 (Exact variance of the PPS estimator)

In the setting of the Hansen–Hurwitz theorem (draws with replacement, selection probabilities $z_k > 0$, $\sum z_k = 1$), show that

(a) $\mathrm{Var}(\hat\tau_{\mathrm{pps}}) = \dfrac{1}{n}\sum_{k=1}^{N} z_k\Big(\dfrac{y_k}{z_k} - \tau\Big)^2$;

(b) if all $y_k > 0$, the variance is zero iff $z_k = y_k/\tau$ for all k;

(c) $\widehat{\mathrm{Var}}(\hat\tau_{\mathrm{pps}})$ of the theorem is unbiased for $\mathrm{Var}(\hat\tau_{\mathrm{pps}})$;

(d) for clusters with $z_i = m_i/M$, $\mathrm{Var}(\hat\tau_{\mathrm{pps}}) = \dfrac{M}{n}\sum_{i=1}^{N} m_i(\bar y_i - \mu)^2$, where $\bar y_i$ is the mean of cluster i.

Solution. [Supplement] Let $t_r = y_{I_r}/z_{I_r}$, $r = 1,\dots,n$, where $I_1,\dots,I_n$ are the independent draws, $P(I_r = k) = z_k$. Then $\hat\tau_{\mathrm{pps}} = \bar t$.

(a) The t_r are iid with mean τ (proof of the theorem) and variance $V_1 = \mathrm{E}(t_1-\tau)^2 = \sum_k z_k(y_k/z_k - \tau)^2$. Hence $\mathrm{Var}(\bar t) = V_1/n$.

(b) V_1 is a sum of nonnegative terms with positive weights z_k, so $V_1 = 0$ iff $y_k/z_k = \tau$ for every k, i.e. $z_k = y_k/\tau$; these z_k are positive and sum to 1. Such a design is impossible in practice (it needs the y_k), but it shows that the variance is small when z_k is nearly proportional to y_k.

(c) For iid t_r the sample variance $s_t^2 = \frac{1}{n-1}\sum_r (t_r - \bar t)^2$ is unbiased for V_1 (the with-replacement case of Ch. 2). Hence $\mathrm{E}[s_t^2/n] = V_1/n = \mathrm{Var}(\hat\tau_{\mathrm{pps}})$, and s_t^2/n is the $\widehat{\mathrm{Var}}(\hat\tau_{\mathrm{pps}})$ of the theorem.

(d) With y_i the cluster total, $y_i/z_i = My_i/m_i = M\bar y_i$ and $\tau = M\mu$. By (a),

$$\mathrm{Var}(\hat\tau_{\mathrm{pps}}) = \frac{1}{n}\sum_i \frac{m_i}{M}\,M^2(\bar y_i - \mu)^2 = \frac{M}{n}\sum_i m_i(\bar y_i - \mu)^2.$$

Only the variation of the cluster *means* enters. By contrast, $N\bar y_t$ (equal probabilities) has variance driven by the cluster *totals* $y_i = m_i\bar y_i$, which vary also because the sizes m_i vary. ■

Chapter 9. Two-Stage Cluster Sampling

1 Introduction

Definition 9.1

A **two-stage cluster sample** is obtained by selecting a probability sample of clusters and then a probability sample of elements from each selected cluster. The clusters are the **primary sampling units** (psu, e.g. classes) and the elements the **secondary sampling units** (ssu, e.g. students).

Measuring every element in a cluster is wasteful when the elements of a cluster are similar; subsampling lets us spread the same budget over more clusters.

Notation. Both stages use SRS without replacement.

N, n	number of clusters in the population / selected
M_i, m_i	number of elements in cluster i / selected from it
$M = \sum_{i=1}^{N} M_i$, $\bar{M} = M/N$	population size and average cluster size
y_{ij}	value of the jth sampled element of cluster i
$\bar{y}_i = \frac{1}{m_i}\sum_{j=1}^{m_i} y_{ij}$	sample mean of cluster i
μ_i, $\tau_i = M_i\mu_i$	true mean and total of cluster i; $\tau = \sum_{i=1}^{N}\tau_i$
$\hat{\tau}_i = M_i\bar{y}_i$	estimated total of cluster i
$S_i^2 = \frac{1}{M_i-1}\sum_{j=1}^{M_i}(y_{ij}-\mu_i)^2$	within-cluster variance
$S_b^2 = \frac{1}{N-1}\sum_{i=1}^{N}(\tau_i-\bar{\tau})^2$, $\bar{\tau} = \tau/N$	between-cluster variance of totals

2 Estimation of the Total and the Mean

Motivation. When clusters are large, it is cheaper to subsample units within each selected cluster than to enumerate them. Two-stage sampling has two sources of error, between clusters and within clusters, and the variance formula reflects both.

Theorem 9.2: Unbiased estimators

$$\hat{\tau} = \frac{N}{n}\sum_{i=1}^{n} M_i\bar{y}_i = \frac{N}{n}\sum_{i=1}^{n} M_i\frac{1}{m_i}\sum_{j=1}^{m_i} y_{ij}, \qquad \bar{y} = \frac{\hat{\tau}}{M}$$

are unbiased for τ and μ respectively.

Proof. Let E_2 denote expectation over the second stage for a fixed set of clusters, and E_1 over the first stage. For a fixed selected cluster, $\mathrm{E}_2(\bar{y}_i) = \mu_i$ (SRS), so $\mathrm{E}_2(\hat{\tau}_i) = M_i\mu_i = \tau_i$ and

$$\mathrm{E}_2(\hat{\tau}) = \frac{N}{n}\sum_{i=1}^{n}\tau_i.$$

This is N times an SRS mean of the cluster totals, so $\mathrm{E}_1\mathrm{E}_2(\hat{\tau}) = N\cdot\frac{\tau}{N} = \tau$. Then $\mathrm{E}(\bar{y}) = \tau/M = \mu$. ■

Theorem 9.3: Variance

With $u_i = M_i^2 \frac{M_i - m_i}{M_i} \frac{S_i^2}{m_i}$ (the second-stage variance of $\hat{\tau}_i$),

$$\mathrm{Var}(\hat{\tau}) = N^2 \frac{N-n}{N} \frac{S_b^2}{n} + \frac{N}{n} \sum_{i=1}^{N} u_i = \underbrace{N^2 \frac{N-n}{N} \frac{S_b^2}{n}}_{\text{between psu}} + \underbrace{\frac{N}{n} \sum_{i=1}^{N} M_i^2 \frac{M_i - m_i}{M_i} \frac{S_i^2}{m_i}}_{\text{within psu}}.$$

2.1 Variance of a two-stage estimator

Theorem 9.4: Conditional variance decomposition

For any estimator $\hat{\theta}$ computed from a two-stage sample,

$$\mathrm{Var}(\hat{\theta}) = \mathrm{Var}_1\big(\mathrm{E}_2(\hat{\theta})\big) + \mathrm{E}_1\big(\mathrm{Var}_2(\hat{\theta})\big).$$

For three stages, $\mathrm{Var}(\hat{\theta}) = \mathrm{Var}_1(\mathrm{E}_2\mathrm{E}_3\hat{\theta}) + \mathrm{E}_1\big(\mathrm{Var}_2(\mathrm{E}_3\hat{\theta})\big) + \mathrm{E}_1\mathrm{E}_2\big(\mathrm{Var}_3\hat{\theta}\big)$.

Proof. Let $\theta = \mathrm{E}(\hat{\theta})$. Conditionally on the first stage,

$$\mathrm{E}_2(\hat{\theta} - \theta)^2 = \mathrm{Var}_2(\hat{\theta}) + \big(\mathrm{E}_2\hat{\theta}\big)^2 - 2\theta\,\mathrm{E}_2\hat{\theta} + \theta^2.$$

Taking E_1 and using $\mathrm{E}_1\mathrm{E}_2\hat{\theta} = \theta$,

$$\mathrm{Var}(\hat{\theta}) = \mathrm{E}_1\mathrm{Var}_2(\hat{\theta}) + \mathrm{E}_1\big(\mathrm{E}_2\hat{\theta}\big)^2 - \theta^2 = \mathrm{E}_1\mathrm{Var}_2(\hat{\theta}) + \mathrm{Var}_1\big(\mathrm{E}_2\hat{\theta}\big).$$

■

Derivation: proof of the variance formula

Between term. $\mathrm{E}_2(\hat{\tau}) = \frac{N}{n}\sum_{i=1}^{n}\tau_i$ is N times the SRS mean of the N cluster totals, so by Ch. 2

$$\mathrm{Var}_1\big(\mathrm{E}_2\hat{\tau}\big) = N^2 \frac{N-n}{N} \frac{S_b^2}{n}.$$

Within term. Given the clusters, the subsamples are independent, and $\mathrm{Var}_2(\hat{\tau}_i) = M_i^2 \mathrm{Var}_2(\bar{y}_i) = M_i^2 \frac{M_i - m_i}{M_i} \frac{S_i^2}{m_i} = u_i$. Thus

$$\mathrm{Var}_2(\hat{\tau}) = \frac{N^2}{n^2} \sum_{i=1}^{n} u_i, \qquad \mathrm{E}_1\mathrm{Var}_2(\hat{\tau}) = \frac{N^2}{n^2} \cdot n \cdot \frac{1}{N} \sum_{i=1}^{N} u_i = \frac{N}{n} \sum_{i=1}^{N} u_i,$$

because each cluster is in the first-stage sample with probability n/N.

Direct expansion (alternative). Writing $\hat{\tau} - \tau = \big(\frac{N}{n}\sum_{i\le n}\tau_i - \tau\big) + \frac{N}{n}\sum_{i\le n}(\hat{\tau}_i - \tau_i)$ and squaring, the cross term has $\mathrm{E}_2(\hat{\tau}_i - \tau_i) = 0$, and the products $(\hat{\tau}_i - \tau_i)(\hat{\tau}_{i'} - \tau_{i'})$, $i \neq i'$, have zero conditional expectation since $\mathrm{E}_2(\hat{\tau}_i\hat{\tau}_{i'}) = \tau_i\tau_{i'}$ by independence of the subsamples. The two remaining squared terms give the between and within components.

2.2 Estimated variance

Theorem 9.5: Unbiased variance estimator

$$\widehat{\mathrm{Var}}(\hat\tau) = N^2\frac{N-n}{N}\frac{s_b^2}{n} + \frac{N}{n}\sum_{i=1}^{n} M_i^2\frac{M_i-m_i}{M_i}\frac{s_i^2}{m_i},$$

$$s_b^2 = \frac{1}{n-1}\sum_{i=1}^{n}(M_i\bar y_i - \bar M\bar y)^2, \qquad s_i^2 = \frac{1}{m_i-1}\sum_{j=1}^{m_i}(y_{ij}-\bar y_i)^2,$$

and $\widehat{\mathrm{Var}}(\bar y) = \widehat{\mathrm{Var}}(\hat\tau)/M^2$. Here s_b^2 is the sample variance of the $\hat\tau_i = M_i\bar y_i$ (note $\bar M\bar y = \hat\tau/N = \frac{1}{n}\sum\hat\tau_i$).

The key fact is that s_b^2 is *not* unbiased for S_b^2: the estimated cluster totals also carry second-stage noise.

Theorem 9.6

$\mathrm{E}(s_b^2) = S_b^2 + \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} u_i$, and consequently $\mathrm{E}[\widehat{\mathrm{Var}}(\hat\tau)] = \mathrm{Var}(\hat\tau)$.

Derivation: proof

Write $\bar\hat\tau = \frac{1}{n}\sum_{i\le n}\hat\tau_i = \hat\tau/N$. Then $(n-1)s_b^2 = \sum_{i\le n}\hat\tau_i^2 - n\bar\hat\tau^2$.
First piece. $\mathrm{E}_2(\hat\tau_i^2) = \mathrm{Var}_2(\hat\tau_i) + \tau_i^2 = u_i + \tau_i^2$, so $\mathrm{E}\big[\sum_{i\le n}\hat\tau_i^2\big] = \frac{n}{N}\sum_{i=1}^{N}(u_i+\tau_i^2)$.
Second piece. $\mathrm{E}(\bar\hat\tau^2) = \frac{\mathrm{Var}(\hat\tau)}{N^2} + \bar\tau^2 = \frac{N-n}{Nn}S_b^2 + \frac{1}{nN}\sum_{i=1}^{N}u_i + \bar\tau^2$.
Combining and using $\frac{1}{N}\sum_{i=1}^{N}\tau_i^2 - \bar\tau^2 = \frac{N-1}{N}S_b^2$,

$$\begin{aligned}\mathrm{E}\big[(n-1)s_b^2\big] &= \frac{n}{N}\sum u_i + n\Big(\frac{1}{N}\sum\tau_i^2 - \bar\tau^2\Big) - \frac{N-n}{N}S_b^2 - \frac{1}{N}\sum u_i\\ &= \frac{n-1}{N}\sum_{i=1}^{N}u_i + \frac{n(N-1)-(N-n)}{N}S_b^2 = (n-1)\Big[S_b^2 + \frac{1}{N}\sum_{i=1}^{N}u_i\Big].\end{aligned}$$

Unbiasedness of $\widehat{\mathrm{Var}}(\hat\tau)$. Since $\mathrm{E}_2(s_i^2) = S_i^2$, the second term has expectation $\frac{N}{n}\cdot\frac{n}{N}\sum_{i=1}^{N}u_i = \sum_{i=1}^{N}u_i$. Hence

$$\mathrm{E}\widehat{\mathrm{Var}}(\hat\tau) = N^2\frac{N-n}{Nn}\Big(S_b^2 + \frac{1}{N}\sum u_i\Big) + \sum u_i = N^2\frac{N-n}{Nn}S_b^2 + \Big(\frac{N-n}{n}+1\Big)\sum u_i = \mathrm{Var}(\hat\tau).$$

Of the within-psu variance $\frac{N}{n}\sum u_i$, the between-psu term of $\widehat{\mathrm{Var}}$ already contributes $\frac{N-n}{n}\sum u_i$, and the explicit within term supplies the remaining $\sum u_i$.

Remark 9.7: Special cases

- All elements of each selected psu are measured ($m_i = M_i$): $\widehat{\mathrm{Var}}(\hat\tau) = N^2\frac{N-n}{Nn}s_b^2$, which is one-stage cluster sampling (Ch. 8).
- All psu are included ($n = N$): $\widehat{\mathrm{Var}}(\hat\tau) = \sum_{i=1}^{N}M_i^2\frac{M_i-m_i}{M_i}\frac{s_i^2}{m_i}$, which is stratified random sampling with the clusters as strata (Ch. 3).

3 Estimating the Mean when M is Unknown

Motivation. If the total number of units M is unknown, the mean per unit is estimated by a ratio of two estimated totals.

$\bar{y} = \hat{\tau}/M$ requires M. Otherwise use the ratio estimator

$$\hat{\mu} = \frac{\sum_{i=1}^{n} M_i \bar{y}_i}{\sum_{i=1}^{n} M_i} = \frac{\sum_{i=1}^{n} \hat{\tau}_i}{\sum_{i=1}^{n} M_i}.$$

Theorem 9.8

$$\widehat{\mathrm{Var}}(\hat{\mu}) = \frac{N^2}{M^2} \cdot \frac{N-n}{N} \cdot \frac{s_r^2}{n} + \frac{N}{nM^2} \sum_{i=1}^{n} M_i^2 \frac{M_i - m_i}{M_i} \frac{s_i^2}{m_i}, \qquad s_r^2 = \frac{1}{n-1} \sum_{i=1}^{n} (M_i \bar{y}_i - M_i \hat{\mu})^2,$$

where, when M is unknown, M is estimated by $\hat{M} = N \cdot \frac{1}{n} \sum_{i=1}^{n} M_i$ (equivalently $\bar{M}$ by the sample average of the M_i).

The only change from $\widehat{\mathrm{Var}}(\bar{y})$ is that the between term uses deviations $M_i \bar{y}_i - M_i \hat{\mu}$ (residuals from a ratio fit) instead of $M_i \bar{y}_i - \bar{M} \bar{y}$.

4 Estimation of a Proportion

For the proportion of students in favor of a law, let $y_{ij} = 1$ or 0 and $\hat{p}_i = \frac{1}{m_i} \sum_j y_{ij}$, the sample proportion in cluster i. With M unknown,

$$\hat{p} = \frac{\sum_{i=1}^{n} M_i \hat{p}_i}{\sum_{i=1}^{n} M_i},$$

$$\widehat{\mathrm{Var}}(\hat{p}) = \frac{N^2}{M^2} \frac{N-n}{N} \frac{s_p^2}{n} + \frac{N}{nM^2} \sum_{i=1}^{n} M_i^2 \frac{M_i - m_i}{M_i} \frac{\hat{p}_i \hat{q}_i}{m_i - 1}, \qquad s_p^2 = \frac{1}{n-1} \sum_{i=1}^{n} (M_i \hat{p}_i - M_i \hat{p})^2,$$

with $\hat{q}_i = 1 - \hat{p}_i$. The within term uses the SRS result $\widehat{\mathrm{Var}}(\hat{p}_i) = \frac{\hat{p}_i \hat{q}_i}{m_i - 1} \frac{M_i - m_i}{M_i}$.

5 Equal-Sized Clusters

Suppose $M_1 = \cdots = M_N = \bar{M}$ and $m_1 = \cdots = m_n = m$. Then

$$\bar{y} = \frac{\hat{\tau}}{M} = \frac{1}{N\bar{M}} \cdot \frac{N}{n} \bar{M} \sum_{i=1}^{n} \bar{y}_i = \frac{1}{n} \sum_{i=1}^{n} \bar{y}_i = \frac{1}{nm} \sum_{i=1}^{n} \sum_{j=1}^{m} y_{ij} = \hat{\mu},$$

the overall sample average. Arrange the data as an $m \times n$ table (columns = psu) with

$$MSB = \frac{m}{n-1} \sum_{i=1}^{n} (\bar{y}_i - \bar{y})^2, \qquad MSW = \frac{1}{n(m-1)} \sum_{i=1}^{n} \sum_{j=1}^{m} (y_{ij} - \bar{y}_i)^2.$$

Theorem 9.9: Equal-size two-stage variance

With $f_1 = n/N$ and $f_2 = m/\bar{M}$,

$$\widehat{\mathrm{Var}}(\bar{y}) = (1 - f_1) \frac{MSB}{nm} + (1 - f_2) \frac{MSW}{Nm}.$$

If N is large, $\widehat{\mathrm{Var}}(\bar{y}) \approx MSB/(nm)$.

Derivation

Between term. $s_b^2 = \frac{1}{n-1}\sum(\bar{M}\bar{y}_i - \bar{M}\bar{y})^2 = \bar{M}^2\frac{MSB}{m}$, and $N^2/M^2 = 1/\bar{M}^2$, so

$$\frac{N^2}{M^2}\frac{N-n}{N}\frac{s_b^2}{n} = (1-f_1)\frac{1}{n}\frac{MSB}{m}.$$

Within term. $\frac{1}{n}\sum_i s_i^2 = MSW$, so

$$\frac{1}{M^2}\frac{N}{n}\sum_{i=1}^{n}\bar{M}^2\frac{\bar{M}-m}{\bar{M}}\frac{s_i^2}{m} = \frac{N\bar{M}^2}{M^2}(1-f_2)\frac{MSW}{m} = \frac{1}{N}(1-f_2)\frac{MSW}{m}.$$

The same computation for $\widehat{\mathrm{Var}}(\hat{\mu})$ gives $s_r^2 = s_b^2$ and $\hat{M} = M$, so $\widehat{\mathrm{Var}}(\hat{\mu}) = \widehat{\mathrm{Var}}(\bar{y})$ here.

Remark 9.10: Limiting cases

- $m = \bar{M}$ ($f_2 = 1$): the within term vanishes and $\widehat{\mathrm{Var}}(\bar{y}) = (1-f_1)\frac{MSB}{nm}$, one-stage cluster sampling.
- $n = N$ ($f_1 = 1$): $\widehat{\mathrm{Var}}(\bar{y}) = (1-f_2)\frac{MSW}{Nm}$, the variance estimate of a stratified random sample with N strata and m observations per stratum.

6 Probabilities Proportional to Size

PPS sampling is useful when cluster sizes vary greatly; it is typically used only at the first stage. Selecting psu with replacement with probabilities $z_i = M_i/M$ and estimating each cluster total by $\hat{\tau}_i = M_i\bar{y}_i$,

$$\hat{\tau}_{\mathrm{pps}} = \frac{1}{n}\sum_{i=1}^{n}\frac{\hat{\tau}_i}{z_i} = \frac{M}{n}\sum_{i=1}^{n}\bar{y}_i, \qquad \hat{\mu}_{\mathrm{pps}} = \frac{\hat{\tau}_{\mathrm{pps}}}{M} = \frac{1}{n}\sum_{i=1}^{n}\bar{y}_i,$$

compared with the equal-probability choice $z_i = 1/N$, which gives $\hat{\tau} = \frac{N}{n}\sum M_i\bar{y}_i$. The variance estimators are

$$\widehat{\mathrm{Var}}(\hat{\tau}_{\mathrm{pps}}) = \frac{M^2}{n(n-1)}\sum_{i=1}^{n}(\bar{y}_i - \hat{\mu}_{\mathrm{pps}})^2, \qquad \widehat{\mathrm{Var}}(\hat{\mu}_{\mathrm{pps}}) = \frac{1}{n(n-1)}\sum_{i=1}^{n}(\bar{y}_i - \hat{\mu}_{\mathrm{pps}})^2,$$

and they already include the second-stage variability, since each $\bar{y}_i$ is an independent draw of a cluster *and* a subsample. With $m_i = m$ we get $\hat{\mu}_{\mathrm{pps}} = \bar{y}$ and $\widehat{\mathrm{Var}}(\hat{\mu}_{\mathrm{pps}}) = \frac{1}{n(n-1)}\sum(\bar{y}_i - \bar{y})^2 = \frac{MSB}{nm}$, the large-$N$ approximation of the previous section.

7 Selected exercises

Motivation. The item below proves the claim above that the PPS variance estimator already captures the second stage.

Exercise 9.1 (Two-stage PPS)

Clusters are drawn n times with replacement, with probabilities $z_i > 0$. Each time cluster i is drawn, an independent SRS of m_i of its M_i elements is taken, giving an unbiased estimate $\hat{\tau}_i = M_i \bar{y}_i$ of τ_i with variance $V_i = M_i^2(1 - m_i/M_i)S_i^2/m_i$. Let $t_r = \hat{\tau}_{(r)}/z_{(r)}$ for the rth draw, $\hat{\tau}_{\text{pps}} = \bar{t}$.

(a) Show that $\hat{\tau}_{\text{pps}}$ is unbiased with

$$\operatorname{Var}(\hat{\tau}_{\text{pps}}) = \frac{1}{n}\Big[\sum_{i=1}^{N} z_i\Big(\frac{\tau_i}{z_i} - \tau\Big)^2 + \sum_{i=1}^{N} \frac{V_i}{z_i}\Big].$$

(b) Show that $\frac{1}{n(n-1)}\sum_r (t_r - \bar{t})^2$ is unbiased for $\operatorname{Var}(\hat{\tau}_{\text{pps}})$.

Solution. [Supplement]

(a)

Step 1. (iid structure.) Each draw consists of a cluster index I_r with $P(I_r = i) = z_i$ and a fresh subsample; draws are independent and identically distributed, so the t_r are iid.

Step 2. (Mean.) Conditionally on $I_r = i$, $\mathrm{E}(t_r \mid I_r = i) = \tau_i/z_i$. Hence $\mathrm{E}t_r = \sum_i z_i \tau_i/z_i = \tau$, and $\mathrm{E}\bar{t} = \tau$.

Step 3. (Variance of one draw.) By the conditional variance decomposition, with $\operatorname{Var}(t_r \mid I_r = i) = V_i/z_i^2$,

$$\operatorname{Var}(t_r) = \operatorname{Var}(\mathrm{E}(t_r \mid I_r)) + \mathrm{E}(\operatorname{Var}(t_r \mid I_r)) = \sum_i z_i\Big(\frac{\tau_i}{z_i} - \tau\Big)^2 + \sum_i z_i \frac{V_i}{z_i^2}.$$

Dividing by n gives the formula: the first term is the one-stage PPS variance (Exercise 8.2), the second the subsampling variance.

(b) For iid t_r the sample variance s_t^2 is unbiased for $\operatorname{Var}(t_r)$, so $\mathrm{E}(s_t^2/n) = \operatorname{Var}(t_r)/n = \operatorname{Var}(\bar{t})$. The estimator involves no s_i^2 at all, yet it estimates both components: this is why the PPS variance estimators of the text need no explicit within-cluster term. The argument requires a new, independent subsample whenever a cluster is drawn again. ■

Chapter 10. Estimating Population Size and Nonresponse

1 Capture–Recapture Estimation of N

To estimate the number N of fish in a lake, catch n_1 fish, mark them and release them; later catch a second sample of n_2 fish and count the number m of marked fish among them.

Definition 10.1: Assumptions

1. The population is closed (no births, deaths or migration between the samples).
2. Each sample is an SRS from the population (marking does not change catchability).
3. The two samples are independent.

The proportion of marked fish in the second sample should resemble that in the population:

$$\frac{m}{n_2} \approx \frac{n_1}{N} \quad \Longrightarrow \quad \hat{N} = \frac{n_1 n_2}{m}$$

(the Lincoln–Petersen estimator), where n_1 is the size of the first (capture) sample, n_2 that of the second (recapture) sample, and m the number of marked elements in the second sample.

1.1 Capture–recapture as a ratio estimator

Theorem 10.2

$\hat{N}$ is a ratio estimator of a population total, and

$$\widehat{\mathrm{Var}}(\hat{N}) = \frac{n_1^2 n_2^2 (n_2 - m)}{m^3 (n_2 - 1)} \approx \frac{n_1^2 n_2 (n_2 - m)}{m^3}.$$

Derivation

Define for every element $y_i = 1$, so $\tau_y = \sum_{i=1}^{N} y_i = N$, and $x_i = 1$ if element i is marked, 0 otherwise, so $\tau_x = \sum_{i=1}^{N} x_i = n_1$ is known. The recapture sample of size n_2 gives

$$r = \frac{\sum_{i=1}^{n_2} y_i}{\sum_{i=1}^{n_2} x_i} = \frac{n_2}{m}, \qquad \hat{N} = \hat{\tau}_y = r\tau_x = \frac{n_2}{m} n_1.$$

From Ch. 5, $\widehat{\mathrm{Var}}(\hat{N}) = \tau_x^2 \widehat{\mathrm{Var}}(r) = n_1^2 \widehat{\mathrm{Var}}(r)$ with

$$\widehat{\mathrm{Var}}(r) = \frac{1}{\bar{x}^2} \cdot \frac{N - n_2}{N} \cdot \frac{1}{n_2} \cdot \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (y_i - r x_i)^2,$$

and we take the fpc ≈ 1. Now

$$\bar{x} = \frac{m}{n_2}, \qquad \sum_{i=1}^{n_2} (y_i - r x_i)^2 = \sum y_i^2 - 2r \sum x_i y_i + r^2 \sum x_i^2 = n_2 - 2\frac{n_2}{m} m + \frac{n_2^2}{m^2} m = \frac{n_2 (n_2 - m)}{m},$$

using $y_i^2 = y_i = 1$ and $x_i^2 = x_i$. Therefore

$$\widehat{\mathrm{Var}}(r) = \frac{n_2^2}{m^2} \cdot \frac{1}{n_2} \cdot \frac{1}{n_2 - 1} \cdot \frac{n_2 (n_2 - m)}{m} = \frac{n_2^2 (n_2 - m)}{m^3 (n_2 - 1)},$$

and multiplying by n_1^2 gives the result.

1.2 Maximum likelihood

Definition 10.3

Let $f(\mathbf{x} \mid \theta)$ be the joint pdf of the sample $\mathbf{X} = (X_1, \ldots, X_n)$. Given $\mathbf{X} = \mathbf{x}$, the function $L(\theta \mid \mathbf{x}) = f(\mathbf{x} \mid \theta)$ of θ is the **likelihood function**. If $L(\theta_1 \mid \mathbf{x}) \geq L(\theta_2 \mid \mathbf{x})$, then θ_1 is more plausible than θ_2 in light of the data. The **maximum likelihood estimator** is $\hat{\theta}(\mathbf{x}) = \arg\max_{\theta \in \Theta} L(\theta \mid \mathbf{x})$.

Example 10.4: Normal mean

For $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, 1)$,

$$\ell(\theta) = \log L(\theta \mid \mathbf{x}) = -\frac{n}{2}\log 2\pi - \frac{1}{2}\sum_{i=1}^{n}(x_i - \theta)^2, \qquad \ell'(\hat{\theta}) = \sum(x_i - \hat{\theta}) = 0 \;\Rightarrow\; \hat{\theta} = \bar{x}.$$

Theorem 10.5: Large-sample properties of the MLE

Under regularity conditions,

1. (consistency) $\lim_{n\to\infty} P(|\hat{\theta}_n - \theta| \geq \varepsilon) = 0$ for every $\varepsilon > 0$;
2. (asymptotic normality) $\sqrt{n}(\hat{\theta}_n - \theta) \to N(0, 1/I(\theta))$, i.e. $\hat{\theta}_n \approx N(\theta, \frac{1}{nI(\theta)})$, where $I(\theta) = \mathrm{E}\big[\big(\frac{\partial}{\partial\theta}\log f(X \mid \theta)\big)^2\big]$ is the Fisher information of one observation.

Theorem 10.6: MLE of N

Model the recapture sample as $X_i = 1$ (marked) or 0, $X_i \overset{\text{iid}}{\sim} \mathrm{Ber}(p)$ with $p = n_1/N$, $i = 1, \ldots, n_2$. Then

$$\hat{N}_{\text{MLE}} = \frac{n_1 n_2}{m}, \qquad \mathrm{Var}(\hat{N}_{\text{MLE}}) \approx \frac{N^2(N - n_1)}{n_1 n_2}, \qquad \widehat{\mathrm{Var}}(\hat{N}_{\text{MLE}}) = \frac{n_1^2 n_2 (n_2 - m)}{m^3}.$$

Derivation

With $\sum_{i=1}^{n_2} X_i = m$,

$$L(N) = \prod_{i=1}^{n_2}\Big(\frac{n_1}{N}\Big)^{X_i}\Big(1 - \frac{n_1}{N}\Big)^{1-X_i} = \Big(\frac{n_1}{N}\Big)^{m}\Big(1 - \frac{n_1}{N}\Big)^{n_2 - m},$$

$$\ell(N) = m\log n_1 - m\log N + (n_2 - m)\log(N - n_1) - (n_2 - m)\log N, \qquad \ell'(N) = \frac{n_2 - m}{N - n_1} - \frac{n_2}{N}.$$

Setting $\ell'(N) = 0$: $N(n_2 - m) = n_2(N - n_1)$, so $\hat{N} = n_1 n_2/m$.

Fisher information. For one observation,

$$\frac{\partial}{\partial N}\log f(x \mid N) = \frac{1 - x}{N - n_1} - \frac{1}{N} = \frac{n_1/N - x}{N - n_1} = \frac{p - x}{N - n_1},$$

$$I(N) = \mathrm{E}\Big(\frac{p - X}{N - n_1}\Big)^2 = \frac{p(1 - p)}{(N - n_1)^2} = \frac{n_1}{N^2(N - n_1)}.$$

Hence $\operatorname{Var}(\hat{N}_{\mathrm{MLE}}) \approx \left[n_2 I(N)\right]^{-1} = \frac{N^2(N-n_1)}{n_1 n_2}$. Plugging in $\hat{N} = n_1 n_2/m$:

$$\frac{(n_1 n_2/m)^2\,(n_1 n_2/m - n_1)}{n_1 n_2} = \frac{n_1^2 n_2^2}{m^2}\cdot\frac{n_1(n_2-m)}{m}\cdot\frac{1}{n_1 n_2} = \frac{n_1^2 n_2 (n_2-m)}{m^3}.$$

The ratio-estimator and likelihood approaches agree: $\widehat{\operatorname{Var}}(\hat{\tau}_y) \approx \widehat{\operatorname{Var}}(\hat{N}_{\mathrm{MLE}})$.

Example 10.7: Illustration

If $n_1 = 200$ fish are marked and a second catch of $n_2 = 100$ contains $m = 20$ marked fish, then $\hat{N} = 200 \cdot 100/20 = 1000$ and

$$\widehat{\operatorname{Var}}(\hat{N}) = \frac{200^2 \cdot 100 \cdot 80}{20^3} = 40{,}000, \qquad 2\sqrt{40000} = 400,$$

so N is estimated as 1000 ± 400.

2 Nonresponse

Nonresponse (missing data) biases the estimation of population parameters. Divide the population into a response stratum R and a nonresponse stratum:

	size	mean
Respondents	N_1	μ_1
Nonrespondents	N_2	μ_2
	$N = N_1 + N_2$	μ

Theorem 10.8: Nonresponse bias

If $\bar{y}_1$ is unbiased for the respondents' mean μ_1, then as an estimator of μ,

$$\mathrm{E}(\bar{y}_1) - \mu = \frac{N_2}{N}(\mu_1 - \mu_2).$$

Proof. $\mu = \frac{N_1}{N}\mu_1 + \frac{N_2}{N}\mu_2 = \frac{N_1+N_2}{N}\mu_1 - \frac{N_2}{N}\mu_1 + \frac{N_2}{N}\mu_2 = \mu_1 - \frac{N_2}{N}(\mu_1 - \mu_2)$. ■

The bias is small only if the nonresponse rate N_2/N is small or respondents and nonrespondents are alike ($\mu_1 \approx \mu_2$). Increasing the sample size does not reduce it.

2.1 A conservative interval for a proportion

Suppose n_1 respondents are an SRS from the N_1 respondents of the population, so a 95% interval for their proportion p_1 is

$$\hat{p}_1 \pm 2\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1 - 1}\cdot\frac{N_1 - n_1}{N_1}}.$$

Nothing is known about the nonrespondents' proportion p_2, except $0 \le p_2 \le 1$. With $w_1 = N_1/N$ and $w_2 = N_2/N$, $p = w_1 p_1 + w_2 p_2$. Take the worst cases $p_2 = 0$ for the lower limit and $p_2 = 1$ for the upper limit:

$$\hat{p}_L = w_1\left(\hat{p}_1 - 2\sqrt{\tfrac{\hat{p}_1\hat{q}_1}{n_1-1}\tfrac{N_1-n_1}{N_1}}\right) + w_2 \cdot 0,$$

$$\hat{p}_U = w_1\Big(\hat{p}_1 + 2\sqrt{\tfrac{\hat{p}_1\hat{q}_1}{n_1-1}\tfrac{N_1-n_1}{N_1}}\Big) + w_2\cdot 1.$$

Whenever the interval for p_1 covers p_1, $[\hat{p}_L, \hat{p}_U]$ covers p whatever p_2 is, so $P(\hat{p}_L \le p \le \hat{p}_U) \ge 0.95$. The width grows by w_2: a high nonresponse rate makes the interval uninformative, which is why reducing nonresponse matters more than increasing n.

3 Selected exercises

Motivation. The Lincoln–Petersen estimator was obtained from a ratio argument and from a Bernoulli likelihood. The items below view it as the completion of a 2×2 table and derive it from the exact hypergeometric likelihood.

Exercise 10.1 (Deer population)

A random sample of $n_1 = 300$ deer is captured, tagged and released. A second sample of $n_2 = 200$ taken two weeks later contains $m = 62$ tagged deer.
(a) Arrange the data as a 2×2 table (caught or not in the first and in the second sample) and estimate the missing cell, assuming that the two captures are independent.
(b) Show that the resulting estimate of N is the Lincoln–Petersen estimator, and place a bound on the error.

Solution.

(a) The table has one unobservable cell x (deer caught in neither sample):

	in 2nd sample	not in 2nd	total
in 1st sample	62	238	300
not in 1st	138	x	
total	200		N

Independence of the two captures means that the odds ratio of the table is 1, i.e. $62\,x = 238\times 138$, so $\hat{x} = 32{,}844/62 = 529.7$.

(b)

Step 1. In general, with cells $n_{11}=m$, $n_{12}=n_1-m$, $n_{21}=n_2-m$ and $\hat{x}=n_{12}n_{21}/n_{11}$,

$$\hat{N} = n_{11}+n_{12}+n_{21}+\frac{n_{12}n_{21}}{n_{11}} = \frac{(n_{11}+n_{12})(n_{11}+n_{21})}{n_{11}} = \frac{n_1 n_2}{m}.$$

Step 2. Here $\hat{N} = 62+238+138+529.7 = 967.7 = 300\times 200/62$.

Step 3. By the MLE theorem, $\widehat{\mathrm{Var}}(\hat{N}) = \dfrac{n_1^2 n_2(n_2-m)}{m^3} = \dfrac{300^2\cdot 200\cdot 138}{62^3} = 10{,}423$, and $2\sqrt{10{,}423} = 204$. The herd is estimated at 968 ± 204 deer. ∎

Exercise 10.2 (The hypergeometric likelihood)

If the second sample is an SRS of n_2 drawn without replacement from N animals of which n_1 are marked, then

$$L(N) = P(m \mid N) = \frac{\binom{n_1}{m}\binom{N-n_1}{n_2-m}}{\binom{N}{n_2}}, \qquad N \ge n_1 + n_2 - m.$$

Assuming $m \geq 1$, show that the MLE of N over the integers is $\lfloor n_1 n_2/m \rfloor$.

Solution. [Supplement]

Step 1. (Likelihood ratio.) For $N - 1 \geq n_1 + n_2 - m$, using $\binom{a}{c}/\binom{a-1}{c} = \frac{a}{a-c}$,

$$\frac{L(N)}{L(N-1)} = \frac{\binom{N-n_1}{n_2-m}}{\binom{N-1-n_1}{n_2-m}} \cdot \frac{\binom{N-1}{n_2}}{\binom{N}{n_2}} = \frac{N-n_1}{N-n_1-n_2+m} \cdot \frac{N-n_2}{N}.$$

Step 2. (Sign.) Both denominators are positive, so $L(N) \geq L(N-1)$ iff $(N-n_1)(N-n_2) \geq N(N-n_1-n_2+m)$. Expanding, the terms N^2 and $-N(n_1+n_2)$ cancel and the condition becomes $n_1 n_2 \geq Nm$, i.e. $N \leq n_1 n_2/m$.

Step 3. (Maximum.) Hence L is nondecreasing for $N \leq n_1 n_2/m$ and strictly decreasing afterwards, so it is maximized at $N = \lfloor n_1 n_2/m \rfloor$ (if $n_1 n_2/m$ is an integer, $n_1 n_2/m - 1$ attains the same value). This value is at least $n_1 + n_2 - m$, since $(n_1 - m)(n_2 - m) \geq 0$ gives $n_1 n_2/m \geq n_1 + n_2 - m$. The exact likelihood therefore gives the Lincoln–Petersen estimate, as the Bernoulli approximation of the text did. ■

Chapter 11. Resampling Methods

1 Training Error and Test Error

Resampling methods repeatedly draw samples from the data at hand to learn about an estimator. The two most common are **cross-validation** and the **bootstrap**; they estimate the prediction error, the standard deviation and the bias of estimates.

Definition 11.1

Let $D_n = \{(x_i, y_i)\}_{i=1}^n$ be the training data, g_{D_n} a prediction rule fitted to it, and L a loss function.

- The **training error** applies the fitted rule back to the training observations, e.g. $\frac{1}{n}\sum(y_i - \hat{y}_i)^2$.
- The **test error** (prediction error) is the expected loss on a new observation (x_0, y_0) independent of D_n:
$$PE = \mathrm{E}[L(y_0, g_{D_n}(x_0))].$$

The training error is easy to compute but usually underestimates the test error, because the rule was tuned to the same data: a curve that zig-zags through every training point has zero training error yet predicts new points $y_i^* = f(x_i) + \varepsilon_i^*$ badly. The smaller the test error, the more suitable the model. Measures such as $MS_E = \frac{SS_E}{n-2}$ describe goodness of fit, not predictive ability.

With an independent test set $\{(x_j, y_j)\}_{j=1}^{n_t}$ we could estimate

$$\widehat{PE} = \frac{1}{n_t}\sum_{j=1}^{n_t} L(y_j, \hat{y}_j), \qquad \hat{y}_j = g_{D_n}(x_j),$$

but a large test set is often unavailable. The methods below hold out part of the training data instead.

2 Cross-Validation

2.1 K-fold cross-validation

Randomly divide the data into K roughly equal parts $C_1, \ldots, C_K$ (C_k holds the indices of part k, n_k observations; $n_k = n/K$ if n is a multiple of K). For each k, fit the model to the other $K-1$ parts and predict part k:

$$\mathrm{MSE}_k = \frac{1}{n_k}\sum_{i \in C_k}(y_i - \hat{y}_i)^2, \qquad CV_{(K)} = \sum_{k=1}^{K}\frac{n_k}{n}\mathrm{MSE}_k,$$

where $\hat{y}_i$ is the fit for observation i obtained with part k removed.

2.2 Leave-one-out cross-validation

$K = n$ gives LOOCV. Let $D_{(i)}$ be the training set with (x_i, y_i) omitted and $\hat{y}_{(i)} = g_{D_{(i)}}(x_i)$. Then

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} L(y_i, \hat{y}_{(i)}),$$

which mimics $\widehat{PE}$ by letting each point serve as a test point for a rule that did not see it.

Advantages of LOOCV over a single validation split:

(i) much less bias, since each fit uses $n-1$ observations, almost the whole data set;

(ii) no randomness, since repeated runs give the same result.

Theorem 11.2: LOOCV shortcut for least squares

For least-squares linear regression $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$, $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$,

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1 - h_{ii}}\right)^2,$$

where $\hat{y}_i$ is the fitted value from the full fit and h_{ii} is the leverage of observation i. The terms $(y_i - \hat{y}_i)/(1-h_{ii}) = y_i - \hat{y}_{(i)}$ are the **PRESS residuals**; only one fit is needed.

Derivation: proof

Let $\mathbf{x}_i^\top$ be the ith row of $\mathbf{X}$, $A = \mathbf{X}^\top\mathbf{X}$, and $h_{ii} = \mathbf{x}_i^\top A^{-1}\mathbf{x}_i$. Deleting row i gives $A_{(i)} = A - \mathbf{x}_i\mathbf{x}_i^\top$ and, by the Sherman–Morrison formula,

$$A_{(i)}^{-1} = A^{-1} + \frac{A^{-1}\mathbf{x}_i\mathbf{x}_i^\top A^{-1}}{1 - h_{ii}}.$$

With $\mathbf{X}_{(i)}^\top\mathbf{y}_{(i)} = \mathbf{X}^\top\mathbf{y} - \mathbf{x}_i y_i$,

$$\hat{y}_{(i)} = \mathbf{x}_i^\top A_{(i)}^{-1}(\mathbf{X}^\top\mathbf{y} - \mathbf{x}_i y_i) = \left(\mathbf{x}_i^\top A^{-1} + \frac{h_{ii}\mathbf{x}_i^\top A^{-1}}{1-h_{ii}}\right)(\mathbf{X}^\top\mathbf{y} - \mathbf{x}_i y_i) = \frac{\mathbf{x}_i^\top A^{-1}(\mathbf{X}^\top\mathbf{y} - \mathbf{x}_i y_i)}{1-h_{ii}}$$
$$= \frac{\hat{y}_i - h_{ii}y_i}{1-h_{ii}}.$$

Hence $y_i - \hat{y}_{(i)} = \dfrac{(1-h_{ii})y_i - \hat{y}_i + h_{ii}y_i}{1-h_{ii}} = \dfrac{y_i - \hat{y}_i}{1-h_{ii}}$.

Remark 11.3: Choice of K

LOOCV does not shake up the data enough: the n fits are trained on nearly identical sets, so their errors are highly correlated and their average can have high variance. $K = 5$ or 10 is usually a better choice.

2.3 Why cross-validation works: model selection

Example 11.4: Choosing the degree of a polynomial

Consider $y_i = f(x_i) + \varepsilon_i$ fitted by $f_p(x) = \beta_0 + \beta_1 x + \cdots + \beta_p x^p$, and let $\hat{f}_p$ be its LS estimate. The goal is to select p. A natural target is

$$\mathrm{MSE}(p) = \frac{1}{n}\sum_{i=1}^{n}\mathrm{E}\big(f(x_i) - \hat{f}_p(x_i)\big)^2,$$

but f is unknown. Consider instead the prediction squared error for new responses $y_i^* =$

$f(x_i) + \varepsilon_i^*$:

$$PSE(p) = \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}(y_i^* - \hat{f}_p(x_i))^2 = \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}[(f(x_i) - \hat{f}_p(x_i)) + \varepsilon_i^*]^2$$
$$= \mathrm{MSE}(p) + \sigma^2,$$

because ε_i^* is independent of $\hat{f}_p$ and has mean zero. So minimizing PSE is equivalent to minimizing MSE. Now let

$$CV(p) = \frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{f}_p^{-i}(x_i))^2,$$

where $\hat{f}_p^{-i}$ is fitted without the ith point. Since y_i is independent of $\hat{f}_p^{-i}$,

$$\mathrm{E}(y_i - \hat{f}_p^{-i}(x_i))^2 = \sigma^2 + \mathrm{E}(f(x_i) - \hat{f}_p^{-i}(x_i))^2,$$

and if $\hat{f}_p^{-i} \approx \hat{f}_p$ (leaving out one point changes little), $\mathrm{E}[CV(p)] \approx \sigma^2 + \mathrm{MSE}(p) = PSE(p)$. Thus $\hat{p} = \arg\min_p CV(p)$ is a sensible rule. The training error $\frac{1}{n}\sum(y_i - \hat{f}_p(x_i))^2$ does not have this property: it keeps decreasing as p grows.

3 The Bootstrap

The bootstrap quantifies the uncertainty of an estimator or a learning method, e.g. the standard error of a coefficient or a confidence interval for it.

3.1 A simple example

We invest a fraction α of our money in an asset with return X and $1-\alpha$ in one with return Y, and choose α to minimize the risk $\mathrm{Var}(\alpha X + (1-\alpha)Y)$.

Theorem 11.5

The risk is minimized by

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}.$$

Proof. $g(\alpha) = \alpha^2\sigma_X^2 + (1-\alpha)^2\sigma_Y^2 + 2\alpha(1-\alpha)\sigma_{XY}$, and

$$g'(\alpha) = 2\alpha\sigma_X^2 - 2(1-\alpha)\sigma_Y^2 + 2(1-2\alpha)\sigma_{XY} = 0 \iff \alpha(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}) = \sigma_Y^2 - \sigma_{XY}.$$

Since $g'' = 2\mathrm{Var}(X - Y) \geq 0$, this is a minimum. ■

The variances are unknown, so we plug in sample estimates to get $\hat{\alpha}$. To learn the standard deviation of $\hat{\alpha}$ in a simulation, we could draw 1,000 independent data sets of 100 pairs from the population, compute $\hat{\alpha}_1, \ldots, \hat{\alpha}_{1000}$, and report

$$\bar{\alpha} = \frac{1}{1000}\sum_{r=1}^{1000} \hat{\alpha}_r, \qquad \sqrt{\frac{1}{1000-1}\sum_{r=1}^{1000} (\hat{\alpha}_r - \bar{\alpha})^2}.$$

3.2 Back to the real world

For real data we cannot draw new samples from the population. The bootstrap instead draws data sets *from the original data set with replacement*, using the computer to mimic the process of

generating new samples.

Definition 11.6: Bootstrap standard error

1. From the original data Z of size n, draw a bootstrap data set Z^{*1} of size n by sampling observations with replacement (some observations appear more than once, some not at all).
2. Compute the estimate $\hat{\alpha}^{*1}$ from Z^{*1}.
3. Repeat B times (say $B = 100$ or 1000) to get $\hat{\alpha}^{*1}, \dots, \hat{\alpha}^{*B}$.
4. Estimate the standard error by

$$SE_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1}\sum_{r=1}^{B}\left(\hat{\alpha}^{*r} - \bar{\alpha}^{*}\right)^2}, \qquad \bar{\alpha}^{*} = \frac{1}{B}\sum_{r'=1}^{B}\hat{\alpha}^{*r'}.$$

Example 11.7: Three observations

With $Z = \{(4.3, 2.4), (2.1, 1.1), (5.3, 2.8)\}$ (observations 1, 2, 3), possible bootstrap samples are $Z^{*1} = \{3, 1, 3\}$, $Z^{*2} = \{2, 3, 1\}$, $\dots$, $Z^{*B} = \{2, 2, 1\}$ (by observation number), each yielding its own $\hat{\alpha}^{*r}$.

In R, `sample(x, replace=T)` draws a bootstrap sample of the indices.

4 Selected exercises

Motivation. The first item, from the resampling chapter of *An Introduction to Statistical Learning*, explains the familiar "63%" of a bootstrap sample. The others make three claims of the chapter exact: what the bootstrap standard error converges to, why LOOCV estimates the prediction error, and how much the training error is too optimistic.

Exercise 11.1 (Which observations enter a bootstrap sample)

A bootstrap sample of size n is drawn with replacement from n observations. (After ISLR, Ch. 5, Exercise 2.)
(a) What is the probability that the first bootstrap observation is not the jth original observation? The second?
(b) Show that the probability that the jth observation is not in the bootstrap sample is $(1 - 1/n)^n$, and find its limit as $n \to \infty$.
(c) Find the probability that the jth observation is in the bootstrap sample for $n = 5, 100, 10{,}000$, and the expected number of distinct original observations in a bootstrap sample.

Solution.

(a) Each bootstrap draw selects one of the n observations with probability $1/n$, so it is not observation j with probability $1 - 1/n$. The second draw is made from the same full set, independently of the first, so the answer is again $1 - 1/n$.

(b)

Step 1. The n draws are independent, so $P(j \notin Z^*) = \prod_{r=1}^{n} P(\text{draw } r \neq j) = (1 - 1/n)^n$.

Step 2. $\log(1-1/n)^n = n\log(1-1/n) = -1-\frac{1}{2n}-\frac{1}{3n^2}-\cdots \to -1$, using $\log(1-u) = -u-u^2/2-\cdots$ for $|u|<1$. Hence $(1-1/n)^n \to e^{-1} = 0.368$.

(c)

Step 1. $P(j \in Z^*) = 1-(1-1/n)^n$ equals $1-0.8^5 = 0.672$ for $n=5$, $1-0.99^{100} = 0.634$ for $n=100$ and 0.632 for $n=10{,}000$, close to the limit $1-e^{-1} = 0.632$.

Step 2. The number of distinct observations is $\sum_{j=1}^{n} \mathbf{1}\{j \in Z^*\}$, so by linearity its mean is $n[1-(1-1/n)^n] \approx 0.632\,n$.

About 37% of the observations are left out of each bootstrap sample; these "out-of-bag" observations can serve as a test set for the rule fitted to that sample. ■

Exercise 11.2 (Ideal bootstrap standard error of the mean)

Let $x_1,\dots,x_n$ be fixed data and $\bar{x}^*$ the mean of a bootstrap sample $x_1^*,\dots,x_n^*$. Writing $\mathrm{E}^*, \mathrm{Var}^*$ for moments under bootstrap resampling, show that

$$\mathrm{E}^*(\bar{x}^*) = \bar{x}, \qquad \mathrm{Var}^*(\bar{x}^*) = \frac{\hat{\sigma}^2}{n}, \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2 = \frac{n-1}{n}s^2,$$

and conclude that $SE_B(\bar{x}) \to \hat{\sigma}/\sqrt{n}$ as $B \to \infty$.

Solution. [Supplement]

Step 1. The x_r^* are iid, each equal to x_i with probability $1/n$. Hence $\mathrm{E}^* x_r^* = \frac{1}{n}\sum_i x_i = \bar{x}$ and $\mathrm{Var}^* x_r^* = \frac{1}{n}\sum_i (x_i-\bar{x})^2 = \hat{\sigma}^2$.

Step 2. By independence, $\mathrm{E}^*\bar{x}^* = \bar{x}$ and $\mathrm{Var}^*\bar{x}^* = \hat{\sigma}^2/n$.

Step 3. The bootstrap replicates $\bar{x}^{*1},\dots,\bar{x}^{*B}$ are iid draws from this resampling distribution, so by the law of large numbers their sample variance converges to $\mathrm{Var}^*(\bar{x}^*)$, and $SE_B(\bar{x}) \to \hat{\sigma}/\sqrt{n}$ almost surely.

The limit is the plug-in standard error $\sqrt{(n-1)/n}\,s/\sqrt{n}$, slightly below the usual $s/\sqrt{n}$: the bootstrap treats the data as the population, with divisor n. The Monte Carlo error from finite B is a separate, controllable source of error. ■

Exercise 11.3 (LOOCV for the sample mean)

Predict every response by the sample mean, $\hat{f} \equiv \bar{y}$.
(a) Show that $y_i - \bar{y}_{(i)} = \frac{n}{n-1}(y_i-\bar{y})$, where $\bar{y}_{(i)}$ is the mean without observation i, and hence $CV_{(n)} = \left(\frac{n}{n-1}\right)^2 \frac{1}{n}\sum_i (y_i-\bar{y})^2$. Check this against the LOOCV shortcut.
(b) If $y_1,\dots,y_n$ are iid with mean μ and variance σ^2, show that $\mathrm{E}\,CV_{(n)} = \sigma^2(1+\frac{1}{n-1})$, the exact prediction error of a mean computed from $n-1$ observations.

Solution. [Supplement]

(a) $\bar{y}_{(i)} = \frac{n\bar{y}-y_i}{n-1}$, so $y_i - \bar{y}_{(i)} = \frac{(n-1)y_i - n\bar{y}+y_i}{n-1} = \frac{n(y_i-\bar{y})}{n-1}$. Squaring and averaging gives $CV_{(n)}$. The mean is LS regression on the constant column $\mathbf{X}=\mathbf{1}$, with $\mathbf{H} = \frac{1}{n}\mathbf{J}$ and $h_{ii} = \frac{1}{n}$; the shortcut gives $(y_i-\bar{y})/(1-\frac{1}{n})$, the same residual.

(b)

Step 1. $\mathrm{E}\sum_i (y_i-\bar{y})^2 = (n-1)\sigma^2$ for iid data (the unbiasedness of s^2). Hence $\mathrm{E}\,CV_{(n)} = \frac{n^2}{(n-1)^2}\cdot\frac{(n-1)\sigma^2}{n} = \frac{n\sigma^2}{n-1} = \sigma^2(1+\frac{1}{n-1})$.

Step 2. For a new y_0 independent of a sample of size $n-1$ with mean $\bar{y}_{n-1}$, $\mathrm{E}(y_0-\bar{y}_{n-1})^2 =$

$\mathrm{Var} y_0 + \mathrm{Var}\bar{y}_{n-1} = \sigma^2 + \frac{\sigma^2}{n-1}$, the same value.

So LOOCV is exactly unbiased for the prediction error of the rule fitted to $n-1$ points, and slightly pessimistic for the rule fitted to all n (whose error is $\sigma^2(1+\frac{1}{n})$), as the polynomial example suggested. ■

Exercise 11.4 (Optimism of the training error)

Let $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}$ an $n \times p$ matrix of rank p, $\mathrm{E}\boldsymbol{\varepsilon} = \mathbf{0}$, $\mathrm{Var}\boldsymbol{\varepsilon} = \sigma^2\mathbf{I}$, and $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ the LS fit. Let $\mathbf{y}^* = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}^*$ be new responses at the same x's, with $\boldsymbol{\varepsilon}^*$ independent of $\boldsymbol{\varepsilon}$ and distributed like it. Show that

$$\mathrm{E}\Big[\frac{1}{n}\|\mathbf{y} - \hat{\mathbf{y}}\|^2\Big] = \sigma^2\Big(1 - \frac{p}{n}\Big), \qquad \mathrm{E}\Big[\frac{1}{n}\|\mathbf{y}^* - \hat{\mathbf{y}}\|^2\Big] = \sigma^2\Big(1 + \frac{p}{n}\Big).$$

Solution. [Supplement]

Step 1. (Facts on $\mathbf{H}$.) $\mathbf{H}$ is symmetric and idempotent, $\mathbf{H}\mathbf{X} = \mathbf{X}$, and $\mathrm{tr}\mathbf{H} = \mathrm{tr}\big[(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}\big] = \mathrm{tr}\mathbf{I}_p = p$. For a random vector $\mathbf{u}$ with mean $\mathbf{0}$ and $\mathrm{Var}\mathbf{u} = \sigma^2\mathbf{I}$ and a fixed matrix A, $\mathrm{E}(\mathbf{u}^\top A\mathbf{u}) = \mathrm{tr}\big(A\,\mathrm{E}\mathbf{u}\mathbf{u}^\top\big) = \sigma^2\mathrm{tr}A$.

Step 2. (Training error.) $(\mathbf{I} - \mathbf{H})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$, so $\mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$ and $\|\mathbf{y} - \hat{\mathbf{y}}\|^2 = \boldsymbol{\varepsilon}^\top(\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$. By Step 1 its mean is $\sigma^2(n-p)$.

Step 3. (Prediction error.) $\hat{\mathbf{y}} = \mathbf{X}\boldsymbol{\beta} + \mathbf{H}\boldsymbol{\varepsilon}$, so $\mathbf{y}^* - \hat{\mathbf{y}} = \boldsymbol{\varepsilon}^* - \mathbf{H}\boldsymbol{\varepsilon}$ and

$$\mathrm{E}\|\mathbf{y}^* - \hat{\mathbf{y}}\|^2 = \mathrm{E}\|\boldsymbol{\varepsilon}^*\|^2 - 2\,\mathrm{E}(\boldsymbol{\varepsilon}^{*\top}\mathbf{H}\boldsymbol{\varepsilon}) + \mathrm{E}(\boldsymbol{\varepsilon}^\top\mathbf{H}\boldsymbol{\varepsilon}) = n\sigma^2 - 0 + p\sigma^2,$$

the middle term vanishing by independence and $\mathrm{E}\boldsymbol{\varepsilon}^* = \mathbf{0}$.

Step 4. Dividing by n gives both formulas. The gap $2p\sigma^2/n$ grows with the number of parameters: the training error decreases as the model grows while the prediction error eventually increases, which is why model selection uses cross-validation (or a penalty such as $2p\hat{\sigma}^2/n$) rather than the training error. ■